

INTRINSIC RECONSTRUCTION FROM NONREDUCED FAILURE SCHEMES

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ABSTRACT. We reconstruct every quadratic pencil in dimension $n \geq 3$ from the abstract $(n^2 + 2n - 4)$ -th neighbourhood of the socle Grassmannian in its nonreduced multiplication-failure scheme. Every lower-order neighbourhood is independent of the pencil, including on the singular-pencil locus. The proof intrinsically recovers a normal cone, its oriented rank-one ruling, and a residual coefficient module carrying the full Pluecker line. The first relation spaces embed the entire pencil parameter scheme and glue to a flat projective family of actual finite neighbourhoods under arbitrary complex base change. They recover the spectral Fitting incidence schemes and realize all Segre-stratum closures. A rank-preserving specialization distinguishes spectral partitions missed by the full discriminant and reduced rank loci. Independently, for every definite real pencil with r distinct eigenvalues, we give explicit real data for which all $2r - 3$ reciprocal likelihood critical points are real and nondegenerate, proving the assertion of Fevola–Mandelstam–Sturmfels, Conjecture 4.5. We also retain the intrinsic reconstruction of smooth-Jacobian four-dimensional webs, the all-ranks contraction-kernel and exterior-support theorems, the polarized K3 and hyperelliptic consequences, and the full boundary and polar-restriction constructions.

1. INTRODUCTION

Which defining equations can be recovered from an abstract degeneracy scheme? The principal result here gives an exact answer for the multiplication-failure schemes of quadratic pencils. For every $n \geq 3$, a pencil in $\mathrm{Sym}^2 \mathbb{C}^n$ is determined up to projective equivalence by the $(n^2 + 2n - 4)$ -th neighbourhood of the socle Grassmannian in its failure scheme. Every lower-order neighbourhood is independent of the pencil. This holds on the entire pencil Grassmannian, including singular pencils. The reduction is a fixed Schubert divisor and does not determine the equations.

The principal theorem chain is *intrinsic coefficient extraction, sharp finite reconstruction, and spectral geometry*. From the unmarked scheme we recover the socle Grassmannian, the degree-one normal bundle and its oriented rank-one ruling. An ideal quotient by the determinant line then recovers the entire Pluecker line of the pencil. These steps, not the formal passage from a homogeneous ideal to its first relation degree, account for intrinsic reconstruction. In fixed tensor coordinates the first relation spaces embed the full pencil parameter scheme. We prove that they glue to a universal family of the actual finite neighbourhoods, compatible with arbitrary base change. This realizes every classical Segre-stratum closure and its scheme-theoretic intersections, and recovers the spectral Fitting incidence schemes.

There is also a concrete consequence independent of the inverse problem. For every definite real pencil with r distinct eigenvalues, we construct a nonempty open set of real data for which all $2r - 3$ reciprocal likelihood critical points are real and nondegenerate. Theorem 11.1 proves the assertion formulated as Conjecture 4.5 in [40]. If the ordered eigenvalues are α_i , choose all the roots of a polynomial Q to their right and take residues $Q(\alpha_i)/D'(\alpha_i)$, where $D = \prod_i (t - \alpha_i)$. Two separated strings of sign changes exhaust the degree of the eliminated score polynomial. The

Date: September 23, 2026.

2020 Mathematics Subject Classification. 14M12, 14J28, 13C40, 14B05, 14D20, 05E05, 62R01.

Key words and phrases. Intrinsic reconstruction, pencils and webs of quadrics, Fitting schemes, Schur functors, exterior support, hyperelliptic curves, nilpotent filtration.

proof also excludes a missing coordinate chart and proves Hessian nondegeneracy. It concerns unrestricted real data, not positivity of the sample matrix or of every critical point.

The web theorem explains the additional recombination mechanism when the coefficient representation is reducible. Every basepoint-free four-dimensional web with smooth Jacobian quartic is recovered without a further genericity hypothesis. An all-ranks contraction-kernel formula provides the exterior-support gap. The polarized Jacobian K3 surface, hyperelliptic correspondence, contact and primary boundary layers, and polar constructions remain supporting results, with their full proofs retained. Neither these auxiliary constructions nor the isolated rank-preserving specialization is used as a substitute for the natural all-pencil family or the external likelihood theorem.

1.1. The multiplication failure scheme. Let V be a complex vector space of dimension $e \in \{3, 4\}$, put $p = e(e - 1)/2$, and let

$$R \subset \operatorname{Sym}^2 V$$

be an e -dimensional relation space. Set

$$S_R = \operatorname{Sym}^2 V / R, \quad B_R = \mathbb{C} \oplus V \oplus S_R,$$

with multiplication induced by the quotient $\operatorname{Sym}^2 V \rightarrow S_R$ and with $(V \oplus S_R)S_R = 0$. On

$$X_R = \operatorname{Gr}(e, V \oplus S_R)$$

let $\mathcal{K}$ be the tautological subbundle. For every $m \geq 2$ the multiplication map defines the same zeroth-Fitting scheme

$$\widehat{D}_R = V(\operatorname{Fitt}_0 \operatorname{coker}[\operatorname{Sym}^m(\mathcal{O} \oplus \mathcal{K}) \rightarrow B_R \otimes \mathcal{O}]). \quad (1.1)$$

Its reduction is the Schubert divisor $\widehat{\Delta}_R$ on which $\mathcal{K} \rightarrow V$ drops rank.

Regard R as a web of quadrics on $\mathbb{P}(V^*)$. On the invariant open G_e° used throughout, the web is basepoint-free and its Jacobian hypersurface

$$Y_R = V(\det(\partial q_i / \partial x_j)) \subset \mathbb{P}(V^*) \quad (1.2)$$

is smooth. For $e = 4$ it is a quartic K3 surface.

1.2. Recovering the original web. The web inverse theorem uses the full nonreduced scheme, including information at the deepest projection-corank stratum. The following conclusion holds without an additional genericity condition and is proved in Section 5.

Theorem 1.1 (Intrinsic web reconstruction). *For every pair $R, R' \in G_4^\circ$,*

$$\widehat{D}_R \simeq \widehat{D}_{R'} \implies R' = gR \text{ for some } g \in \operatorname{PGL}(V).$$

Consequently the abstract failure scheme determines B_R throughout the smooth Jacobian locus and separates every finite web-to-K3 packet contained in this locus.

The proof first recovers $\operatorname{Gr}(4, S_R)$ by iterated reduced singular loci. Its nonreduced normal cone identifies the two components of

$$\bigwedge^4 \operatorname{Sym}^2 V = \mathbb{S}_{(4,3,1,0)} V \oplus \mathbb{S}_{(5,1,1,1)} V.$$

The second component is the Jacobian covariant; the first is not retained by the quartic alone. The two projective components define a line in the Pluecker space. The contraction-kernel theorem (Lemma 5.12) computes the exact exterior support of the Jacobian component. For a smooth quartic that support has dimension ten. A secant or tangent four-vector on the Grassmannian has support at most eight. This excludes every second point and every infinitesimal ambiguity, so the line meets the Grassmannian in one reduced point. In fact three essential variables already

suffice. The global tensor rulings of the normal cone exclude the transposition ambiguity of an isolated square-matrix cone. No extension of an abstract scheme isomorphism to the ambient Grassmannian is assumed.

The exceptional fibres of the two-component invariant are described in Theorem S4.1. A proper intersection of the component line with the Grassmannian has length at most two. Its second point can coincide with the first, lie at a discarded pure endpoint, or give a distinct candidate exchanged by a regular residual involution. A positive-dimensional fibre is exactly a Grassmann line specified by a three-plane inside a five-plane. The tangent condition $Qw_R \in \wedge^3 R \wedge W$ distinguishes the double-point case from a reduced secant pair. All these alternatives are excluded on G_4° by Theorem 5.15. The ambient obstruction is confined to the binary-Jacobian boundary, defined invariantly by the first catalecticant. Thus the smooth family has no residual secant involution, flag-line ambiguity, or ramification divisor to resolve. Outside that family the component-fibre classification is not promoted to a classification of abstract failure-scheme isomorphisms.

1.3. The mechanism in arbitrary dimension. There is a second inverse theorem, for a family that is not confined to the two-summand plethysm in dimension four. If $\dim V = n$, polarization and the Jacobian determinant give natural maps

$$j_n : \det V \otimes \operatorname{Sym}^n V \longrightarrow \bigwedge^n \operatorname{Sym}^2 V, \quad C_n : \bigwedge^n \operatorname{Sym}^2 V \longrightarrow \det V \otimes \operatorname{Sym}^n V, \quad C_n j_n = 2 \operatorname{id}.$$

Put $Q_n = \frac{1}{2} j_n C_n$ and $P_n = 1 - Q_n$. Theorems 2.1–2.6 prove the exact all-ranks contraction kernel, the resulting support formula, and the following consequence. For every $n \geq 4$, on the nonempty open set of relation spaces whose Jacobians have at least three essential variables, the pair $([P_n w_R], [Q_n w_R])$ determines R . The corresponding pencil intersection is a single section even after nonreduced base change. The complement $\ker C_n$ need not be irreducible.

The numerical mechanism is uniform: for $d < n$ essential variables, the exterior support is $\binom{n+1}{2} - \binom{n-d+1}{2}$, and for $d = n$ it is full except for a one-dimensional annihilator of a nondegenerate quadratic power when n is even. The resulting gap above $2n$ excludes secants and tangents. The skew-flattening support criterion itself is classical [35, 36]; the calculation here is its restriction to the specified Jacobian component, including all singular and nondegenerate contraction kernels. In dimension four the intrinsic failure scheme supplies the required component pair, thereby turning the uniform inverse mechanism into Theorem 1.1. The smooth application has four essential variables; the algebraic exclusion is stronger and already applies with three.

1.4. Natural pencil reconstruction and discriminant fibres. Let $n \geq 3$, $N = n(n+1)/2$, and let $R \subset \operatorname{Sym}^2 V$ now have dimension two. Apply the same multiplication construction to the algebra with Hilbert function $(1, n, N-2)$. Theorem 7.3 proves

$$\hat{D}_R \simeq \hat{D}_{R'} \iff R' = gR \text{ for some } g \in \operatorname{PGL}(V).$$

Theorem 7.1 derives every hypothesis of the coefficient criterion for the wider family $R \in \operatorname{Gr}(r, \operatorname{Sym}^2 V)$, $1 \leq r < N-n$. Thus the pencil result is an application of the original multiplication failure scheme, not of an auxiliary map carrying a coefficient block.

For regular pencils, the Weierstrass–Segre classification uses root-labelled elementary divisors, not just the determinant [40, Theorem 1.1 and Corollary 2.1]. Example 7.5 exhibits two pencils with identical discriminants and distinct partitions over the unique double root. Their failure schemes are nonisomorphic, although their reductions and discriminant double covers agree. For $n = 2g + 2$, $g \geq 2$, the simple-discriminant locus recovers the classical hyperelliptic moduli correspondence [41, §4.2]; Corollary 7.6 gives an injective scheme invariant of the resulting $(2g-1)$ -dimensional family of curve classes. The new assertion is recovery from the unmarked failure scheme, not a new classification of pencils or curves.

For webs, the polarized K3 construction has a nine-dimensional image and is generically finite (Section 13). We do not determine its generic degree or assume it exceeds one. Our web theorem separates any fibre on the stated smooth locus. The actual loss of information by a classical invariant, used here to illustrate the strength of nonreduced reconstruction, is the explicit pencil discriminant example rather than a conjectural nontrivial K3 packet.

1.5. The first infinitesimal order that remembers the pencil. There is an exact bound on the amount of nonreduced structure needed for pencil reconstruction. Put $N = \binom{n+1}{2}$, $p = N - 2$, and $d = n + 2p = n^2 + 2n - 4$. If $\mathfrak{b}$ is the ideal of the intrinsic socle Grassmannian $B = \mathrm{Gr}(n, S_R)$ in $\widehat{D}_R$, let $Z_R^{[k]}$ denote its neighbourhood defined by $\mathfrak{b}^{k+1}$. Theorem 8.1 proves that every $Z_R^{[k]}$ with $k < d$ is independent of R , whereas the abstract scheme $Z_R^{[d]}$ determines R up to projective equivalence. The first nonzero homogeneous relation in its intrinsic conormal algebra recovers the entire normal cone. The bound is uniformly sharp for the entire family. It does not assert pointwise minimality against every comparison pencil. Once the homogeneous ideal and its intrinsic tensor structure are known, the first visible degree is formal; recovering that structure from the unmarked scheme is the substantive inverse step.

Theorem 8.3 strengthens the coefficient calculation at the level of parameter schemes: in fixed tensor coordinates the degree- d relation spaces define a closed immersion of the whole $\mathrm{Gr}(2, \mathrm{Sym}^2 V)$. It detects infinitesimal variations on singular as well as regular pencils, with no auxiliary coefficient block added to the original multiplication map. For hyperelliptic curves of genus g , Corollary 8.6 gives the exact order $4g^2 + 12g + 4$. The intrinsic assertion uses a neighbourhood of the entire socle Grassmannian; the parameter-family assertion retains its tensor coordinates. Neither asserts an equivalence of unmarked moduli stacks.

1.6. Fixed rank data and a changing spectral sheaf. The distinction persists even when all pointwise ranks are fixed. Theorem 9.3 recovers the torsion cokernel of the tautological quadratic form on the pencil's parameter line from the order- d neighbourhood. In Theorem 9.4, a flat projective family of these neighbourhoods has a constant full discriminant and constant reduced rank degeneracy loci, but the spectral partition at one root changes from $(3, 1)$ to $(2, 2)$. All lower-order neighbourhoods are constant families. The local lengths modulo successive powers of the spectral maximal ideal detect the change that the ranks and discriminant multiplicities miss. In dimension four the reciprocal curve is a conic on the punctured family and a line on the special fibre (Corollary 9.5). Thus the twentieth neighbourhood detects an explicit change in independently studied pencil geometry, while every preceding neighbourhood forgets it.

1.7. Universal families and a real spectral consequence. Theorem 10.1 proves that the first relation subbundles glue under right frame changes to the *actual* ideal-adic neighbourhoods. Their finite flat rank over the relative socle Grassmannian is $\binom{n^2+d}{d} - \binom{N}{2}$. The theorem applies to nonreduced parameter schemes while retaining the relative socle subscheme; it does not confuse its ideal with the absolute nilradical of a nonreduced base. Theorem 10.4 then recovers the schemes defined locally by $I_{n-h+1}((A^{-1}B - zI)^a)$, not just their closed points.

For a definite real pencil, the spectral points are real after normalization. Theorem 11.1 uses them to prove the all-real reciprocal likelihood assertion above, for arbitrary eigenvalue multiplicities. This real assertion has its own proof and its own specified real structure; no real form is silently recovered from an unmarked complex scheme.

1.8. Further coefficient constructions. For $\dim V = n \geq 2$, $d \geq 1$, and $0 \neq L \subset \mathrm{Sym}^d V^*$, consider $B = \mathrm{Gr}(n, H)$, where $\dim H > n$. On $\mathrm{Tot} \mathrm{Hom}(\mathcal{U}, V)$, the block map $\det T \oplus \mathrm{ev}_{L,T}$ combines the determinant with the coefficients of the pulled-back system. Theorem 6.3 proves that its zeroth-Fitting scheme Z_L determines L up to projective equivalence, from an abstract

isomorphism of schemes. Every reduction is the same determinant hypersurface. There is no smoothness restriction on L , and no supplied marking of the zero section or tensor factors. The structural criterion of Theorem 6.1 accounts for this family as well as the quadratic multiplication families. The polar construction is useful independently of recombination, but the natural pencil theorem, not this auxiliary block map, supplies the application to classical moduli in the preceding subsection.

1.9. Intrinsic reconstruction. On the projection-corank-at-most-one locus let D_R be the restriction of $\widehat{D}_R$, let $\mathcal{N}_R$ be its nilradical, and let E_R be the annihilator scheme of the first nilpotent layer. The first-layer reconstruction is the following result proved in Section 12.

Theorem 1.2 (Polarized reconstruction). *For $R \in G_e^\circ$, put*

$$\mathcal{L}_R = \omega_{E_R} \otimes \mathcal{N}_R^{\otimes -(p+e-1)}.$$

Then

$$\bigoplus_{n \geq 0} H^0(E_R, \mathcal{L}_R^{\otimes n}) \simeq \bigoplus_{n \geq 0} H^0(Y_R, \mathcal{O}_{Y_R}(en))$$

as graded algebras. In particular, for $e = 4$ the abstract failure scheme determines the polarized quartic K3 $(Y_R, \mathcal{O}_{Y_R}(1))$.

1.10. Organization and relation to prior work. Section 2 proves the dimension-uniform contraction and component-pair inverse theorems. Sections 3 and 4 construct the algebra and its Fitting scheme. Section 5 contains the complete reconstruction argument: the deepest stratum, its oriented normal cone, the universal coefficient readout, one common projective transformation, and the contraction-kernel and exterior-support theorems. The scheme structure of the cone is used directly; no primary decomposition of its ideal is needed. Section 7 proves automatic coefficient extraction, the all-pencil theorem, and the discriminant and hyperelliptic consequences. Section 8 proves the sharp finite-order and scheme-theoretic parameter results. Sections 10 and 11 supply the universal relative construction, spectral incidence readout, and the independent real likelihood theorem. Section 6 proves the abstract coefficient criterion and the polar-restriction theorem. Sections 12–14 compare this invariant with the polarized Jacobian surface and its moduli. Section 15 identifies the classical representation and secant tools and the precise additional contraction calculation.

The technical supplement is a separate reading object, not part of the proof of the main inverse theorem. It contains the full nilpotent-depth, primary-boundary, collision, relative-filtration, and exact-coordinate results, as well as the detailed Reye comparison and the ambient component-pencil classification. Its rank-defect schemes are defined by actual minors; their support is only contained in the binary-Jacobian boundary. No complete embedded-primary atlas of the highest-corank residual layers is used or asserted.

Finite exact checks accompany the calculations, but do not formally verify the written structural proofs. The historically relevant Ballico paper [2] remains a specific documentary issue: its publisher first page has been inspected, but its complete text has not been obtained. No theorem-level nonanticipation conclusion is drawn from that page, its title, or later citations. This does not change the hypotheses or conclusion of the reconstruction theorem proved here.

2. A UNIFORM CONTRACTION THEOREM AND ITS INVERSE CONSEQUENCE

The support mechanism is not confined to quartics. In this section let $\dim V = n \geq 2$, $W = \text{Sym}^2 V$, $N = \dim W = n(n+1)/2$, and $D = \det V$. Polarization defines a natural injection

$$j_n : D \otimes \text{Sym}^n V \longrightarrow \bigwedge^n W, \quad j_n(\epsilon \otimes \ell^n) = (\ell x_1) \wedge \cdots \wedge (\ell x_n), \quad \epsilon = x_1 \wedge \cdots \wedge x_n. \quad (2.1)$$

Changing the basis multiplies both sides by the same determinant. The map is nonzero and its source is irreducible, so it is injective. For $q \in W^* = \text{Sym}^2 V^*$ put $\kappa_{n,q} = \iota_q \circ j_n$. We regard q as the symmetric map $V \rightarrow V^*$; when it is invertible, $q^{-1} : V^* \rightarrow V$ defines an element of $\text{Sym}^2 V$. The pairing convention is $\langle q, xy \rangle = q(x, y)$.

Theorem 2.1 (The contraction kernel in every dimension). *For every nonzero $q \in \text{Sym}^2 V^*$, the following equalities hold:*

$$\ker \kappa_{n,q} = \begin{cases} D \otimes \text{Sym}^n(\text{rad } q), & q \text{ singular,} \\ 0, & q \text{ nondegenerate and } n \text{ odd,} \\ D \otimes \mathbb{C}(q^{-1})^{n/2}, & q \text{ nondegenerate and } n \text{ even.} \end{cases} \quad (2.2)$$

Thus for $\text{rank } q = r < n$ the kernel dimension is $\binom{2n-r-1}{n}$. For full rank it is zero or one according to the parity of n .

2.1. Shears and the singular kernel.

Lemma 2.2 (Shear descent in a fixed weight). *Let $V = A \oplus B$, with $B \neq 0$, and let $m \geq 1$. Suppose that $N \subset D \otimes \text{Sym}^m V$ is invariant under $\text{id}_A \oplus t \text{id}_B$, $t \in \mathbb{C}^*$, and under all shears*

$$u_L(a + b) = a + b + L(a), \quad L \in \text{Hom}(A, B).$$

If $N \not\subset D \otimes \text{Sym}^m B$, then

$$N \cap (D \otimes A \otimes \text{Sym}^{m-1} B) \neq 0.$$

More precisely, a nonzero vector in the A -degree- k weight space, $k > 0$, has a nonzero image in the degree-one space under a product of $k - 1$ infinitesimal rank-one shears. The same fixed nonzero vector of B can be used in every shear.

Proof. Put $s = \dim B$. The displayed torus acts on $D \otimes \text{Sym}^k A \otimes \text{Sym}^{m-k} B$ with weight $s + m - k$. These weights are distinct. Since a complex torus acts semisimply, its invariant subspace N is the direct sum of its intersections with these weight spaces. This justifies projecting a vector of N to one degree before differentiating; no cancellation between degrees is possible.

Choose a basis $a_1, \dots, a_r$ of A , a nonzero $b \in B$, and a basis vector ϵ of D . Write a nonzero degree- k vector as $\epsilon \otimes F$, where

$$F = \sum_{|\beta|=k} a^\beta f_\beta, \quad f_\beta \in \text{Sym}^{m-k} B.$$

Choose β with $f_\beta \neq 0$, and i with $\beta_i > 0$. Set $\alpha = \beta - e_i$. In $\partial_A^\alpha F$, the coefficient of a_i is exactly $\beta_i f_\beta$. Indeed a monomial that contributes to this coefficient must have exponent $\alpha + e_i = \beta$. This coefficient is nonzero in characteristic zero, so the derivative is a nonzero element of $A \otimes \text{Sym}^{m-k} B$.

The shear sending a_j to $a_j + tb$ and fixing the other basis vectors has determinant one. Its infinitesimal action on $D \otimes \text{Sym}^m V$ is consequently $1_D \otimes b \partial_{a_j}$, with no extra action on D . Invariance under the shear implies invariance under its infinitesimal action: the orbit is a polynomial in t , and all its coefficients belong to the invariant linear subspace. These differential operators commute, because every ∂_{a_j} kills b . Their product with multiplicities α_j sends our vector to

$$\epsilon \otimes b^{k-1} \partial_A^\alpha F.$$

This vector belongs to N . It is nonzero, since multiplication by b^{k-1} is injective in the polynomial algebra $\text{Sym } B$. Its exact bidegree is $(1, m - 1)$; thus its image in the corresponding weight-graded piece is also nonzero. The proof uses neither a special choice of b nor a restriction on r or s . When $k = 1$ the product of shears is the identity. This proves all assertions, including the low-dimensional cases. $\square$

Proof of Theorem 2.1 for singular q . Choose $V = A \oplus B$, where $B = \text{rad } q$ and $q|_A$ is nondegenerate. Pure powers from B are killed by contraction, so $D \otimes \text{Sym}^n B \subset \ker \kappa_{n,q}$. The stabilizer of q contains the B -scaling torus, all shears $a + b \mapsto a + b + L(a)$, and $O(A, q|_A) \times GL(B)$. Both the shears and the torus preserve q : adding a radical vector to either argument does not change its value. On $D \otimes \text{Sym}^n V$ the torus has the ordinary polynomial weights shifted by the constant $\dim B$, its determinant weight; no weight spaces are identified by that shift. Consequently any additional kernel vector would, by Lemma 2.2, give a nonzero kernel vector in $D \otimes A \otimes \text{Sym}^{n-1} B$. This module is irreducible for the last product group. Here the standard module of the full O_2 , not just SO_2 , is used: a reflection exchanges its two weight lines. The rank-one case is one-dimensional; in higher rank the standard orthogonal module is irreducible. The determinant character only twists this assertion.

Choose $a = x_1 \in A$ with $q(a, a) = 1$, a nonzero $b \in B$, and complete to a basis with $q(a, x_i) = 0$ for $i > 1$. Differentiate (2.1) at $\ell = b + ta$ and contract. Only its first factor contributes in degree one in t , giving

$$n\kappa_{n,q}(\epsilon \otimes ab^{n-1}) = (bx_2) \wedge \cdots \wedge (bx_n) \neq 0. \quad (2.3)$$

Multiplication by b is injective on V , which proves the nonvanishing. The restriction to the irreducible degree-one module is therefore injective, a contradiction. This proves the singular case, including all radical dimensions. $\square$

2.2. A cofactor identity for the scalar kernel. The exceptional even-dimensional line admits a proof that does not require a decomposition of the exterior-power target. The cofactor divergence cancellation is the exterior-valued version of the classical Piola identity, rather than a new divergence identity; see [39, equation (1.1), §3.1]. Its use below combines that cancellation with the algebraic Gaussian moment recurrence.

Lemma 2.3 (Algebraic cofactor cancellation). *For $q(x_i, x_j) = \delta_{ij}$ and $\rho = \sum_i x_i^2$, $\kappa_{n,q}(\epsilon \otimes \rho^{n/2}) = 0$ when n is even.*

Proof. Introduce indeterminates $t_1, \dots, t_n$, write $\ell(t) = \sum_i t_i x_i$, and set

$$a_i(t) = \ell(t)x_i, \quad A_i(t) = (-1)^{i-1}a_1 \wedge \cdots \wedge \widehat{a_i} \wedge \cdots \wedge a_n.$$

The symmetry $\partial_{t_i} a_j = x_i x_j = \partial_{t_j} a_i$ implies the polynomial identity

$$\sum_i \partial_{t_i} A_i = 0. \quad (2.4)$$

Indeed the term differentiating a_j in A_i cancels with the term differentiating a_i in A_j : for $i < j$, moving their common factor to the same position changes the sign by $(-1)^{j-i-1}$, opposite to their cofactor signs.

Define a linear functional $G : \mathbb{C}[t_1, \dots, t_n] \rightarrow \mathbb{C}$ by $G(t^\alpha) = 0$ if some α_i is odd, and otherwise $G(t^\alpha) = \prod_i (\alpha_i - 1)!!$, with $(-1)!! = 1$. The monomial recurrence gives $G(t_i F) = G(\partial_{t_i} F)$; this is an algebraic identity, not an appeal to convergence of a Gaussian integral. If $n = 2r$, the multinomial formula gives $G(\ell(t)^n) = (n-1)!! \rho^r$. On the other hand contraction of (2.1) gives

$$\kappa_{n,q}(\epsilon \otimes \ell(t)^n) = \sum_i t_i A_i(t).$$

Apply G coefficientwise and use (2.4). The result is $(n-1)!! \kappa_{n,q}(\epsilon \otimes \rho^r) = 0$, as required in characteristic zero. $\square$

Lemma 2.4 (Harmonics for the full orthogonal group). *Let $n \geq 2$, $\rho = x_1^2 + \cdots + x_n^2$, and $\mathcal{H}_a = \ker(\Delta : \text{Sym}^a \mathbb{C}^n \rightarrow \text{Sym}^{a-2} \mathbb{C}^n)$. For every $k \geq 0$,*

$$\text{Sym}^k \mathbb{C}^n = \bigoplus_{t=0}^{\lfloor k/2 \rfloor} \rho^t \mathcal{H}_{k-2t}. \quad (2.5)$$

Each $\mathcal{H}_a$ is irreducible for the full complex group O_n , and distinct nonnegative a give inequivalent modules. These statements remain true after tensoring every summand by the same determinant character. In particular they include $n = 2$.

Proof. For a harmonic homogeneous polynomial h of degree a ,

$$\Delta(\rho^t h) = 2t(2a + 2t + n - 2)\rho^{t-1}h.$$

Induction on k proves (2.5): the formula makes Δ an isomorphism from $\rho \text{Sym}^{k-2} \mathbb{C}^n$ to $\text{Sym}^{k-2} \mathbb{C}^n$ after decomposing the latter by the induction hypothesis. Its complementary kernel is $\mathcal{H}_k$.

Here is a proof of the group assertion that also treats the disconnected case. Restriction to the real unit sphere gives $\mathcal{H}_a$ a positive definite $O(n, \mathbb{R})$ -invariant Hermitian inner product. It is positive definite because a homogeneous polynomial vanishing on that sphere vanishes on $\mathbb{R}^n$, hence identically. The stabilizer of e_1 is the full $O(n-1, \mathbb{R})$. A homogeneous polynomial of degree a fixed by this stabilizer has the form

$$\sum_{j=0}^{\lfloor a/2 \rfloor} c_j x_1^{a-2j} (x_2^2 + \cdots + x_n^2)^j.$$

Indeed, expand in powers of x_1 ; invariance makes each homogeneous coefficient constant on the real spheres in the other variables, and invariance under their simultaneous negation kills its odd degrees. The harmonic equation is the recurrence

$$2(j+1)(2j+n-1)c_{j+1} = -(a-2j)(a-2j-1)c_j.$$

All denominators are nonzero, so the fixed subspace in $\mathcal{H}_a$ is one-dimensional. This argument includes $n = 2$, when the stabilizer is $O(1) = \{1, -1\}$.

Let K_v represent evaluation at a point v of the sphere under this inner product. The vector K_{e_1} is nonzero, as witnessed by the harmonic polynomial $\frac{1}{2}((x_1 + \sqrt{-1}x_2)^a + (x_1 - \sqrt{-1}x_2)^a)$. The K_v span $\mathcal{H}_a$ and form one orthogonal-group orbit. If $M \subset \mathcal{H}_a$ is invariant, its orthogonal projection commutes with the group and sends K_{e_1} into its one-dimensional stabilizer-fixed space. Idempotence makes that scalar zero or one. The orbit spans, so the projection is zero or the identity. This proves irreducibility for the real compact group, and therefore for its containing complex orthogonal group.

Finally, writing $E = \sum_i x_i \partial_i$, the invariant operator

$$\Omega = \sum_{i < j} (x_i \partial_j - x_j \partial_i)^2 = \rho \Delta - E(E + n - 2)$$

acts on $\mathcal{H}_a$ by $-a(a + n - 2)$. These eigenvalues are distinct for $a \geq 0$ and $n \geq 2$, which proves inequivalence. Tensoring by a common one-dimensional character is an invertible operation on representations. In dimension two the two nonzero SO_2 -weight lines are exchanged by a reflection; we have not asserted their irreducibility for SO_2 . For clarity, a complex subspace invariant under $O_n(\mathbb{C})$ is invariant under its compact real subgroup $O(n, \mathbb{R})$. The irreducibility just proved for that subgroup therefore gives irreducibility for the full complex group as asserted. $\square$

The Fischer decomposition and harmonic irreducibility are classical; compare [38, §1, equations (1)–(3)]. The proof above supplies the full-group and low-dimensional conventions needed here, rather than importing them from a stable-range statement.

Proof of Theorem 2.1 for nondegenerate q . Work in an orthonormal basis, and put $\mathcal{H}_m = \ker(\sum_i \partial_{x_i}^2 : \text{Sym}^m V \rightarrow \text{Sym}^{m-2} V)$. The classical harmonic decomposition is

$$D \otimes \text{Sym}^n V = \bigoplus_{t=0}^{\lfloor n/2 \rfloor} D \otimes \rho^t \mathcal{H}_{n-2t}. \quad (2.6)$$

The summands are irreducible and pairwise inequivalent for the full orthogonal group by Lemma 2.4. One can obtain the direct sum by applying the Laplacian to powers of ρ ; the nonzero coefficient on $\rho^t h_m$ is $2t(2m + 2t + n - 2)$. The harmonic summand has highest weight $(m, 0, \dots, 0)$ for $n \geq 3$. For $n = 2$ and $m > 0$ its two SO_2 -weight lines are exchanged by the other component of O_2 , so it is again irreducible. For $m = 0$ it is the scalar module. These facts also specify the determinant twists in (2.6).

It suffices to show that every summand with $m = n - 2t > 0$ has nonzero image. Put

$$h_m = \frac{1}{2}((x_1 + \sqrt{-1}x_2)^m + (x_1 - \sqrt{-1}x_2)^m), \quad \tau = (x_1 x_2) \wedge \cdots \wedge (x_1 x_n).$$

Then $h_m \in \mathcal{H}_m$. Polarization in (2.1) gives, for a degree- n polynomial f ,

$$[\tau] \kappa_{n,q}(\epsilon \otimes f) = [x_1^n] f - \binom{n}{2}^{-1} \sum_{i=2}^n [x_1^{n-2} x_i^2] f. \quad (2.7)$$

To check the coefficient, a contracted factor must be a square. Contracting the first column leaves τ only for x_1^n . Contracting column $i > 1$ leaves it only when that column is x_i^2 and the first is $x_1 x_i$; its wedge sign is minus. The source monomial has multinomial coefficient $\binom{n}{2}$. No other placement can leave all the factors of τ .

For $f = \rho^t h_m$, the coefficient of x_1^n is one and the sum in (2.7) is $(n-1)t - \binom{m}{2}$. Hence

$$[\tau] \kappa_{n,q}(\epsilon \otimes \rho^t h_m) = \frac{m(n+m-2)}{n(n-1)} \neq 0 \quad (m > 0). \quad (2.8)$$

Equivariance and irreducibility imply injectivity on each such summand. Pairwise inequivalence and complete reducibility prevent cancellation between them. For odd n this exhausts the source. For even n , the remaining line is killed by Lemma 2.3. A change of basis treats every nondegenerate q , and proves the theorem. $\square$

2.3. Exact support and reconstruction in all dimensions at least four. For $0 \neq z \in \wedge^n W$ its exterior support is the smallest subspace $S \subset W$ with $z \in \wedge^n S$. Its dimension is the rank of the first skew-flattening $c_z : W^* \rightarrow \wedge^{n-1} W$, $q \mapsto \iota_q z$. Indeed coordinates along $\ker c_z$ show that $S = (\ker c_z)^\perp$. This is the classical enclosing-space construction, not a new notion of rank [35, §10.1][36, §2.4]. The essential-variable space of a form f , in contrast, is a subspace of V , the annihilator of $\{\eta \in V^* : \partial_\eta f = 0\}$ [31].

Theorem 2.5 (Uniform support table). *Let $0 \neq f \in \text{Sym}^n V$ have d essential variables. For $0 \neq \epsilon \in D$, the exterior support of $j_n(\epsilon \otimes f)$ has dimension*

$$s_n(f) = \begin{cases} N - \binom{n-d+1}{2}, & d < n, \\ N - 1, & d = n, \text{ } n \text{ even, and } f \in \mathbb{C}^* h^{n/2} \text{ for a nondegenerate quadratic } h, \\ N, & \text{otherwise.} \end{cases} \quad (2.9)$$

For every $n \geq 4$, every form with at least three essential variables satisfies $s_n(f) > 2n$. Consequently its image is outside the closed second secant variety of $\text{Gr}(n, \text{Sym}^2 V)$.

Proof. If the essential-variable space is $H \neq V$, Theorem 2.1 says that the annihilating forms q are exactly $\text{Sym}^2(V/H)^*$. A nondegenerate annihilator would make f a power of a nondegenerate

quadratic, which has essential-variable space V : differentiating that power and cancelling a nonzero polynomial proves the assertion. The first support formula follows by taking the codimension of $\text{Sym}^2(V/H)^*$ in W^* .

If $H = V$, there is no nonzero singular annihilator. The nondegenerate case of the same theorem leaves precisely the stated even-dimensional exception. The projective quadratic root of $h^{n/2}$ is unique, by factorization in a polynomial ring. Thus all nonzero annihilators in the exceptional case are proportional, and the annihilator space has dimension one.

For $3 \leq d < n$, the first formula is at least $3n - 3 > 2n$ when $n \geq 4$. For $d = n$, even its lower bound $N - 1$ is greater than $2n$. A sum of two decomposable n -vectors has support at most $2n$. The locus defined by this support bound is closed, being a minors locus, so it contains the closed second secant variety. This last containment is classical [34, §6.1][35, §10.1]. $\square$

The determinant of the gradients defines a natural Jacobian map

$$C_n : \bigwedge^n \text{Sym}^2 V \longrightarrow D \otimes \text{Sym}^n V.$$

For $q_1 \wedge \cdots \wedge q_n$, in the basis used above, it is $\epsilon \otimes \det(\partial q_i / \partial x_j)$. Evaluating on $(x_1^2) \wedge (x_1 x_2) \wedge \cdots \wedge (x_1 x_n)$ and then using pure powers proves $C_n j_n = 2 \text{id}$. Hence $Q_n = \frac{1}{2} j_n C_n$ and $P_n = 1 - Q_n$ are canonical complementary projections. There is no assumption that the complement $\ker C_n$ is irreducible.

Theorem 2.6 (A family of inverse problems beyond dimension four). *Let $n \geq 4$. On $X = \text{Gr}(n, \text{Sym}^2 V)$, let U_n be the open locus where the Jacobian has at least three essential variables. This locus is nonempty. Both $P_n w$ and $Q_n w$ are nonzero there, and*

$$\mathbb{P}(\mathbb{C}P_n w \oplus \mathbb{C}Q_n w) \cap X = \{[w]\} \quad (2.10)$$

as schemes. The analogous relative intersection over U_n is the tautological section and remains so after every base change, including nonreduced base change. In particular the pair $([P_n w], [Q_n w])$ determines $[w]$, and an element of $\text{PGL}(V)$ transporting both members of the pair transports the underlying relation space.

Proof. The first-catalecticant rank condition shows that U_n is open. For $R = \langle x_1^2, \dots, x_n^2 \rangle$, the Jacobian is $2^n x_1 \cdots x_n$, with n essential variables, so this open is nonempty in every dimension under consideration. By Theorem 2.5, $Q_n w$ has support greater than $2n$, whereas w has support n . Neither projection can therefore vanish.

Every quadratic Pluecker relation restricts on the component line as $F(aP_n w + bQ_n w) = (b - a)\ell_F(a, b)$. If the linear forms ℓ_F span a two-dimensional space, their ideal sheaf on $\mathbb{P}^1$ is the unit ideal, and the intersection is the reduced point $[1 : 1]$. If their span has dimension one, the intersection is a degree-two divisor: either it contains a second decomposable point or it is doubled at $[w]$. A second point would express $Q_n w$ as a linear combination of two decomposable vectors. A double point would put $Q_n w$ in $\hat{T}_w X = \bigwedge^{n-1} R \wedge W$. Every vector in that affine tangent space is a sum of at most n terms, one for each omitted basis vector of R ; it is supported on R and at most n further directions. Both cases thus contradict support greater than $2n$. If all ℓ_F vanish, the entire line is contained in X , giving the same tangent contradiction. This proves (2.10).

For the relative assertion, the tautological Pluecker line and its two projections identify the component-line bundle with $\mathbb{P}^1 \times U_n$, with tautological section $[1 : 1]$. In local frames the coefficients of the ℓ_F form a matrix with two columns. Its rank is two on every geometric fibre, so its two-by-two minors generate the unit ideal on U_n . Locally two rows are invertible; hence the restricted ideal sheaf is exactly $(b - a)$. That calculation is preserved under every base change,

with no reducedness assumption on the new base. The final equivariance statement follows from the naturality of j_n and C_n . $\square$

Example 2.7 (A nondiagonal family in every dimension). For $n \geq 4$ and $t \neq 0$, put

$$R_t = \langle x_1^2 + tx_2x_3, x_2^2, \dots, x_n^2 \rangle.$$

Its Jacobian is $2^n x_1 \cdots x_n$, by expansion along the first column of the gradient matrix. The n first partials have distinct monomial supports, so the essential-variable dimension is n . The product is squarefree, hence not a positive power of a quadratic. Theorem 2.5 gives full exterior support $n(n+1)/2 > 2n$, and Theorem 2.6 applies.

This web is not a projective change of the diagonal web. A quadric $a_1(x_1^2 + tx_2x_3) + \sum_{i>1} a_i x_i^2$ with $a_1 \neq 0$ has rank at least two: its isolated nonzero x_1^2 block and its nonzero x_2x_3 block contribute separately. For $a_1 = 0$ it has rank one exactly at the $n-1$ square points $[x_2^2], \dots, [x_n^2]$. The diagonal web has n such points, an invariant distinction. This example belongs to the uniform component-reconstruction open, not to the smooth-Jacobian locus: its Jacobian is a normal-crossing hypersurface. For $n = 5, t = 1$ it gives an explicit nondiagonal example with exterior support fifteen.

Remark 2.8 (What is, and is not, dimension-specific). The kernel, support, and component-pair inverse theorems hold in all specified dimensions. For $n = 4$, the intrinsic normal cone of the multiplication-failure scheme supplies precisely that pair, which is the additional geometric input proved below. We do not assert that the same failure-scheme readout has been proved for $n > 4$. Thus the uniform result is an actual nonempty family of inverse problems, not a claim that every step of the four-dimensional geometric theorem automatically generalizes. The classical skew-flattening containment is used only after the restricted kernel has been calculated. A separate intrinsic extension, for a different family of actual Fitting failure schemes in every dimension, is proved in Section 6.

3. QUADRATIC RELATIONS AND A RELATIVE RAMIFICATION FAMILY

Put $N = e(e+1)/2$ and $P = \mathbb{P}(V^*)$. Let $\mathcal{H} \subset V \otimes \mathcal{O}_P$ be the tautological hyperplane bundle. Contraction is normalized by $c_\alpha(vw) = \alpha(v)w + \alpha(w)v$.

Lemma 3.1 (Restriction and contraction). *There is a locally split exact sequence*

$$0 \longrightarrow \mathrm{Sym}^2 \mathcal{H} \longrightarrow \mathrm{Sym}^2 V \otimes \mathcal{O}_P \xrightarrow{c} V \otimes \mathcal{O}_P(1) \longrightarrow 0.$$

The cokernels of $\mathrm{Sym}^2 \mathcal{H} \rightarrow S_R \otimes \mathcal{O}_P$ and $R \otimes \mathcal{O}_P \rightarrow V \otimes \mathcal{O}_P(1)$ are naturally isomorphic. Their determinant ideals define the same degree- e hypersurface when nonzero.

Proof. Locally choose ℓ with $\alpha(\ell) = 1$. The decomposition $\mathrm{Sym}^2 V = \mathrm{Sym}^2 H \oplus H\ell \oplus \mathbb{C}\ell^2$ identifies contraction with zero, the identity, and multiplication by two on the three summands, respectively. The twist accounts for rescaling α . Both cokernels are the quotient by $R + \mathrm{Sym}^2 \mathcal{H}$; hence their zeroth Fitting ideals agree. Finally, $\det c_R$ is a section of $\det V \otimes (\det R)^* \otimes \mathcal{O}_P(e)$. $\square$

Proposition 3.2. *If R is basepoint-free, its quadratic morphism $\phi_R : P \rightarrow \mathbb{P}(R^*)$ is finite and surjective of degree 2^{e-1} . Its ramification Fitting scheme is Y_R in (1.2). Moreover,*

$$R \text{ is basepoint-free} \iff \gamma_R(HV) = S_R \text{ for every hyperplane } H \subset V. \quad (3.1)$$

Proof. The pullback of $\mathcal{O}(1)$ is $\mathcal{O}_P(2)$. A positive-dimensional projective fibre contains a curve on which this ample line bundle would be trivial. Thus the proper morphism is quasi-finite, and hence finite. Its image has dimension $e-1$ and is the whole target. Intersecting $e-1$ pulled-back hyperplanes computes its degree.

For $J = (\partial q_i / \partial x_j)$, Euler's identity reads $J(x)x = 2(q_1(x), \dots, q_e(x))^t$. On charts where a source coordinate and a target coordinate are nonzero, choose the radial vectors first. The radial differential is a unit; its quotient is the projective differential. Removing that invertible block identifies the two ramification Fitting ideals. The contraction matrix is J^t , up to bases, so Lemma 3.1 applies.

A functional on S_R vanishing on $\gamma_R(HV)$ pulls back to one vanishing on $R + HV$. If $H = \ker \alpha$, then $(HV)^\perp = C\alpha^2$. Thus such a nonzero functional exists exactly when all $q \in R$ vanish at $[\alpha]$. $\square$

Proposition 3.3 (The universal smooth open). *The set G_e° is nonempty, open, and $\mathrm{PGL}(V)$ -invariant. The determinant construction over it is a relative effective Cartier divisor $\mathcal{Y} \subset G_e^\circ \times P$, proper and smooth over G_e° . It is obtained by descent from the frame bundle of the universal quotient, and its formation commutes with arbitrary base change. Each fibre is integral, with trivial canonical bundle; for $e = 4$ it also has $H^1(\mathcal{O}) = 0$.*

Proof. Fix a p -dimensional space S and set $T = \mathrm{Hom}(\mathrm{Sym}^2 V, S)$. Restriction over $T \times P$ to $\mathrm{Sym}^2 \mathcal{H}$ is a surjective evaluation map to the rank- p^2 bundle of square matrices. The incidence of corank at least two has codimension four on each fibre over P , hence dimension $\dim T + e - 5 < \dim T$. Its projection to T is proper and has proper closed image. Similarly, HV has dimension $p + e - 1$; nonsurjectivity of its restriction has codimension e and universal dimension $\dim T - 1$. Remove these two images and nonsurjective tensors. Also require invertibility of the square restriction at one fixed hyperplane. Each is a nonempty open condition on the irreducible space T , so their intersection T_1 is nonempty.

On $T_1 \times P$ the universal determinant section is nonzero on every fibre. Since a nonzero homogeneous polynomial is a non-zero-divisor on projective space over a field, its zero scheme Z is a relative Cartier divisor and is flat over T_1 . More explicitly, apply the fibrewise flatness criterion to $0 \rightarrow \mathcal{O}(-e) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_Z \rightarrow 0$: the first two terms are base-flat and the first map stays injective on every fibre. This also proves that the same sequence stays exact after any base change.

The total space Z is smooth over $\mathbb{C}$. Indeed, evaluation is a submersion in the tensor directions, and the determinant hypersurface of square matrices is smooth at matrices of corank one; the corank-two incidence was removed. Every fibre is a nonempty degree- e hypersurface. The generic fibre over T_1 is regular, since its local rings are localizations of the regular local rings of Z . In characteristic zero it is geometrically regular, hence smooth. The nonsmooth locus of $Z \rightarrow T_1$ is closed, and its image is closed by properness and misses the generic point. Deleting that image gives a nonempty open with smooth fibres.

Surjections $\mathrm{Sym}^2 V \twoheadrightarrow S$ form the frame bundle of the rank- p universal quotient on $\mathrm{Gr}(e, \mathrm{Sym}^2 V)$. The conditions of basepoint-freeness and smooth determinant fibre depend only on the kernel. Their bad loci have closed proper projections from the corresponding projective incidences. They therefore define the asserted open downstairs. The determinant ideal, relative Cartier property, and flatness descend along the faithfully flat frame bundle; flatness with smooth geometric fibres gives smoothness. The cokernel and Fitting constructions themselves commute with arbitrary base change. The extra fixed-hyperplane condition above is used only to produce a nonempty open in T ; it is not imposed in the definition downstairs.

A smooth positive-degree hypersurface in $\mathbb{P}^{e-1}$ is integral here: distinct positive-degree components would meet and give a singular point, and repeated components are not smooth. Adjunction gives $\omega_{Y_R} = \mathcal{O}_{Y_R}$. The hypersurface sequence gives genus one when $e = 3$ and $H^1(Y_R, \mathcal{O}) = 0$ when $e = 4$. Thus the latter is a K3 surface. Smoothness of the determinant also excludes corank at least two in the square restriction, since all cofactors then vanish. $\square$

Lemma 3.4. *The algebras B_R and $B_{R'}$ are isomorphic exactly when R and R' lie in the same $\mathrm{PGL}(V)$ -orbit.*

Proof. An isomorphism preserves the maximal ideal and its square. On their associated graded spaces it induces an isomorphism of V carrying the kernel of quadratic multiplication to the other kernel. Conversely such a linear isomorphism gives an isomorphism of the graded algebras. Scalar matrices act trivially on relation subspaces. $\square$

4. THE MULTIPLICATION IDEAL AND INTRINSIC TORSION

We first calculate on the whole Grassmannian. In this section $R \in G_e^\circ$ is fixed, and the subscript R is omitted.

Proposition 4.1 (The full Fitting presentation). *On a frame chart write $\mathcal{K} \rightarrow V \oplus S$ as $\begin{pmatrix} M \\ A_0 \end{pmatrix}$. For every $m \geq 2$,*

$$\mathcal{I}_{\widehat{D}} = (\det M)I_p(\gamma \operatorname{Sym}^2 M). \quad (4.1)$$

The reduction of $\widehat{D}$ is the integral divisor $\widehat{\Delta}$. Its smooth locus is the rank- $e - 1$ stratum. Consequently D is the restriction of $\widehat{D}$ to the smooth locus of its reduction, and this restriction is intrinsic.

Proof. Cube-zero multiplication and the unit show, over the coefficient ring and not only on fibres, that the image of every degree $m \geq 2$ is $\mathbb{C}1 + \mathcal{K} + \mathcal{K}^2$. Its presentation is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \operatorname{Sym}^2 M \end{pmatrix}.$$

Every nonzero maximal minor uses the scalar column and all e columns meeting the V -rows. Expanding on these rows proves (4.1). The graph block A_0 has no effect on this ideal. Off $\det M = 0$, the quadratic block is surjective because γ is surjective, so the ideal is the unit ideal. On the divisor it vanishes. The divisor is the usual irreducible Schubert divisor: the rank- $e - 1$ stratum is dense and integral, and every lower-rank plane is a specialization of that stratum. Equivalently the standard Schubert charts give its reduced determinantal equations.

Near a point with projection rank r , an invertible r -minor reduces the projection to $\operatorname{diag}(I_r, T)$, where the entries of the $(e - r) \times (e - r)$ block T are independent transverse coordinates. One can see their independence by varying the V -components of a basis of $K \cap S$ while keeping an invertible S -frame block fixed. The determinant equation is $\det T$. Its gradient is nonzero exactly at residual corank one. This proves the smooth-locus claim, including its assertion on the Grassmannian rather than on a chosen frame space. $\square$

On Δ put $H = \operatorname{im}(K \rightarrow V)$ and $\Lambda = K \cap S$. These give a morphism to $P \times \mathbb{P}(S)$. Define $E = \Delta \times_P Y$. The graph of K/Λ yields isomorphisms

$$\Delta \simeq \operatorname{Tot}_{P \times \mathbb{P}(S)} \operatorname{Hom}(\mathcal{H}, S/\Lambda), \quad (4.2)$$

$$E \simeq \operatorname{Tot}_{Y \times \mathbb{P}(S)} \operatorname{Hom}(\mathcal{H}, S/\Lambda). \quad (4.3)$$

Indeed the inverse construction takes the inverse image of a graph $H \rightarrow S/\Lambda$ in $H \oplus S$. Both constructions commute with base change. In particular E and Δ are smooth integral, of dimensions $ep - 2$ and $ep - 1$.

Theorem 4.2 (Ramification normal form). *On all of U the ideal of D is*

$$\mathcal{I}_D = \mathcal{I}_\Delta \mathcal{I}_E = \mathcal{I}_\Delta \cap \mathcal{I}_E^2. \quad (4.4)$$

The associated supports are exactly Δ and E . The nilradical is the invertible $\mathcal{O}_E$ -module $\mathcal{N} = \mathcal{O}_E(-\Delta)$ and is square-zero. Its annihilator recovers E as a scheme.

Proof. Near projection corank one, Gaussian operations make $M = \text{diag}(I_{e-1}, t)$. The quadratic block becomes $[A, tB, t^2c]$, with $A|_{t=0} = \gamma|_{\text{Sym}^2 H}$. Equation (3.1) says $[A, B]$ is onto locally; choose $AU + BV = I_p$. Multiplication by t then shows that $[A, tB, t^2c]$ and $[A, tI_p]$ have exactly the same column module. Their zeroth Fitting ideals agree. Off Y , A is invertible and (4.1) is (t) . On Y , the restriction has rank $p - 1$ by Proposition 3.3. A Schur complement gives a residual row with ideal (t, f) , where $f \bmod t$ is a local ramification equation. Thus the full ideal is

$$(t^2, tf) = t(t, f). \quad (4.5)$$

The pair t, f is a regular parameter pair cutting out E . The identity $(t) \cap (t, f)^2 = t(t, f)$ proves (4.4). These are primary ideals because their associated graded rings are polynomial rings over the integral smooth supports. At the generic point of E , neither primary component can be omitted. Their radicals are distinct, giving exactly the stated associated points. Finally the nilradical is $(t)/t(t, f)$; cancellation of t gives its module identification and annihilator $(t, f)/t(t, f)$. Both are intrinsic, so the calculation glues. $\square$

Lemma 4.3 (Generic torsion as a sheaf). *Let Z be noetherian with integral reduction and generic point η . For a coherent module $\mathcal{F}$ define*

$$\text{tor}_\eta(\mathcal{F}) = \ker(\mathcal{F} \longrightarrow i_* \widetilde{\mathcal{F}}_\eta), \quad i : \text{Spec } \mathcal{O}_{Z, \eta} \longrightarrow Z.$$

This kernel is coherent and is the largest coherent submodule with zero generic stalk. It is preserved by isomorphisms of schemes.

Proof. On an affine open meeting η this is the localization kernel at the prime of η . It is a submodule of a finite module over a noetherian ring, hence finite. Localization of that kernel on smaller affine opens is the corresponding kernel, so these modules glue. Every submodule with zero generic stalk lies in the kernel, and the kernel itself has zero generic stalk. The definition involves no ambient embedding or chosen primary decomposition. $\square$

5. THE DEEPEST NORMAL CONE AND RECONSTRUCTION OF THE WEB

The polarized Jacobian retains only one of two irreducible components of the Pluecker vector of a web. The full failure scheme retains both. In this section we recover these two projective components intrinsically and prove that they determine every web in the smooth Jacobian open. Exterior support excludes every exceptional pencil in this geometric family. Thus the deeper boundary separates the finite ambiguity of the polarized-K3 invariant. The normal cone used here is along the deepest Schubert stratum, not along the reduction of the failure scheme.

Throughout this section $\dim V = 4$, $W = \text{Sym}^2 V$, and $E = \wedge^4 W$. There is a multiplicity-free decomposition

$$E = E_{175} \oplus E_{35}, \quad E_{175} = \mathbb{S}_{(4,3,1,0)} V, \quad E_{35} = \mathbb{S}_{(5,1,1,1)} V \simeq \det V \otimes \text{Sym}^4 V. \quad (5.1)$$

Write P and Q for its two equivariant projections, in that order. For a web R , write $[w_R] \in \mathbb{P}(E)$ for its Pluecker point. On the locus where both projections are nonzero, put

$$\mathcal{L}_R = \mathbb{P}(\mathbb{C}Pw_R \oplus \mathbb{C}Qw_R).$$

The quadratic ideal of $\text{Gr}(4, W)$ is denoted by $\mathcal{I}_{P1,2}$. Each of its quadrics satisfies

$$F(aPw_R + bQw_R) = (b - a)\ell_{F,R}(a, b), \quad F \in \mathcal{I}_{P1,2}, \quad (5.2)$$

for a linear form $\ell_{F,R}$, since w_R is decomposable. Put $X = \text{Gr}(4, W)$, and let $X^\times$ be the open where both projections are nonzero. Set $H_w = \langle \ell_{F,w} : F \in \mathcal{I}_{P1,2} \rangle$, and let $\pi : X^\times \rightarrow \mathbb{P}(E_{175}) \times \mathbb{P}(E_{35})$ send $[w]$ to its pair of component lines. Define $G_4^{\text{rec}} \subset G_4^{\circ}$ by the conditions

$$Pw_R \neq 0, \quad Qw_R \neq 0, \quad \dim \langle \ell_{F,R} : F \in \mathcal{I}_{P1,2} \rangle = 2. \quad (5.3)$$

These conditions are independent of a representative of $[w_R]$ and invariant under $\text{PGL}(V)$.

Theorem 5.1 (Intrinsic reconstruction on the smooth Jacobian locus). *The reconstruction open is the whole smooth geometric locus: $G_4^{\text{rec}} = G_4^\circ$. Let $R \in G_4^\circ$ and $R' \in G_4^\circ$. Every isomorphism of complex schemes*

$$\widehat{D}_R \simeq \widehat{D}_{R'}$$

implies $R' = gR$ for some $g \in \text{PGL}(V)$. In particular, the abstract nonreduced failure scheme determines the cube-zero algebra B_R , up to isomorphism, throughout G_4° .

More precisely, the normal cone of $\widehat{D}_R$ along its intrinsically determined deepest Schubert stratum recovers $([Pw_R], [Qw_R])$, up to the same projective change of coordinates in V . The webs with this pair of projective components are exactly

$$(\mathcal{L}_R \cap \text{Gr}(4, W)) \setminus \{[Pw_R], [Qw_R]\}. \quad (5.4)$$

On G_4^{rec} , this intersection is the single reduced point $[w_R]$, before taking projective-coordinate orbits.

5.1. The intrinsic deepest stratum and its oriented tensor cone. For clarity, set

$$\Sigma_r(R) = \{K \in \text{Gr}(4, V \oplus S_R) : \text{rank}(K \rightarrow V) \leq r\}, \quad 0 \leq r \leq 3,$$

with reduced structure. Thus $\Sigma_3(R) = \widehat{\Delta}_R$ and $\Sigma_0(R) = \text{Gr}(4, S_R)$.

Lemma 5.2 (The deepest normal cone). *Starting with the reduction of $\widehat{D}_R$, three iterations of taking the reduced singular locus give $\Sigma_0(R)$. Consequently $\Sigma_0(R)$ and $\mathcal{C}_R = C_{\Sigma_0(R)/\widehat{D}_R}$ are intrinsic. Let $\mathcal{K}_0$ denote the tautological rank-four bundle on $\Sigma_0(R)$. The ambient vector bundle determined by the degree-one part is $\text{Hom}(\mathcal{K}_0, V)$. At a point $K_0 \subset S_R$, choose bases and write $T : K_0 \rightarrow V$. The fibre cone is*

$$\text{Spec } \mathbb{C}[t_{ij}]/(d(T)J_R(T)), \quad d(T) = \det T, \quad J_R(T) = I_6(\gamma_R \text{Sym}^2 T). \quad (5.5)$$

Its ideal is generated in degree sixteen and its reduction is the determinant hypersurface. The two rulings of its rank-one locus are intrinsically distinguishable over $\Sigma_0(R)$: the V -ruling is the projectively trivial one.

Proof. At a point of projection rank r , invert an r -minor and take the Schur complement. The remaining $(4-r)^2$ entries are independent coordinates and the rank conditions become the usual matrix rank conditions, times smooth auxiliary coordinates. The Jacobian criterion for minors gives

$$(\text{Sing } \Sigma_r(R))_{\text{red}} = \Sigma_{r-1}(R), \quad 1 \leq r \leq 3.$$

This also follows directly by differentiating the minors: the rank- r stratum is smooth, whereas at rank at most $r-1$ the tangent dimension exceeds the local dimension. Since reduction and reduced singular locus are preserved by abstract isomorphisms, $\Sigma_0(R)$ is intrinsic.

Choose a splitting $S_R = K_0 \oplus S'$. A Grassmann chart near K_0 consists of graphs

$$(T, B) : K_0 \longrightarrow V \oplus S'.$$

The deepest stratum is $T = 0$; B provides its eight smooth coordinates. The Fitting factorization gives precisely $d(T)J_R(T)$, independently of B . All generators of J_R have degree twelve in T , so passing to the (t_{ij}) -adic associated graded gives the same homogeneous ideal. This proves (5.5). There are no degree-one relations. The dual of the conormal bundle is therefore $\text{Hom}(\mathcal{K}_0, V)$, as also follows by quotienting the Grassmannian tangent bundle along $\text{Gr}(4, S_R)$. If T is invertible, $\gamma_R \text{Sym}^2 T$ is onto, so its maximal minors do not vanish simultaneously. Thus $V(dJ_R) = V(d)$; the determinant is irreducible, and the Nullstellensatz gives $\sqrt{dJ_R} = (d)$. This argument does not require any higher-colon or primary-boundary calculation.

The rank-one subcone is recovered from the reduced determinant cone by iterated reduced singular loci in each fibre. Its projectivization is the relative Segre variety

$$\mathbb{P}(V) \times \mathbb{P}(\mathcal{K}_0^*) \longrightarrow \Sigma_0(R).$$

The two components of the relative parameter space of its maximal linear $\mathbb{P}^3$'s are $\mathbb{P}(V) \times \Sigma_0(R)$ and $\mathbb{P}(\mathcal{K}_0^*)$. Their identification as schemes is Lemma 5.4 below. On points, note that a vector space of rank-one tensors cannot contain both $v_1 \otimes u_1$ and $v_2 \otimes u_2$ with both pairs independent: their sum has rank two. Thus a maximal such space fixes one tensor factor.

The first parameter bundle is projectively trivial. The second is not. Restrict to a Schubert line in $\mathrm{Gr}(4, 6)$, obtained by varying K_0 between a fixed three-plane and five-plane. On this line

$$\mathcal{K}_0^*|_{\mathbb{P}^1} \simeq \mathcal{O}_{\mathbb{P}^1}^{\oplus 3} \oplus \mathcal{O}_{\mathbb{P}^1}(1).$$

If its projectivization were a trivial $\mathbb{P}^3$ -bundle, the vector bundle would be a direct sum of four copies of one line bundle. This follows by comparing the relative $\mathcal{O}(1)$ on a projective-bundle isomorphism and pushing forward. Its determinant degree would then be divisible by four, contrary to the degree one displayed above. Triviality is invariant under an isomorphism of the base. Since the base is connected, an isomorphism of the cone cannot interchange these two parameter bundles on some fibres but not others. It must preserve the V -ruling. $\square$

Remark 5.3 (Why the ruling argument is needed). A single determinant cone of square matrices permits transposition. The preceding relative argument removes that ambiguity using only intrinsic data on $\Sigma_0(R)$; it does not assume that an isomorphism of failure schemes extends to the ambient Grassmannian. A linear isomorphism of tensor spaces preserving the two rank-one rulings has the form $g \otimes h$. Indeed, its restrictions to the two Segre factors are projective linear, and after lifting those maps the remaining linear map fixes every decomposable line. Applying it to sums with one common factor shows that its scalar is constant. Thus the coordinate change in the recovered V factor is an ordinary projective linear change, as required below.

5.2. Regular descent of the oriented ruling.

Lemma 5.4 (The relative scheme of maximal rank-one spaces). *Let B be any complex scheme and E, F vector bundles of rank $n \geq 2$. In $\mathrm{Gr}_B(n, E \otimes F)$, let $Z(E, F)$ be the zero scheme of the restriction of the quadratic rank-one equations to the universal subbundle $\mathcal{T}$. Equivalently it is the zero scheme of*

$$\mathrm{Sym}^2 \mathcal{T} \longrightarrow \bigwedge^2 E \otimes \bigwedge^2 F.$$

The natural morphism

$$\mathbb{P}_B(E) \sqcup \mathbb{P}_B(F) \longrightarrow Z(E, F), \quad \ell \mapsto \ell \otimes F, \quad m \mapsto E \otimes m, \quad (5.6)$$

is an isomorphism. It commutes with arbitrary change of complex base, including change to a nonreduced base.

Proof. We first work over $\mathbb{C}$. Each displayed map to the Grassmannian is a closed immersion. Its composition with the Pluecker embedding is the degree- n Veronese embedding on $\ell^{\otimes n} \otimes \det F$, respectively $\det E \otimes m^{\otimes n}$, followed by its natural linear inclusion into $\bigwedge^n(E \otimes F)$. This linear inclusion is injective: polarization of the nonzero pure-power map gives the corresponding irreducible symmetric-power summand. The rank-one equations vanish identically on both universal families, so the closed immersions factor through $Z(E, F)$.

Every geometric point is in one of the two images. Indeed the sum of two rank-one tensors is rank at most one only if their first factors or their second factors are proportional. Starting with two independent tensors, this observation forces a linear space consisting of rank-one tensors

to fix the same factor throughout. A space of dimension n is then exactly one of the displayed spaces. The two images are disjoint: a space fixing both factors has dimension one, not n .

We verify that these pointwise statements omit no infinitesimal structure. At $\ell \otimes F$, choose a generator of ℓ and a complement in E . A Grassmannian tangent vector is a map $\phi : F \rightarrow (E/\ell) \otimes F$. For each coordinate of E/ℓ , write its component as $A : F \rightarrow F$. Linearizing the restricted minors gives

$$u \wedge A(u) = 0 \quad \text{for every } u \in F.$$

Polarizing this diagonal identity gives

$$u \wedge A(v) + v \wedge A(u) = 0 \quad (u, v \in F).$$

In characteristic zero the diagonal and polarized identities are equivalent. These are precisely all coefficients of the bilinear linearization of the restricted quadratic map on $\text{Sym}^2 F$, so no infinitesimal equation is omitted by testing the diagonal. Taking the basis vectors shows that A is diagonal; taking their pairwise sums makes all its diagonal entries equal. Conversely a scalar A satisfies all the linearized equations. Thus the Zariski tangent space to $Z(E, F)$ is precisely $\text{Hom}(\ell, E/\ell)$, of dimension $n - 1$. The other image has the symmetric calculation.

At every closed point the closed immersed projective space gives local dimension at least $n - 1$, whereas the tangent calculation gives embedding dimension exactly $n - 1$. The local ring is therefore regular. A finite-type complex scheme whose closed-point local rings are regular is reduced: a nonempty support of its nilradical would contain a closed point. In particular there is no nilpotent thickening or embedded associated structure at either image. The disjoint union of the two closed immersions is itself a closed immersion, surjective on geometric points. Its ideal in the now reduced target vanishes at every closed point, so it is zero. This proves the scheme isomorphism over $\mathbb{C}$.

For general B , trivialize E, F on a common open cover. The Grassmannian and the zero scheme of the displayed universal quadratic map are, on such an open, the base change of the constant construction just proved. Thus (5.6) is an isomorphism there even if the base has nilpotents. The maps are natural under changes of frame and hence glue. This argument is an identification by equations and base change, not an inference of relative flatness from a fibrewise tangent calculation. $\square$

Lemma 5.5 (Oriented relative Segre descent). *Let B be a connected projective complex variety with $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and let V, V' have the same dimension $n \geq 2$. Let $\mathcal{U}, \mathcal{U}'$ be vector bundles of rank n on B . Suppose a linear bundle isomorphism*

$$\alpha : V \otimes \mathcal{U}^* \xrightarrow{\sim} V' \otimes \mathcal{U}'^*$$

preserves the rank-one cones and the ruling whose parameter bundle is $\mathbb{P}(V) \times B$. The induced isomorphism $\mathbb{P}(V) \times B \simeq \mathbb{P}(V') \times B$ is regular and is induced by one constant projective isomorphism $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$. Locally $\alpha(T) = gTh^{-1}$, with the same $[g]$ on every chart. The possible line-bundle twists of vector-bundle lifts do not change the projective left coefficient lines of any homogeneous functorial coefficient construction.

Proof. Lemma 5.4, applied to $E = V \otimes \mathcal{O}_B$ and $F = \mathcal{U}^*$, identifies the closed relative Fano scheme with $(\mathbb{P}(V) \times B) \sqcup \mathbb{P}_B(\mathcal{U}^*)$. In particular its two components carry their displayed scheme structures, with no additional nilpotent or embedded components. The isomorphism α induces a regular map of these relative parameter schemes, and the orientation hypothesis preserves the first component.

For completeness, an isomorphism of two trivial $\mathbb{P}^{n-1}$ -bundles over any base gives a regular projective-matrix-valued map, not merely a fibrewise collection. Its pullback of the relative $\mathcal{O}(1)$ has the form $\mathcal{O}(1) \otimes p^*L$ for a line bundle L on the base. To see this, tensor by $\mathcal{O}(-1)$; the

resulting line bundle is trivial on every projective-space fibre, and its pushforward and evaluation identify it with a pullback from the base. Pushing forward $\mathcal{O}(1)$ now gives an isomorphism of the corresponding trivial rank- n bundles, with that line-bundle twist. On charts trivializing L , it is an invertible matrix g_i of regular functions. On overlaps, $g_j = u_{ij}g_i$ for an invertible regular function u_{ij} . The classes $[g_i]$ thus glue to a regular morphism to the projective-isomorphism scheme, identified after choosing fixed bases with PGL_n . This accounts explicitly for the possible absence of a global linear lift.

The group PGL_n is affine: it is the determinant-nonzero principal open in the projective space of matrices. A morphism from B to an affine scheme factors through $\mathrm{Spec} H^0(B, \mathcal{O}_B) = \mathrm{Spec} \mathbb{C}$, so the projective map is constant. Choose a constant linear lift g . Each local g_i then differs from g by a scalar unit. After removing its two projective-factor actions, a linear map preserving every decomposable line is scalar: apply it to sums of two tensors with one common factor. Consequently the remaining action is a right bundle map, and the scalar units may be absorbed in its local lifts h . This proves the local tensor formula.

Finally a scalar multiplying a lift acts on a homogeneous functor of degree a by its scalar to the power a (or $-a$ on its dual). It cannot alter a projective coefficient line. For the degree-twelve construction below, both left Cauchy factors have the same degree twelve; the exterior-duality determinant multiplier is common as well. Thus neither line-bundle transition functions nor right changes of frame can introduce two different projective transformations on the left. $\square$

The lemma applies after pulling the target bundles back along the intrinsic base isomorphism $\sigma : \Sigma_0(R) \rightarrow \Sigma_0(R')$. Here the orientation is supplied by Lemma 5.2; it is not an extra marking of the failure scheme. Nor is a canonical basis of the trivial ruling required: different choices alter its representative only by constant projective coordinate changes.

5.3. The intrinsic residual ideal.

Lemma 5.6 (Residual graded data without a determinant equation). *On $B = \Sigma_0(R)$, put $\mathcal{A} = \mathrm{Sym}(V^* \otimes \mathcal{K}_0)$, and let $\mathcal{I} \subset \mathcal{A}$ be the homogeneous ideal of $\mathcal{C}_R$ in its intrinsic degree-one ambient bundle. Let $\mathcal{D} \subset \mathcal{A}$ be the ideal of the reduced cone. Its degree-four part is the line bundle*

$$L = \mathcal{D}_4 = (\det V)^* \otimes \det \mathcal{K}_0 \subset \mathcal{A}_4, \quad \mathcal{D} = L \otimes \mathcal{A}(-4). \quad (5.7)$$

The graded ideal

$$\mathcal{J} = (\mathcal{I} : \mathcal{D}) = \{a \in \mathcal{A} : a\mathcal{D} \subset \mathcal{I}\} \quad (5.8)$$

is intrinsic, and its degree-twelve part is the maximal-minor coefficient space used in the Schur readout. Multiplication gives canonical isomorphisms

$$L \otimes \mathcal{J}_{12} \xrightarrow{\sim} \mathcal{I}_{16}, \quad \mathcal{J}_{12} \simeq L^* \otimes \mathcal{I}_{16}. \quad (5.9)$$

The construction is preserved by graded cone isomorphisms. Its local residual-ideal calculation also commutes with arbitrary base change when $\mathcal{D}$ is transported as the determinant ideal.

Proof. The graded cone algebra determines $\mathcal{A}$ and its homogeneous kernel $\mathcal{I}$. Its reduction determines $\mathcal{D}$. In frames for V and $\mathcal{K}_0$, Lemma 5.2 says $\mathcal{D} = (d)$ and $\mathcal{I} = dJ$, where $d = \det T$ and $J = I_6(\gamma_R \mathrm{Sym}^2 T)$. The determinant coefficient spans $\wedge^4 V^* \otimes \wedge^4 \mathcal{K}_0$ inside $\mathrm{Sym}^4(V^* \otimes \mathcal{K}_0)$, proving (5.7) with its precise twist.

Multiplication by the generic determinant is injective over any coefficient ring. For example, choose a diagonal term order; its leading coefficient is a unit, so the leading term of df cannot vanish for any nonzero polynomial f . Therefore $(dJ : d) = J$. The ideal J is generated in degree twelve, which gives (5.9). If a frame changes d to ud , with u a unit, then $(dJ : d) = (dJ : ud)$. The local residual ideals hence glue in $\mathcal{A}$; the inverse line-bundle twist in (5.9) is exactly what cancels the transition functions of the determinant generators.

Every graded isomorphism transports $\mathcal{I}$, $\mathcal{D}$, and their colon, so it preserves this operation before any scalar equation has been chosen. After arbitrary base change the same identity $I_T = d_T J_T$ holds for the image ideals in the new polynomial algebra. The determinant remains a nonzerodivisor there, so $(I_T : d_T) = J_T$ again. No general claim that colons commute with base change is being used: it is the factorization and this universal nonzerodivisor that prove it here. On a nonreduced new base one transports the determinant ideal; one does not recompute the absolute reduction of the total cone, which need not commute with base change. $\square$

5.4. The universal Schur coefficient map. The distinction between an isotypic component and a specified line in its multiplicity space is essential here. We establish the coefficient identity before fixing the web. All vector spaces in this subsection are complex and four-dimensional. Put

$$\mathcal{F}(X) = \bigwedge^6 \text{Sym}^2 X, \quad \Lambda = \{(5, 4, 2, 1), (4, 4, 4, 0)\}.$$

For a linear map T write $\mathcal{F}(T) = \bigwedge^6 \text{Sym}^2 T$.

Lemma 5.7 (Universal coefficient identity). *The natural degree-twelve map*

$$\begin{aligned} \Phi_{V,U} : \mathcal{F}(V)^* \otimes \mathcal{F}(U) &\longrightarrow \text{Sym}^{12}(V^* \otimes U), \\ \Phi_{V,U}(\beta \otimes \xi)(T) &= \beta(\mathcal{F}(T)\xi) \end{aligned} \tag{5.10}$$

is $GL(V) \times GL(U)$ -equivariant. Under the functorial multiplicity-free decompositions

$$\mathcal{F}(X) = \mathbb{S}_{(5,4,2,1)} X \oplus \mathbb{S}_{(4,4,4,0)} X \tag{5.11}$$

and the Cauchy decomposition

$$\text{Sym}^{12}(V^* \otimes U) = \bigoplus_{\substack{\lambda \vdash 12 \\ \ell(\lambda) \leq 4}} \mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U,$$

the restriction of $\Phi_{V,U}$ to $\mathbb{S}_\lambda V^ \otimes \mathbb{S}_\mu U$ is zero if $\lambda \neq \mu$. For $\lambda = \mu \in \Lambda$, it is a nonzero scalar multiple of the canonical Cauchy inclusion. With compatible functorial normalizations this scalar is one. Consequently, for every fixed $\beta = \sum_\lambda \beta_\lambda$,*

$$\Phi_{V,U}(\mathbb{C}\beta \otimes \mathcal{F}(U)) = \bigoplus_{\lambda \in \Lambda} \mathbb{C}\beta_\lambda \otimes \mathbb{S}_\lambda U, \tag{5.12}$$

where each tensor product on the right is placed in its Cauchy summand. A zero component contributes the zero subspace.

Proof. The action on matrices is $T \mapsto gTh^{-1}$; the action on functions is $f(T) \mapsto f(g^{-1}Th)$. Functoriality of $\mathcal{F}$ shows directly that (5.10) intertwines this action with $\beta \otimes \xi \mapsto (\mathcal{F}(g^{-1})^*\beta) \otimes \mathcal{F}(h)\xi$. Thus equivariance is a statement about the universal map, not about a fixed- β slice.

For completeness, the exterior character identity used here is

$$[z^6] \prod_{1 \leq i \leq j \leq 4} (1 + zx_i x_j) = s_{(5,4,2,1)}(x) + s_{(4,4,4,0)}(x).$$

Equivalently, it follows by exterior duality from (5.1) and $\det(\text{Sym}^2 X) = (\det X)^5$. The two dimensions are 175 and 35, whose sum is $\binom{10}{6}$. The characteristic-zero Schur decomposition is functorial; the Cauchy formula is multiplicity-free as a representation of the product group [18, 8]. The source of Φ is the direct sum of the four irreducible product representations indexed by $\Lambda \times \Lambda$. Only the two diagonal ones occur in the target. Schur's lemma kills the off-diagonal blocks and makes each diagonal block a scalar multiple of its Cauchy inclusion. This also proves that no other left multiplicity line can enter the formula.

Here is an explicit nonvanishing check for both scalars. For an ordered basis $x_1, \dots, x_4$, order the monomials of $\text{Sym}^2 X$ as

$$(b_1, \dots, b_{10}) = (x_1^2, x_1x_2, x_1x_3, x_1x_4, x_2^2, x_2x_3, x_2x_4, x_3^2, x_3x_4, x_4^2).$$

Highest-weight vectors for the two summands are

$$\xi_{5421} = b_1 \wedge b_2 \wedge b_3 \wedge b_4 \wedge b_5 \wedge b_6, \quad \xi_{4440} = b_1 \wedge b_2 \wedge b_3 \wedge b_5 \wedge b_6 \wedge b_8.$$

Their weights are respectively $(5, 4, 2, 1)$ and $(4, 4, 4, 0)$. Every raising operator $x_i \partial / \partial x_j$, $i < j$, either gives zero or repeats a wedge factor, so both are highest vectors. Complete reducibility and the dimension sum above show that these two generated irreducible submodules exhaust $\mathcal{F}(X)$; this also proves (5.11) directly. Choose β_λ in the dual summand with $\beta_\lambda(\xi_\lambda) = 1$, and extend it by zero on the other summand. After identifying the ordered bases of U and V , evaluation at $T = \text{diag}(t_1, t_2, t_3, t_4)$ gives

$$\begin{aligned} \Phi(\beta_{5421} \otimes \xi_{5421})(T) &= t_1^5 t_2^4 t_3^2 t_4, \\ \Phi(\beta_{4440} \otimes \xi_{4440})(T) &= t_1^4 t_2^4 t_3^4. \end{aligned} \tag{5.13}$$

In particular both evaluations are one at the identity. This proves nonvanishing without a computed rank or a choice of numerical web. The canonical coefficient realization of the Cauchy inclusion is $\beta \otimes \xi \mapsto [T \mapsto \beta(\mathbb{S}_\lambda(T)\xi)]$. Compatible functorial splittings identify the restriction of $\mathcal{F}(T)$ with $\mathbb{S}_\lambda(T)$, giving scalar one; other scalar normalizations have the same image. Finally let ξ vary independently in its two summands. The off-diagonal vanishing and the nonzero diagonal scalars give (5.12). $\square$

Lemma 5.8 (Exterior duality and the common determinant twist). *Set $W = \text{Sym}^2 V$. The canonical isomorphism*

$$\begin{aligned} \bigwedge^4 W \otimes (\det W)^* &\longrightarrow \bigwedge^6 W^*, \\ w \otimes \omega &\longmapsto [\eta \mapsto \omega(w \wedge \eta)] \end{aligned} \tag{5.14}$$

sends the Pluecker line of $R = \ker \gamma_R$ to the line of $\beta_R = \gamma_R^(\text{vol}_{S_R})$. It identifies $E_{175} \otimes (\det V)^{-5}$ with $\mathbb{S}_{(5,4,2,1)} V^*$, and $E_{35} \otimes (\det V)^{-5}$ with $\mathbb{S}_{(4,4,4,0)} V^*$.*

Proof. Extend a basis of R to a basis of W . Both sides of the first assertion are nonzero alternating six-forms annihilating a wedge containing a vector of R , and on the six complementary basis vectors they agree after one scalar normalization. This proves the line identity. The map in (5.14) is natural for all isomorphisms of W . The highest weights of the two twisted summands are $(-1, -2, -4, -5)$ and $(0, -4, -4, -4)$, respectively: these are the negatives of the reversed partitions $(5, 4, 2, 1)$ and $(4, 4, 4, 0)$. Thus both labels and the common twist are as stated. Changing the volume multiplies both components by the same scalar. $\square$

5.5. Reading both Pluecker components from the residual ideal.

Lemma 5.9 (The two-component readout). *At any fibre of the oriented cone in Lemma 5.2, its degree-sixteen ideal recovers the projective pair $([Pw_R], [Qw_R])$ whenever both components are nonzero. It detects the vanishing of either component as well.*

Proof. Lemma 5.6 recovers the residual ideal as $\mathcal{J} = (\mathcal{I} : \mathcal{D})$, and its canonical degree-twelve coefficient module as $\mathcal{J}_{12} \simeq L^* \otimes \mathcal{I}_{16}$. In a fibre this is exactly $(J_R)_{12}$. This uses the nonreduced ideal, not just its support, and is independent of a local generator of the determinant line.

Let $U = K_0$ and choose a volume on S_R . Its pullback $\beta_R \in \wedge^6(\mathrm{Sym}^2 V)^*$ pairs with $\mathcal{F}(T)$ to give exactly the maximal minors of $\gamma_R \mathrm{Sym}^2 T$. Lemma 5.7 now gives, as an equality of subspaces in the Cauchy decomposition,

$$(J_R)_{12} = \bigoplus_{\lambda \in \Lambda} \mathbb{C}\beta_{R,\lambda} \otimes \mathbb{S}_\lambda U. \quad (5.15)$$

This is the fixed-slice consequence of the universal block identity. It does not treat the fixed slice as a $GL(V)$ -representation. Intersecting with a Cauchy summand and contracting its right factor with all elements of $(\mathbb{S}_\lambda U)^*$ recovers exactly $\mathbb{C}\beta_{R,\lambda}$, including its possible vanishing. Lemma 5.8 identifies these two lines with $[Pw_R]$ and $[Qw_R]$, with their labels and one common determinant twist. Right coordinate changes preserve the left lines; left coordinate changes act on both by the same representation of the coordinate transformation. Naturality under an abstract isomorphism is isolated in Lemma 5.10 below. $\square$

5.6. Naturality under an abstract isomorphism.

Lemma 5.10 (One projective transformation for both coefficient lines). *Let $\varphi : \widehat{D}_R \xrightarrow{\sim} \widehat{D}_{R'}$ be an abstract isomorphism, with $R, R' \in G_4^\circ$. It induces an isomorphism $\sigma : \Sigma_0(R) \rightarrow \Sigma_0(R')$ and a grading-preserving isomorphism of their normal cones. There is one $g \in \mathrm{PGL}(V)$, independent of the point of $\Sigma_0(R)$, which transports both recovered nonzero Pluecker component lines:*

$$[Pw_{R'}] = [P((\bigwedge^4 \mathrm{Sym}^2 g)w_R)], \quad [Qw_{R'}] = [Q((\bigwedge^4 \mathrm{Sym}^2 g)w_R)].$$

The corresponding vanishing statements hold for zero components. No extension of φ to either ambient Grassmannian is required.

Proof. The deepest stratum is intrinsic by Lemma 5.2. If $\mathfrak{b}$ and $\mathfrak{b}'$ are its ideals in the two failure schemes, pullback by φ identifies every power of these ideals. Hence it induces the graded algebra isomorphism

$$\sigma^* \left(\bigoplus_{m \geq 0} \mathfrak{b}'^m / \mathfrak{b}'^{m+1} \right) \simeq \bigoplus_{m \geq 0} \mathfrak{b}^m / \mathfrak{b}^{m+1}. \quad (5.16)$$

The symmetric algebra of degree one surjects canonically onto either algebra. Thus (5.16) also identifies its homogeneous kernel, not merely the cone's reduction; this is the functorial normal-cone construction [19]. Put $\mathcal{G} = \mathrm{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R}$, where $\mathfrak{b}$ is the deepest-stratum ideal. Its degree-one part is the conormal bundle $\mathcal{G}_1 = V^* \otimes \mathcal{K}_0$; its dual, not $\mathcal{G}_1$ itself, is $\mathrm{Hom}(\mathcal{K}_0, V)$. Thus the vector bundle is recovered before choosing either tensor factor.

Here is the whole construction in one diagram. The arrows $\rightsquigarrow$ mean assignments functorial under abstract isomorphisms, not morphisms or chosen retractions between the displayed schemes. Write $\mathcal{A} = \mathrm{Sym} \mathcal{G}_1$, with homogeneous kernel $\mathcal{I}$, and $\mathcal{D} = \sqrt{\mathcal{I}}$.

$$\begin{aligned} \widehat{D}_R &\rightsquigarrow B = \Sigma_0(R) \rightsquigarrow \mathcal{G} \rightsquigarrow (\mathcal{G}_1, \mathcal{A}, \mathcal{I}, \mathcal{D}), \\ \mathrm{Spec}_B(\mathcal{A}/\mathcal{D}) &\rightsquigarrow \{\mathrm{rank} \leq 1\} \rightsquigarrow \mathrm{Fano}_4 \rightsquigarrow \mathbb{P}(V) \times B, \\ (\mathcal{I}, \mathcal{D}) &\rightsquigarrow L = \mathcal{D}_4 \rightsquigarrow (\mathcal{I} : \mathcal{D})_{12} = L^* \otimes \mathcal{I}_{16} \rightsquigarrow \{\mathbb{C}\beta_{R,\lambda}\}_{\lambda \in \Lambda} \rightsquigarrow ([Pw_R], [Qw_R]). \end{aligned} \quad (5.17)$$

The first row uses the reduced-singular-locus construction and powers of its ideal. The second uses reduced rank strata, the closed relative Fano scheme of Lemma 5.4, and its uniquely projectively trivial component. The third uses the intrinsic colon, Cauchy projection and contraction of the entire right factor, followed by exterior duality. It involves no selected generator of L . Every row is transported by φ ; comparison of the two trivial ruling bundles, after pullback along σ , is what produces $[g]$. There is no coordinate map $[g]$ attached to an individual scheme before comparing its projective factors.

The relative rank-one Segre variety and its two ruling parameter bundles are preserved. Their distinction by projective triviality was proved in Lemma 5.2; in particular the V -ruling cannot be exchanged with the other ruling. Lemma 5.5, applied after pullback by σ , gives a regular isomorphism of the trivial $\mathbb{P}(V)$ -bundles. It explicitly accounts for the line-bundle ambiguity of its local linear lifts. Since $\Sigma_0(R)$ has only constant global regular functions, the resulting $\mathrm{PGL}(V)$ -valued morphism is constant. Denote its value by g .

Fix a point $U = K_0$ and its image $U' = K'_0$. A linear isomorphism preserving both rank-one rulings has, after a lift of g , the form $\alpha(T) = gTh^{-1}$ for some isomorphism $h : U \rightarrow U'$. Indeed the two Segre factors give projective linear maps. After lifting them, the remaining linear map fixes each decomposable line; sums of tensors with a common factor force its scalar to be constant. This scalar can be absorbed in h .

The only frame-dependent display is the following naturality square; all its objects were already recovered in (5.17). A change of a local lift g_i to $u_{ij}g_i$ is a line-bundle transition, not another projective coordinate map. A homogeneous degree-twelve coefficient functor acquires the common factor u_{ij}^{12} (or its inverse on a dual). It therefore preserves the recovered projective lines. The right-frame action is shown on the right tensor factor in the square:

$$\begin{array}{ccc} \mathcal{F}(V)^* \otimes \mathcal{F}(U') & \xrightarrow{\Phi_{V,U'}} & \mathrm{Sym}^{12}(V^* \otimes U') \\ \mathcal{F}(g)^* \otimes \mathcal{F}(h^{-1}) \downarrow & & \downarrow \alpha^* \\ \mathcal{F}(V)^* \otimes \mathcal{F}(U) & \xrightarrow{\Phi_{V,U}} & \mathrm{Sym}^{12}(V^* \otimes U). \end{array}$$

In fact both routes evaluate at T as $\beta'(\mathcal{F}(g)\mathcal{F}(T)\mathcal{F}(h^{-1})\xi')$. The induced graded map transports the determinant ideal line and the degree-sixteen cone ideal. By Lemma 5.6 it therefore transports $(\mathcal{I}' : \mathcal{D}')_{12}$ to $(\mathcal{I} : \mathcal{D})_{12}$, that is, $(J_{R'})_{12}$ to $(J_R)_{12}$ in the chosen fibres. This is the intrinsic residual operation, not a choice of scalar determinant equations. The universal diagonal-block identity gives

$$\mathbb{C}\beta_{R,\lambda} = \mathbb{C}\mathcal{F}(g)^*\beta_{R',\lambda} \quad (\lambda \in \Lambda)$$

in the respective left factors; h acts only on the right factor and cannot change either left line. Exterior duality in Lemma 5.8 now gives the displayed Pluecker identities. More explicitly the last arrow is the natural map

$$\bigwedge^4 \mathrm{Sym}^2 V \otimes (\det \mathrm{Sym}^2 V)^* \longrightarrow \bigwedge^6 (\mathrm{Sym}^2 V)^*, \quad (\det \mathrm{Sym}^2 V)^* = (\det V)^{-5}.$$

The factor $(\det V)^{-5}$ is common to both summands, so it disappears projectively; it cannot introduce two independent left transformations. Every step arose from the isomorphism of pairs, establishing the last assertion. $\square$

5.7. Exterior support of the Jacobian component. The obstruction to recombining two projective Schur components has a simple linear-algebraic description. A secant or tangent vector of $\mathrm{Gr}(4, W)$ uses at most eight directions in W . The Jacobian component of a smooth quartic uses all ten. We prove a sharper statement which also applies to singular quartics.

Fix a volume $\epsilon = x_1 \wedge x_2 \wedge x_3 \wedge x_4$ and suppress it when writing quartics. Define the equivariant map

$$j : \det V \otimes \mathrm{Sym}^4 V \longrightarrow \bigwedge^4 \mathrm{Sym}^2 V$$

by polarization of

$$j(\epsilon \otimes \ell^4) = (\ell x_1) \wedge (\ell x_2) \wedge (\ell x_3) \wedge (\ell x_4). \quad (5.18)$$

The right side changes by the determinant of a change of the x_i , so this definition does not depend on the chosen volume. More explicitly,

$$j(\epsilon \otimes v_1 v_2 v_3 v_4) = \frac{1}{24} \sum_{\sigma \in \mathfrak{S}_4} (v_{\sigma(1)} x_1) \wedge \cdots \wedge (v_{\sigma(4)} x_4). \quad (5.19)$$

This is nonzero. Its irreducible source and (5.1) identify its image with E_{35} . For the Jacobian map C , evaluation at $\ell = x_1$ gives $Cj(x_1^4) = 2x_1^4$; hence

$$Cj = 2 \text{ id}, \quad Q = \frac{1}{2} jC. \quad (5.20)$$

These identities also give $j = C^*/24$ in the Fischer normalization used below.

For $q \in W^* = \text{Sym}^2 V^*$, regard q as the symmetric bilinear functional $q(uv)$, and let ι_q denote exterior contraction. A quartic has *essential-variable space* $H \subset V$ if H is the smallest subspace for which it lies in $\text{Sym}^4 H$. Equivalently, $H^\perp$ is the kernel of its first differentiation map $V^* \rightarrow \text{Sym}^3 V$. This is the first-catalecticant characterization of essential variables [31, Proposition 1]; it also follows by choosing coordinates along the kernel and using characteristic zero. This subspace of V is distinct from the exterior-support subspace of $W = \text{Sym}^2 V$ defined below.

5.7.1. An independent four-dimensional character calculation. The general shear lemma is Lemma 2.2. The following character calculation gives a second proof of the even-dimensional scalar-kernel step when $n = 4$. Write $D = \det V$.

For the next lemma write $[a, b] = \text{Sym}^a U \otimes \text{Sym}^b U'$, where U, U' are the two standard two-dimensional representations of $SL_2 \times SL_2$. Only even weights occur below, so all representations descend through its double cover of SO_4 .

Lemma 5.11 (The exterior-cube character). *Let q be nondegenerate on the four-dimensional space V , and put $W = \text{Sym}^2 V$. Under $SL_2 \times SL_2 \rightarrow SO(V, q)$, choose $V = [1, 1]$. Then*

$$\begin{aligned} \bigwedge^3 W &\simeq [6, 0] \oplus [0, 6] \oplus [4, 4] \oplus 2[4, 2] \oplus 2[2, 4] \\ &\quad \oplus [2, 2] \oplus 2[2, 0] \oplus 2[0, 2]. \end{aligned} \quad (5.21)$$

In particular $(\bigwedge^3 W)^{SO(V, q)} = 0$, and $\text{Hom}_{O(V, q)}(D, \bigwedge^3 W) = 0$.

Proof. Put $A_2 = \text{Sym}^2 U$ and $B_2 = \text{Sym}^2 U'$. The symmetric Cauchy formula gives

$$W = (A_2 \otimes B_2) \oplus \mathbb{C}\rho, \quad \rho = q^{-1},$$

where ρ is invariant. Write $T = A_2 \otimes B_2$. For any two three-dimensional vector spaces, antisymmetrization in $(A_2 \otimes B_2)^{\otimes r}$, for $r = 2, 3$, gives

$$\begin{aligned} \bigwedge^2 T &= (\text{Sym}^2 A_2 \otimes \bigwedge^2 B_2) \oplus (\bigwedge^2 A_2 \otimes \text{Sym}^2 B_2), \\ \bigwedge^3 T &= (\text{Sym}^3 A_2 \otimes \bigwedge^3 B_2) \oplus (\mathbb{S}_{(2,1)} A_2 \otimes \mathbb{S}_{(2,1)} B_2) \oplus (\bigwedge^3 A_2 \otimes \text{Sym}^3 B_2). \end{aligned}$$

For clarity, the reason for the conjugate partitions in these formulas is that tensoring a symmetric-group representation with the sign representation conjugates its partition. Schur–Weyl decomposition of the two tensor powers therefore gives exactly the displayed summands. This is the usual skew Cauchy formula, not an additional hypothesis about this particular q .

Here are all the required one-factor calculations:

functor on $\text{Sym}^2 U$	restriction to SL_2
Sym^2	$\text{Sym}^4 U \oplus \mathbb{C}$
$\bigwedge^2$	$\text{Sym}^2 U$
$\bigwedge^3$	$\mathbb{C}$
Sym^3	$\text{Sym}^6 U \oplus \text{Sym}^2 U$
$\mathbb{S}_{(2,1)}$	$\text{Sym}^4 U \oplus \text{Sym}^2 U$

(5.22)

The weights of $\text{Sym}^2 U$ are $2, 0, -2$. Taking unordered pairs or triples of these weights verifies the symmetric-power rows; taking distinct weights verifies the exterior rows. Complete reducibility identifies the resulting characters with the stated representations. Alternatively the final row follows from $A_2 \otimes \bigwedge^2 A_2 = \mathbb{S}_{(2,1)} A_2 \oplus \bigwedge^3 A_2$: the tensor product $\text{Sym}^2 U \otimes \text{Sym}^2 U$ is $\text{Sym}^4 U \oplus \text{Sym}^2 U \oplus \mathbb{C}$, and the exterior cube removes the scalar summand.

Substitution gives

$$\begin{aligned} \bigwedge^2 T &= [4, 2] \oplus [0, 2] \oplus [2, 4] \oplus [2, 0], \\ \bigwedge^3 T &= [6, 0] \oplus [2, 0] \oplus [4, 4] \oplus [4, 2] \oplus [2, 4] \oplus [2, 2] \oplus [0, 6] \oplus [0, 2]. \end{aligned}$$

Since $\bigwedge^3 W = \bigwedge^3 T \oplus (\bigwedge^2 T) \wedge \rho$, these equalities prove (5.21). The dimensions are $84 + 36 = 120 = \binom{10}{3}$. There is no summand $[0, 0]$. Finally the determinant character of $O(V, q)$, namely its representation on D , restricts to the trivial character of $SO(V, q)$. Any equivariant image of D would therefore be a nonzero $SO(V, q)$ -invariant vector, which the decomposition excludes. This tracks the determinant twist without choosing an extension of each nontrivial SO_4 summand to O_4 . $\square$

Lemma 5.12 (The contraction kernel). *For $0 \neq q \in \text{Sym}^2 V^*$, let $\kappa_q : D \otimes \text{Sym}^4 V \rightarrow \bigwedge^3 \text{Sym}^2 V$ be the map $\epsilon \otimes f \mapsto \iota_q j(\epsilon \otimes f)$. If q is singular, then*

$$\ker \kappa_q = \det V \otimes \text{Sym}^4(\text{rad } q). \quad (5.23)$$

If q is nondegenerate, its inverse is a quadratic tensor $q^{-1} \in \text{Sym}^2 V$, and

$$\ker \kappa_q = \det V \otimes \mathbb{C}(q^{-1})^2. \quad (5.24)$$

Proof. For each fixed q , the map in the statement is explicitly

$$\kappa_q : D \otimes \text{Sym}^4 V \longrightarrow \bigwedge^3 \text{Sym}^2 V, \quad \epsilon \otimes f \longmapsto \iota_q j(\epsilon \otimes f).$$

Naturality of polarization and contraction makes this map equivariant for the stabilizer of q in $GL(V)$. The source carries the determinant factor D ; the target has its ordinary exterior representation. We use that equivariance in both cases.

Suppose first that q has rank $r \in \{1, 2, 3\}$. Choose $V = A \oplus B$, where $B = \text{rad } q$ and $q|_A$ is nondegenerate. Every pure power b^4 , $b \in B$, is killed by contraction in (5.18), since $q(bx) = 0$ for all $x \in V$. Pure powers span $\text{Sym}^4 B$, so $D \otimes \text{Sym}^4 B \subset \ker \kappa_q$.

The stabilizer contains $O(A, q|_A) \times GL(B)$, the B -scaling torus, and all shears in Lemma 2.2. On the degree-one weight space

$$D \otimes A \otimes \text{Sym}^3 B$$

the product-group representation is irreducible. Indeed its untwisted factors are the standard representation of the full orthogonal group and the irreducible symmetric cube for $GL(B)$. The determinant factor restricts to the product character $\det_A \det_B$, which does not change irreducibility. For $r = 1$ the standard factor is one-dimensional; for $r = 2$ its two isotropic weight

lines for SO_2 are exchanged by a reflection in O_2 , so neither is invariant under the full group. The usual standard representation is irreducible also for $r = 3$.

The restriction of κ_q to this space is not zero. Choose an orthogonal basis with $a = x_1 \in A$, $q(a, a) = 1$, and $b = x_4 \in B$, and set $\epsilon = x_1 \wedge \cdots \wedge x_4$. In the coefficient of st^3 in $\iota_q j(\epsilon \otimes (sa + tb)^4)$, only the contraction of the first factor contributes. Thus

$$4\kappa_q(\epsilon \otimes ab^3) = (bx_2) \wedge (bx_3) \wedge b^2 \neq 0.$$

The three factors are independent because multiplication by b is injective on V . Irreducibility now gives injectivity on the degree-one space. If the kernel contained any other positive A -degree, Lemma 2.2, with $m = 4$, would give a nonzero kernel vector in that degree-one space, a contradiction. This proves (5.23) for every singular rank, without a genericity condition on q .

Now let q be nondegenerate. After a change of coordinates, $q(x_i x_j) = \delta_{ij}$ and $\rho = q^{-1} = x_1^2 + x_2^2 + x_3^2 + x_4^2$. The harmonic decomposition is

$$D \otimes \text{Sym}^4 V = (D \otimes \mathcal{H}_4) \oplus (D \otimes \rho \mathcal{H}_2) \oplus (D \otimes \mathbb{C} \rho^2). \quad (5.25)$$

Here $\mathcal{H}_m$ is the kernel of the Laplacian in degree m . To check the decomposition directly, the symmetric Cauchy formula for $V = U \otimes U'$ in degree four has partitions (4), (3, 1), (2, 2), giving SO_4 -types $[4, 4]$, $[2, 2]$, $[0, 0]$ of dimensions 25, 9, 1. Multiplication by the invariant ρ identifies the degree-two and scalar pieces; differentiation verifies the harmonic description. The three summands are irreducible and pairwise nonisomorphic for SO_4 . The determinant factor restricts trivially to this group.

The last summand is killed: as an $O(V, q)$ -representation, $D \otimes \mathbb{C} \rho^2 \simeq D$, and Lemma 5.11 proves that the target of κ_q has no such character. It remains to show that the other two restrictions are nonzero. Put

$$h_4 = x_1^4 - 6x_1^2 x_2^2 + x_2^4 \in \mathcal{H}_4, \quad h_2 = x_1^2 - x_2^2 \in \mathcal{H}_2.$$

In the basis triple $\tau = (x_1 x_2) \wedge (x_1 x_3) \wedge (x_1 x_4)$, formula (5.19) gives

$$[\tau] \kappa_q(\epsilon \otimes h_4) = 2, \quad [\tau] \kappa_q(\epsilon \otimes \rho h_2) = \frac{2}{3}.$$

For the first equality the terms x_1^4 and $-6x_1^2 x_2^2$ contribute 1, 1. For the second the terms $x_1^4, x_1^2 x_3^2, x_1^2 x_4^2$ contribute 1, $-1/6, -1/6$; all other terms contribute zero. These are coefficient identities, not numerical rank assumptions. Each nonzero restriction from an irreducible summand is injective, and the images of the two nonisomorphic summands cannot intersect. The kernel is therefore precisely $D \otimes \mathbb{C} \rho^2$. Equivariance transports this conclusion back to the original form q , proving (5.24). $\square$

For $z \in \wedge^4 W$, write $s(z)$ for the dimension of its smallest exterior support: the smallest subspace S with $z \in \wedge^4 S$. In any basis,

$$s(z) = 10 - \dim\{q \in W^* : \iota_q z = 0\}. \quad (5.26)$$

Indeed the vanishing of contraction by a dual basis vector is exactly the absence of all wedge monomials containing its corresponding basis vector. Applying this after a change of basis proves both the formula and the existence of the smallest support.

Proposition 5.13 (Exact support of the Schur embedding). *Let $0 \neq f \in \text{Sym}^4 V$ have essential-variable dimension d . With the determinant factor understood,*

d	condition on f	$s(j(f))$
1	arbitrary	4
2	arbitrary	7
3	arbitrary	9
4	f not the square of a nondegenerate quadratic	10
4	f the square of a nondegenerate quadratic	9

(5.27)

In particular, $s(j(f)) \geq 9$ if $d \geq 3$, and it is 10 if the projective quartic $V(f)$ is smooth.

Proof. If $d < 4$ and H is the essential-variable space, the contraction kernel lemma says that the annihilating bilinear forms are exactly those with $H \subset \text{rad } q$. They form $\text{Sym}^2(V/H)^*$, of dimension $(4-d)(5-d)/2$. No nondegenerate form can annihilate $j(f)$, since the square of a nondegenerate quadratic has four essential variables. This gives the first three entries of (5.27).

If $d = 4$, no singular form annihilates $j(f)$. A nondegenerate form does so exactly when f is proportional to its inverse square. In this case its projective class is unique: proportional squares of nonzero quadratics over $\mathbb{C}$ have proportional square roots, and inversion is projectively unique. The annihilator therefore has dimension one in the square case and zero otherwise. Formula (5.26) gives the last two entries.

A quartic with fewer than four essential variables is a cone and is singular at its vertex. A square of a quadratic is nonreduced and has vanishing gradient along that quadratic. Neither defines a smooth quartic surface, proving the last assertion. $\square$

Lemma 5.14 (The eight-dimensional secant and tangent bound). *A sum of two decomposable four-vectors in $\bigwedge^4 W$ has exterior support at most eight. Every vector in the affine tangent space $\bigwedge^3 R \wedge W$ to the Pluecker cone at $w_R \neq 0$ also has exterior support at most eight.*

Proof. For a sum, take the sum of the two supporting four-planes. For a tangent vector, choose a basis $r_1, \dots, r_4$ of R . Every element of $\bigwedge^3 R \wedge W$ has the form

$$\sum_{i=1}^4 r_1 \wedge \dots \wedge r_{i-1} \wedge u_i \wedge r_{i+1} \wedge \dots \wedge r_4$$

for four vectors $u_i \in W$, and is supported in $R + \langle u_1, \dots, u_4 \rangle$. $\square$

Theorem 5.15 (No exceptional recombination above three essential variables). *Let $R \in \text{Gr}(4, \text{Sym}^2 V)$ have nonzero Jacobian quartic $f_R = Cw_R$ with at least three essential variables. Then both Pw_R, Qw_R are nonzero and*

$$\dim H_{w_R} = 2, \quad \mathcal{L}_R \cap \text{Gr}(4, W) = \{[w_R]\}$$

as schemes. In particular,

$$G_4^{\text{rec}} = G_4^{\circ}. \quad (5.28)$$

The secant, double-point, pure-endpoint and flag-line strata of the component pencil are all disjoint from G_4° .

Proof. By (5.20) and Proposition 5.13, the vector Qw_R has support at least nine. It is nonzero and cannot be proportional to the decomposable vector w_R ; hence $Pw_R \neq 0$. On the component line, the quadratic Pluecker ideal restricts to $(b-a)H_{w_R}$, where $H_{w_R} = \langle \ell_{F,R} \rangle \subset H^0(\mathbb{P}^1, \mathcal{O}(1))$. If this space has dimension two, its ideal sheaf is $(b-a)$. If it has dimension one, the intersection is the degree-two divisor of $(b-a)\ell$, including multiplicities. If it is zero, the entire line is contained in the Grassmannian. These assertions follow directly from the ideal generated by the Pluecker quadrics; no classification of an ambient boundary is used.

If the closed pencil has a distinct second decomposable point $z = aPw_R + bQw_R$ with $a \neq b$, then $z = aw_R + (b - a)Qw_R$. Thus Qw_R is a linear combination of two decomposable vectors, contradicting Lemma 5.14. This argument includes a pure endpoint. If the residual point coincides with $[w_R]$, then the line is tangent at $[w_R]$: its restricted equations have zero first derivative there. Its direction is Qw_R modulo $\mathbb{C}w_R$, so $Qw_R \in \wedge^3 R \wedge W$, giving the same contradiction. A whole Grassmann line also has that tangent property. All alternatives to rank two are excluded.

Smoothness of Y_R implies four essential variables, so every $R \in G_4^\circ$ satisfies this conclusion. The equality in (5.28) holds as an equality of open subschemes. Here G_4° has the reduced open-subscheme structure inherited from $\text{Gr}(4, 10)$. The complement is defined by the two-by-two minors of the two-column coefficient matrix of the residual linear forms. It is a closed subscheme of finite type over $\mathbb{C}$ with no geometric points, hence is empty, regardless of possible nilpotents in that determinantal presentation. The statement for all the named exceptional strata follows at the same time. $\square$

Corollary 5.16 (An invariant location of the ambient obstruction). *On $X^\times$, the support of the rank-defect locus $\dim H_w \leq 1$ is contained in the inverse image under the Jacobian covariant of the subspace variety of binary quartics. That variety is defined set-theoretically by rank at most two of the first catalecticant $V^* \rightarrow \text{Sym}^3 V$. In particular, it produces no exceptional divisor or ramification locus in the smooth geometric family, and no pair of non-projectively-equivalent smooth webs with isomorphic full failure schemes.*

Proof. The containment is the contrapositive of Theorem 5.15; the catalecticant characterization is the essential-variable characterization above. The inverse-scheme assertion follows from the intrinsic readout and Theorem 5.1. No equality with the whole binary-Jacobian inverse image, or classification of its ambient irreducible components, is asserted. $\square$

Proof of Theorem 5.1. Theorem 5.15 proves that the rank-two condition holds on all of G_4° , rather than merely on a nonempty subopen. The coefficient-rank condition in (5.3) defines this open algebraically. Nonemptiness of the geometric locus is independently witnessed by the smooth integral web C_* of Lemma 13.1; Appendix S14 retains its exact separating quadrics and the square-web normalization.

Indeed the restricted quadrics are $(b - a)H_{w_R}$. When H_{w_R} is the full space of linear forms, those forms generate the unit ideal sheaf on the line, and the intersection has ideal sheaf $(b - a)$. It is the single reduced point $[w_R]$. Without that rank condition, the points with the same two nonzero projective components are precisely the decomposable points $[aPw_R + bQw_R]$ with $ab \neq 0$, which proves (5.4). This is an exact characterization of the ambiguity of the *two-component invariant*; it need not describe every additional invariant of the full failure scheme on an exceptional locus. If the line is not contained in the Grassmannian, its scheme-theoretic intersection has length at most two, because the Grassmannian ideal is generated by quadrics.

An abstract isomorphism $\widehat{D}_R \simeq \widehat{D}_{R'}$ induces the graded normal-cone map of Lemma 5.10. Its commutative coefficient square and Lemma 5.7 give one $g \in \text{PGL}(V)$ for which the pairs $([Pw_{R'}], [Qw_{R'}])$ and $([P(gw_R)], [Q(gw_R)])$ agree. The two components are transported by the same g ; the source-factor change acts only on the right Cauchy factors. Replace R' by $g^{-1}R'$. Its Pluecker point belongs to the pencil $\mathcal{L}_R$, and both components are nonzero. The reduced one-point intersection gives $[w_{R'}] = [w_R]$, hence $R' = R$. The algebra-orbit equivalence of Lemma 3.4 gives the assertion for B_R . $\square$

Corollary 5.17 (Separation of the Torelli packet). *On the finite etale locus of the web-to- $K3$ map in Theorem S3.3, every fibre is separated by the isomorphism class of the full failure scheme. No further restriction to a reconstruction subopen is necessary. Distinct projective web classes in*

one such fibre have nonisomorphic failure schemes. On the regular Reye locus, the full failure scheme therefore also determines the web's Reye congruence and its Enriques datum.

Proof. Every web in the indicated finite étale cover belongs to $G_4^\circ = G_4^{\text{rec}}$. Theorem 5.1 therefore separates all its projective web classes directly. The Reye and Enriques constructions on the regular web locus are functorial under projective equivalence, which proves the last assertion. $\square$

The result concerns the actual degree-sixteen ideal of the deepest transverse failure cone. It uses information that is lost on passing to the reduced incidence image or to the Jacobian quartic alone. It neither identifies an incidence-resolution decomposition with a primary decomposition of W_3 or W_4 , nor asserts that all higher embedded primary components have been classified. Its geometric conclusion is instead an intrinsic inverse theorem: for every smooth Jacobian web, the deeper nonreduced boundary recovers the original relation space.

6. AN INTRINSIC COEFFICIENT PRINCIPLE

The web problem combines two operations: extracting coefficient data from an abstract scheme and reconstructing a relation space from those data. The first operation admits a formulation independent of webs or quadratic multiplication. We state the criterion and retain its polar-restriction application for linear systems in every dimension. Theorem 7.1 subsequently derives all of its geometric hypotheses directly for a full Grassmannian family of the original quadratic multiplication schemes.

6.1. The structural criterion. Let X be a finite-type complex scheme with a closed subvariety B specified by a construction preserved by every abstract scheme isomorphism. Write $\mathfrak{b} \subset \mathcal{O}_X$ for its ideal and

$$\mathcal{G} = \bigoplus_{r \geq 0} \mathfrak{b}^r / \mathfrak{b}^{r+1}, \quad \mathcal{A} = \text{Sym}_{\mathcal{O}_B}(\mathcal{G}_1), \quad \mathcal{I} = \ker(\mathcal{A} \twoheadrightarrow \mathcal{G}).$$

Assume the following conditions, for fixed integers $n \geq 2, m \geq 1$.

- (i) B is connected and projective, $H^0(B, \mathcal{O}_B) = \mathbb{C}$, and $\mathcal{G}_1$ is locally free. Its dual admits a presentation $V \otimes \mathcal{U}^*$, where $\dim V = \text{rank } \mathcal{U} = n$.
- (ii) The reduction of $\text{Spec}_B \mathcal{G}$ in its degree-one ambient bundle is the generic determinant cone. The bundle $\mathbb{P}_B(\mathcal{U}^*)$ is not projectively trivial. Thus precisely one of the two ruling parameter bundles of its rank-one locus is projectively trivial.
- (iii) If $\mathcal{D} \subset \mathcal{A}$ is the ideal of that reduced cone and $\mathcal{L} = \mathcal{D}_n$, then $\mathcal{I} = \mathcal{D}\mathcal{J}$, where $\mathcal{J}$ is generated in degree m . Under the Cauchy decomposition in any local frame of $\mathcal{U}$,

$$\mathcal{J}_m = \bigoplus_{\lambda \in \Lambda} L_\lambda \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^m(V^* \otimes \mathcal{U}), \quad L_\lambda \subset \mathbb{S}_\lambda V^*, \quad (6.1)$$

where Λ is a specified collection of partitions of m of length at most n , and the left subspaces are constant over B . Zero subspaces are permitted. The Cauchy inclusions, not just their representation labels, are the coefficient realizations $\beta \otimes \xi \mapsto [T \mapsto \beta(\mathbb{S}_\lambda(T)\xi)]$.

These conditions are hypotheses on the normal cone and its residual coefficient module. They do not assume that an abstract isomorphism extends to an ambient space, preserves a chosen frame, or already acts on all the left subspaces through a common projective transformation.

Theorem 6.1 (Intrinsic coefficient reconstruction). *Suppose X, B satisfy (i)–(iii). The abstract scheme X determines the collection $(L_\lambda)_{\lambda \in \Lambda}$ up to one common projective coordinate transformation on V . More precisely, an isomorphism between two such schemes induces a constant projective isomorphism $[g] : \mathbb{P}(V) \rightarrow \mathbb{P}(V')$ such that*

$$L_\lambda = \mathbb{S}_\lambda(g)^* L'_\lambda \quad (\lambda \in \Lambda).$$

The equality means equality of subspaces; scalar changes of a linear lift of $[g]$ do not affect it.

Proof. An isomorphism transports B , every power of $\mathfrak{b}$, $\mathcal{G}_1$, and the surjection $\mathcal{A} \rightarrow \mathcal{G}$. In particular it transports $\mathcal{I}$ in the intrinsic degree-one ambient bundle. Iterated reduced singular loci of the determinant cone give its rank-one locus. Lemma 5.4 reconstructs its two ruling parameter schemes, including their scheme structure. Condition (ii) distinguishes the trivial ruling intrinsically. Lemma 5.5 then gives one constant projective map $[g]$. After choosing a constant linear lift and local frames on both bases, the ambient bundle map has the form $T \mapsto gTh^{-1}$. These local choices are made only after constructing the graded algebra, its degree-one part, and the ruling.

Locally $\mathcal{D} = (d)$ with $d = \det T$. Its leading coefficient is a unit, so it is a nonzerodivisor over every coefficient ring. Hence $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$, and

$$\mathcal{J}_m \simeq \mathcal{L}^* \otimes \mathcal{I}_{n+m}.$$

This recovers the residual module without choosing a scalar determinant equation. Under $T \mapsto gTh^{-1}$, the Cauchy coefficient map transports a tensor by $\mathbb{S}_\lambda(g)^* \otimes \mathbb{S}_\lambda(h^{-1})$. Projection to the specified Cauchy summand, followed by contraction against its whole right dual, therefore transports exactly L'_λ to L_λ . The right map is invertible and has no effect on that left image. Both determinant-line transition functions and changes of a lift of $[g]$ act by scalar units. Since every partition has size m , this ambiguity is common in degree and disappears on subspaces. This proves the assertion. $\square$

Corollary 6.2 (The web instance of the criterion). *For $R \in G_4^\circ$, the scheme $\widehat{D}_R$ satisfies (i)–(iii) with*

$$B = \mathrm{Gr}(4, S_R), \quad n = 4, \quad m = 12, \quad \Lambda = \{(5, 4, 2, 1), (4, 4, 4, 0)\}.$$

The left subspaces are the two lines $\mathbb{C}\beta_{R,\lambda}$. Their intrinsic recovery, exterior duality, and the support gap imply Theorem 1.1.

Proof. The intrinsic base and determinant normal cone are Lemma 5.2. The nontriviality of the other ruling is its Schubert-line test. The residual factorization is Lemma 5.6, and (6.1) is exactly Lemma 5.7. Exterior duality is Lemma 5.8. The final uniqueness is Theorem 5.15; the smooth Jacobian has four essential variables. Thus every hypothesis is verified on the full stated smooth locus, not on a newly restricted generic subset. $\square$

6.2. Determinantal failure for polar restrictions. Fix $n \geq 2$, $d \geq 1$, a vector space H of dimension $s > n$, and an n -dimensional vector space V . Put $B = \mathrm{Gr}(n, H)$, with tautological bundle $\mathcal{U}$, and $M = \mathrm{Tot} \mathrm{Hom}(\mathcal{U}, V)$. Let $0 \neq L \subset \mathrm{Sym}^d V^*$ be a linear system, not furnished with a preferred basis. The tautological map $T : \mathcal{U} \rightarrow V$ on M gives the polar-restriction evaluation

$$\mathrm{ev}_{L,T} : L \otimes \mathrm{Sym}^d \mathcal{U} \longrightarrow \mathcal{O}_M, \quad f \otimes \xi \longmapsto f(\mathrm{Sym}^d(T)\xi).$$

Here and below bundles from B are pulled back to M . Define a vector-bundle map and its zeroth-Fitting failure scheme by

$$\Psi_L = \begin{pmatrix} \det T & 0 \\ 0 & \mathrm{ev}_{L,T} \end{pmatrix} : \det \mathcal{U} \oplus (L \otimes \mathrm{Sym}^d \mathcal{U}) \longrightarrow \det V \oplus \mathcal{O}_M, \quad (6.2)$$

$$Z_L = V(\mathrm{Fitt}_0 \mathrm{coker} \Psi_L) \subset M. \quad (6.3)$$

In local frames, if $f_1(T), \dots, f_a(T)$ are the entries of the evaluation row and $\delta = \det T$, this is literally

$$\Psi_L = \begin{pmatrix} \delta & 0 & \cdots & 0 \\ 0 & f_1(T) & \cdots & f_a(T) \end{pmatrix}.$$

Its only nonzero maximal minors are $\delta f_i(T)$. The two blocks ask simultaneously whether T is invertible and whether the restricted system is nonzero. This is a different failure map from cube-zero multiplication; neither a relation web nor its Jacobian covariant is used in its definition.

Theorem 6.3 (Reconstruction of linear systems from polar failure). *For fixed n, d, s as above and any nonzero systems $L \subset \text{Sym}^d V^*$, $L' \subset \text{Sym}^d V'^*$,*

$$Z_L \simeq Z_{L'} \iff g^* L' = L \text{ for a linear isomorphism } g : V \rightarrow V'. \quad (6.4)$$

The isomorphism on the left is abstract: it need not preserve the projection to B , the zero section, the grading, or an ambient embedding. All reductions $(Z_L)_{\text{red}}$ are the same determinant hypersurface in M . Thus the nonreduced scheme, not its reduction, separates the projective-equivalence classes of these linear systems. No smoothness hypothesis on the hypersurfaces or their base locus is required.

Proof. In a frame, write J_L for the ideal generated by all coefficients of $f \circ T$, for $f \in L$. The maximal minors of the block map (6.2) give

$$I(Z_L) = (\det T) J_L. \quad (6.5)$$

No zero of J_L has invertible T : an invertible substitution cannot annihilate a nonzero polynomial. Therefore $V(J_L)$ is contained in the determinant hypersurface, and the reduction in (6.5) is precisely that hypersurface with its reduced structure. This also follows locally from the primality of the generic determinant and the Nullstellensatz, since B is smooth.

The reduced singular locus of the rank-at-most- r matrix variety, for $0 < r < n$, is the rank-at-most- $r - 1$ variety. For clarity, invert an r -minor and use block elimination: the remaining lower-right block is zero, giving a smooth chart. At smaller rank a similar block reduction leaves minors of size at least two, with zero linear terms and tangent dimension exceeding the local dimension. These charts prove the assertion, also after product with a smooth base. Applying it repeatedly to $(Z_L)_{\text{red}}$ recovers its zero section B intrinsically. Its normal-cone algebra is the original graded algebra $\text{Sym}(V^* \otimes \mathcal{U})/((\det T) J_L)$: the ideal is homogeneous and generated in degree $n + d$. In particular its degree-one conormal is $V^* \otimes \mathcal{U}$, before any factorization is chosen by an isomorphism.

The two ruling parameter bundles are $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Only the first is projectively trivial. To check the second, restrict to a Schubert line in $\text{Gr}(n, H)$. There $\mathcal{U}^* = \mathcal{O}^{n-1} \oplus \mathcal{O}(1)$. If its projectivization were trivial, the vector bundle would be a direct sum of n copies of one line bundle: pull back the relative $\mathcal{O}(1)$ and push it down to see this assertion. Its determinant would have degree divisible by n , whereas the displayed splitting has determinant degree one. This is impossible for $n \geq 2$.

Finally the homogeneous coefficient map

$$\text{Sym}^d V^* \otimes \text{Sym}^d \mathcal{U} \longrightarrow \text{Sym}^d(V^* \otimes \mathcal{U}) \quad (6.6)$$

is the (d) -Cauchy inclusion. It is injective even before the Cauchy decomposition is invoked. Evaluate on a rank-one map $T = v \otimes a$, with $a \in \mathcal{U}^*$: a tensor $f \otimes \xi$ evaluates to $f(v)a^d(\xi)$. These products identify the tensor product with the polynomials bihomogeneous of degree (d, d) in (v, a) , so a zero coefficient polynomial has zero tensor. Consequently

$$(J_L)_d = L \otimes \text{Sym}^d \mathcal{U} \quad (6.7)$$

as an actual subspace of the specified Cauchy component. The ideal has no lower-degree generators. All hypotheses of Theorem 6.1 are now verified, with $m = d$ and $\Lambda = \{(d)\}$, proving the forward implication. For the reverse implication the bundle map $T \mapsto gT$ transports both block maps, up to invertible changes in their domain and target. It therefore transports their Fitting schemes. Scalar choices of g do not alter the system. $\square$

For a single equation $L = \mathbb{C}f$, this yields an intrinsic scheme-valued invariant of a degree- d hypersurface in $\mathbb{P}(V)$, up to projective equivalence. For example $n = 3, d = 4$ gives all plane quartics, smooth or singular, using a degree-seven Fitting ideal over $\text{Gr}(3, H)$. Higher-dimensional systems give Grassmannian families of equations rather than isolated examples. The support is fixed throughout each such family, and the varying coefficient system is recovered solely from its nonreduced structure. This construction concerns arbitrary homogeneous systems. The original multiplication construction has the different natural extension proved in Section 7. Nor does (6.4) assert an equivalence of moduli stacks or a classification of all automorphisms of Z_L ; it is the stated separation of abstract isomorphism classes.

Example 6.4 (An explicit polar nilpotent layer). For $n = d = 2$, $f = x_1^2 + x_2^2$, and $T = \begin{pmatrix} a & b \\ c & e \end{pmatrix}$, put $\delta = ae - bc$, $A = a^2 + c^2$, $B_0 = ab + ce$, and $C = b^2 + e^2$. The polar ideal is $J = (A, 2B_0, C)$, and $I(Z_{\mathbb{C}f}) = \delta J$. The identity $AC - B_0^2 = \delta^2$ gives $\delta^3 \in I$. But $\delta \notin J_2 = \langle A, B_0, C \rangle$, since the monomials ae, bc do not occur in that span. Cancellation in the polynomial domain gives $\delta^2 \notin I$. Since $\sqrt{I} = (\delta)$, the nilradical therefore has nilpotency index exactly three on this chart. The example displays scheme structure lost by the common reduced determinant support.

7. QUADRATIC RELATION SPACES AND INTRINSIC RECONSTRUCTION OF PENCILS

The multiplication construction itself supplies a wider natural class for the coefficient criterion. In this section the number of quadratic relations is allowed to vary independently of the embedding dimension. For two relations the resulting theorem recovers every pencil of quadrics, including singular pencils. No extra block is added to the multiplication map.

7.1. Automatic coefficient extraction for quadratic multiplication. Let $\dim V = n \geq 3$, $W = \text{Sym}^2 V$, $N = n(n+1)/2$, and $R \in \text{Gr}(r, W)$, where $1 \leq r < N - n$. Put $p = N - r > n$, $S = W/R$, and

$$A_R = \mathbb{C} \oplus V \oplus S, \quad X_R = \text{Gr}(n, V \oplus S).$$

The only nonzero product in the maximal ideal is the quotient map $\gamma : \text{Sym}^2 V \rightarrow S$. Write $\mathcal{K}$ for the tautological bundle on X_R , and use the same definition (1.1) of $\hat{D}_R$, with A_R in place of B_R . Thus $\hat{D}_R$ is the failure scheme of generating A_R by a varying n -plane and the unit. The notation in this section agrees with the web construction when $(n, r) = (4, 4)$.

Let $\Lambda_{n,p}$ denote the partitions occurring in the classical multiplicity-free decomposition

$$F(V) := \bigwedge^p \text{Sym}^2 V = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{S}_\lambda V. \quad (7.1)$$

Every partition has size $2p$; terms of length greater than n are omitted. Choose a nonzero volume form on S and let $\beta_R \in F(V)^*$ be its pullback by $\bigwedge^p \gamma$. Its component in $\mathbb{S}_\lambda V^*$ is denoted by $\beta_{R,\lambda}$. Only the lines of the nonzero components will be used, so the volume choice is immaterial.

Theorem 7.1 (Automatic intrinsic coefficient extraction). *For every $R \in \text{Gr}(r, \text{Sym}^2 V)$ in the stated range, the abstract scheme $\hat{D}_R$ determines*

$$(\mathbb{C}\beta_{R,\lambda})_{\lambda \in \Lambda_{n,p}}$$

up to one common projective transformation of V , with zero components recorded as zero subspaces. More precisely, the intrinsic construction recovers the base $B = \text{Gr}(n, S)$, its degree-one normal-cone bundle, and the residual ideal generated in degree $2p$ with coefficient module

$$\mathcal{J}_{2p} = \bigoplus_{\lambda \in \Lambda_{n,p}} \mathbb{C}\beta_{R,\lambda} \otimes \mathbb{S}_\lambda \mathcal{U} \subset \text{Sym}^{2p}(V^* \otimes \mathcal{U}), \quad (7.2)$$

where $\mathcal{U}$ is the tautological rank- n bundle on B . Consequently all geometric hypotheses of Theorem 6.1 hold for this entire Grassmannian family, not just for smooth webs.

Proof. On a frame chart of $\mathcal{K}$, write its inclusion as $\binom{M}{A_0}$, with M an $n \times n$ matrix. Since A_R is cube-zero, columns of degree greater than two vanish. After retaining the scalar, linear, and quadratic columns, multiplication has the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & A_0 & \gamma \operatorname{Sym}^2 M \end{pmatrix}. \quad (7.3)$$

A nonzero maximal minor must use the scalar column and all n linear columns: these are the only columns meeting the first $1+n$ rows. Expanding along those rows gives the identity of ideals

$$I_{\widehat{D}_R} = (\det M) I_p(\gamma \operatorname{Sym}^2 M). \quad (7.4)$$

The same expansion works for every $m \geq 2$. Where M is invertible, the quadratic map is surjective. Hence the reduction of $\widehat{D}_R$ is exactly the Schubert divisor where M is singular, independent of R after identifying the vector spaces S .

Successive reduced singular loci of that divisor are the projection rank loci. Their last nonempty member is $B = \operatorname{Gr}(n, S)$. For clarity, the usual determinantal assertion here is local: at a projection-rank- k point, an invertible k -minor splits away, leaving a generic square $(n-k)$ -matrix and smooth coordinates. Every remaining entry can vary independently by moving the kernel of the projection in V modulo its image. Thus its reduced singular locus is the next rank locus. Since $p > n$, the final base is nonempty, connected and projective, with $H^0(B, \mathcal{O}_B) = \mathbb{C}$.

Near B , a Grassmannian chart consists of a varying plane in S and its graph $T : \mathcal{U} \rightarrow V$. Formula (7.4) is independent of the other graph coordinates and homogeneous in T . It therefore gives its actual normal-cone ideal, not merely an initial reduced support:

$$C_B \widehat{D}_R \subset \operatorname{Tot} \operatorname{Hom}(\mathcal{U}, V), \quad \mathcal{I} = (\det T) \mathcal{J}, \quad \mathcal{J} = I_p(\gamma \operatorname{Sym}^2 T). \quad (7.5)$$

There are no degree-one equations. The ambient bundle in (7.5) is consequently the dual of the intrinsically recovered degree-one conormal. Its reduced ideal is the determinant ideal $\mathcal{D}$. The determinant polynomial is monic for a suitable monomial order, so cancellation gives $(\mathcal{I} : \mathcal{D}) = \mathcal{J}$. This is an intrinsic ideal quotient; choosing a generator of the determinant line is not part of the recovered datum.

The rank-one locus of the reduced cone determines the two ruling parameter schemes by Lemma 5.4. They are $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial. Indeed a Schubert line in $\operatorname{Gr}(n, S)$ restricts $\mathcal{U}^*$ to $\mathcal{O}(1) \oplus \mathcal{O}^{n-1}$. A trivial projective bundle on that line would make this bundle a line-bundle twist of $\mathcal{O}^n$; its determinant would then have degree divisible by n , rather than degree one. Thus the trivial ruling is intrinsically oriented. Lemma 5.5 applies: an abstract isomorphism gives a morphism $B \rightarrow \operatorname{PGL}(V)$, constant because B is connected projective and $\operatorname{PGL}(V)$ is affine. Local linear lifts and right changes of frame give $T \mapsto gTh^{-1}$ for this one fixed $[g]$.

It remains to prove (7.2), rather than assume it as a condition on the example. The minors are the image, for fixed β_R , of the universal coefficient map

$$F(V)^* \otimes F(U) \longrightarrow \operatorname{Sym}^{2p}(V^* \otimes U), \quad \beta \otimes \xi \longmapsto [T \mapsto \beta(F(T)\xi)]. \quad (7.6)$$

In characteristic zero, the polynomial functor F has the multiplicity-free decomposition (7.1); see [37, Theorem 3.5]. The Cauchy decomposition of the target contains each diagonal $\mathbb{S}_\lambda V^* \otimes \mathbb{S}_\lambda U$ once, and no off-diagonal pair. Equivariance kills every off-diagonal pair. On each diagonal pair the coefficient map is nonzero: identify U with V , evaluate at $T = 1$, and choose β, ξ in that summand with $\beta(\xi) \neq 0$. Schur's lemma then identifies its image with the specified Cauchy

summand. This is the actual matrix-coefficient inclusion, not merely a dimension count. Keeping β_R fixed proves (7.2). All generators have degree $2p$, and at least one minor is nonzero since γ is surjective.

Under $T \mapsto gTh^{-1}$, the right map acts within the full factor $\mathbb{S}_\lambda U$. Contracting against its dual therefore leaves exactly the left coefficient line transformed by $\mathbb{S}_\lambda(g)^*$. Determinant-line scalars and local lift ambiguities do not change these lines. This proves the theorem. $\square$

The theorem does not assert that separately projectivized Schur components always determine their relative scalars. For webs this is the recombination problem solved above. For pencils that issue disappears altogether.

7.2. Every pencil, without a regularity hypothesis.

Lemma 7.2 (The exterior square of the quadratic representation). *For $n \geq 2$, $\bigwedge^2 \text{Sym}^2 V = \mathbb{S}_{(3,1)} V$ is irreducible. Consequently $\bigwedge^{N-2} \text{Sym}^2 V$ is irreducible as well.*

Proof. The vector $x_1^2 \wedge x_1 x_2$ is a highest-weight vector of weight $(3, 1)$. By complete reducibility it generates a copy of $\mathbb{S}_{(3,1)} V$. The hook formula gives

$$\dim \mathbb{S}_{(3,1)} V = \frac{n(n-1)(n+1)(n+2)}{8} = \binom{N}{2}.$$

This exhausts the exterior square. The last assertion follows by exterior duality. These are classical representation identities. $\square$

Theorem 7.3 (Intrinsic reconstruction of all quadratic pencils). *Let $n \geq 3$ and let R, R' be two-dimensional subspaces of $\text{Sym}^2 V$. For the natural multiplication-failure schemes of their cube-zero algebras of Hilbert function $(1, n, N-2)$,*

$$\widehat{D}_R \simeq \widehat{D}_{R'} \iff R' = gR \text{ for some } g \in \text{PGL}(V). \quad (7.7)$$

No pencil, base locus, or discriminant is required to be smooth or regular. The isomorphism on the left is abstract and carries no supplied ambient embedding, base, or tensor-factor marking. All the reduced schemes are isomorphic to the same Schubert divisor.

Proof. Here $p = N - 2 > n$. By Theorem 7.1 and Lemma 7.2, the normal cone has only one left coefficient line to recover: the entire line $\mathbb{C}\beta_R$. The canonical duality

$$\bigwedge^{N-2} W^* \simeq \bigwedge^2 W \otimes (\det W)^* = \bigwedge^2 W \otimes (\det V)^{-(n+1)} \quad (7.8)$$

sends that line to the Pluecker line of R , with the same projective transformation g . Indeed the pullback of a quotient volume is the annihilator-volume dual of the kernel R . The common determinant character is a scalar on projective space. A nonzero decomposable two-vector recovers its plane as $\{w \in W : w \wedge w_R = 0\}$. Thus an abstract scheme isomorphism implies $R' = gR$.

Conversely such a linear coordinate change induces the quotient isomorphism $S_R \rightarrow S_{R'}$, the algebra isomorphism $A_R \rightarrow A_{R'}$, and the corresponding Grassmannian isomorphism. It intertwines multiplication and hence preserves its zeroth Fitting ideal. This proves the reverse implication. The assertion about reductions was proved in (7.4). $\square$

7.3. Discriminants, elementary divisors, and hyperelliptic moduli. A regular pencil means one containing an invertible symmetric matrix; it need not have a reduced discriminant. The classical invariant of such a pencil consists of its discriminant points together with their Jordan-block partitions, usually called its Weierstrass–Segre data. We use the root-labelled partitions, not a pairing of an unlabelled list with an independently unlabelled divisor. Their classification is classical; compare [40, Theorem 1.1, Corollary 2.1 and §2]. The following consequence identifies precisely what the intrinsic failure scheme remembers beyond the discriminant.

Corollary 7.4 (Intrinsic recovery of elementary divisors). *For a regular pencil in any dimension $n \geq 3$, the abstract scheme $\widehat{D}_R$ determines all root-labelled elementary divisors of a matrix pencil, up to projective reparametrization of its parameter line. Conversely these data determine its abstract failure scheme. In particular the invariant distinguishes different Jordan partitions over a fixed multiple discriminant root.*

Proof. The first implication follows by reconstructing R , then choosing an invertible member A and a complementary member B . For completeness, the relevant classical sufficiency can be seen directly. Put $C = A^{-1}B$; it is self-adjoint for A . If $hC = C'h$ identifies the Jordan forms for another pair (A', B') , set

$$H = A^{-1}h^t A' h.$$

Then H is invertible, A -self-adjoint, and commutes with C . There is a polynomial k in H with $k^2 = H$: on each nonzero eigenvalue choose a square root and truncate its Taylor series to the size of the largest Jordan block, then combine by the Chinese remainder theorem. Thus k is invertible, self-adjoint, and commutes with C . The map hk^{-1} is an isometry from A to A' and intertwines C, C' , so it simultaneously identifies A, B with A', B' .

Changing the basis of the pencil replaces C by a fractional linear expression with invertible denominator. At each eigenvalue its derivative is nonzero; hence each Jordan-block size is preserved while its root moves by the same projective transformation. A parameter change can avoid infinity for both finite spectra. Root-labelled elementary divisors therefore suffice, and (7.7) proves the converse for the failure schemes. This argument recalls the classical classification; the new input is recovery from an unmarked abstract failure scheme. $\square$

Example 7.5 (Identical discriminant, distinct intrinsic failure schemes). Choose distinct nonzero $\mu_3, \dots, \mu_n$, and put

$$S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Consider the symmetric pairs

$$\begin{aligned} A_s &= I_n, & B_s &= \text{diag}(0, 0, \mu_3, \dots, \mu_n), \\ A_j &= S_2 \oplus I_{n-2}, & B_j &= S_2 J_2 \oplus \text{diag}(\mu_3, \dots, \mu_n). \end{aligned}$$

Their determinant binary forms agree up to a nonzero scalar:

$$\det(sA_s + tB_s) = -\det(sA_j + tB_j) = s^2 \prod_{i=3}^n (s + \mu_i t). \quad (7.9)$$

At the unique double root the ranks are respectively $n-2$ and $n-1$, so no simultaneous projective equivalence exists. The two root partitions are $(1, 1)$ and (2) . Their abstract failure schemes are therefore nonisomorphic although both their reduced failure schemes and their discriminant divisors, including multiplicities, agree. This realizes inside the same multiplication construction the classical loss of elementary-divisor information under passage to the determinant.

Corollary 7.6 (Hyperelliptic isomorphism classes). *Fix $g \geq 2$ and $n = 2g + 2$. On the locus of pencils with simple discriminant, $\widehat{D}_R$ gives an injective invariant of the isomorphism classes of smooth hyperelliptic curves of genus g . All these failure schemes have the same reduction; their nonreduced structures distinguish a $(2g - 1)$ -dimensional family of curve classes. The pencil invariant also distinguishes the two regular boundary pencils in Example 7.5, where their singular discriminant double covers alone do not.*

Proof. The classical curve associated with a simple-discriminant pencil is the double cover of its parameter line branched at $\det(sA + tB) = 0$; see [41, §4.2, equation (4.3)]. Choose an invertible member. Distinct eigenvalues of $A^{-1}B$ have mutually A -orthogonal eigenlines, each nonisotropic,

so simultaneous diagonalization identifies the pencil with $\langle \sum x_i^2, \sum \lambda_i x_i^2 \rangle$. The unordered branch divisor modulo PGL_2 therefore determines the pencil, and every reduced divisor of degree $2g + 2$ arises. A smooth curve of genus at least two has a unique degree-two map to a rational curve up to automorphism of that curve: two distinct such maps would give a birational map to a curve of bidegree $(2, 2)$ in $\mathbb{P}^1 \times \mathbb{P}^1$, whose arithmetic genus is one. Thus curve isomorphism is exactly branch-divisor equivalence. Applying Theorem 7.3 gives the asserted injectivity and well-definedness.

Ordered distinct branch points have dimension $2g + 2$. Normalizing three of them removes the three-dimensional projective action; taking the finite permutation quotient leaves dimension $2g - 1$. Equality of all reductions follows from (7.4). Finally the two binary forms in (7.9) differ only by a scalar, a square over $\mathbb{C}$, and define isomorphic double covers, whereas their failure schemes have already been separated. $\square$

These are statements about complex isomorphism classes. They do not identify moduli stacks, assert a universal family over a coarse moduli space, or eliminate the different automorphism groups of pencils and curves. Nor is the classical pencil–hyperelliptic correspondence a new Torelli theorem. The conclusion here is reconstruction of those objects and of the elementary-divisor information on their boundary from the abstract nonreduced multiplication-failure scheme, while all reduced failure data are fixed.

8. THE EXACT INFINITESIMAL ORDER OF PENCIL RECONSTRUCTION

The all-pencil theorem admits a finite-order strengthening. It identifies the first infinitesimal neighbourhood that retains the pencil, and shows that every preceding neighbourhood is independent of it. We use ideal-adic neighbourhoods of the deepest stratum; these are not arc spaces or jet schemes.

Throughout this section let $n \geq 3$, $W = \mathrm{Sym}^2 V$, and

$$N = \binom{n+1}{2}, \quad p = N - 2, \quad d = n + 2p = n^2 + 2n - 4. \quad (8.1)$$

For a pencil $R \subset W$, put $S_R = W/R$ and $B = \mathrm{Gr}(n, S_R) \subset \widehat{D}_R$. This notation for the socle Grassmannian is local to this section; the cube-zero algebra is denoted by A_R , as in Section 7. Write $\mathfrak{b} \subset \mathcal{O}_{\widehat{D}_R}$ for the ideal of B , and define

$$Z_R^{[k]} = (B, \mathcal{O}_{\widehat{D}_R} / \mathfrak{b}^{k+1}) \quad (k \geq 0). \quad (8.2)$$

Thus $Z_R^{[0]} = B$. No embedding or identification of its reduction is supplied with the abstract scheme $Z_R^{[k]}$.

Theorem 8.1 (Sharp finite-neighbourhood reconstruction). *For any two pencils $R, R' \subset \mathrm{Sym}^2 V$, the following conditions are equivalent:*

$$\begin{aligned} Z_R^{[d]} \simeq Z_{R'}^{[d]} &\iff \widehat{D}_R \simeq \widehat{D}_{R'} \\ &\iff R' = gR \text{ for some } g \in \mathrm{PGL}(V). \end{aligned} \quad (8.3)$$

In contrast, for every $0 \leq k < d$, all the schemes $Z_R^{[k]}$ are isomorphic. Consequently d is the smallest uniform order at which these neighbourhoods classify all pencils up to projective equivalence. Singular pencils are included.

8.1. The first relation in the conormal algebra. Let $\mathcal{U}$ be tautological on B . The open set of X_R where projection to S_R is injective is canonically $\text{Tot } E$, where $E = \text{Hom}(\mathcal{U}, V)$: a plane is the graph of a unique map from its image in S_R to V . In particular this is one open neighbourhood of all of B , not merely a collection of unrelated formal charts. Let $\mathfrak{a}$ be its zero-section ideal. The exact multiplication calculation (7.4) gives

$$\mathcal{I}_R = (\det T)I_p(\gamma_R \text{Sym}^2 T) \subset \text{Sym } E^*. \quad (8.4)$$

It is generated in degree d , with a nonzero degree- d piece K_R . Indeed each quadratic minor has degree $2p$, and an invertible T makes $\gamma_R \text{Sym}^2 T$ surjective. There are no terms of lower degree, even on special pencils. Homogeneity is an equality of ideals in the actual graph neighbourhood, not just an equality of initial supports.

Lemma 8.2 (Intrinsic recovery from the first relation). *The abstract scheme $Z_R^{[d]}$ determines, as a graded $\mathcal{O}_B$ -algebra, the complete normal-cone algebra $\text{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R}$.*

Proof. The reduction of $Z_R^{[d]}$ is B , since its zero-section ideal is nilpotent and B is reduced. Hence its nilradical $\mathfrak{n}$ is intrinsic. Put $L = \mathfrak{n}/\mathfrak{n}^2$. Formula (8.4) and $d > 1$ identify $L = E^*$. Multiplication in the given finite scheme recovers

$$K_R = \ker[\text{Sym}^d L \longrightarrow \mathfrak{n}^d/\mathfrak{n}^{d+1}]. \quad (8.5)$$

The last denominator is zero, but retaining it displays the grading. No splitting of the reduction map is needed to form this kernel. Because $\mathcal{I}_R$ is generated in this single degree,

$$\text{gr}_{\mathfrak{b}} \mathcal{O}_{\widehat{D}_R} = \text{Sym}_{\mathcal{O}_B} L/(K_R). \quad (8.6)$$

Here (K_R) denotes the homogeneous ideal generated by K_R , not its radical. The right side is intrinsic to the finite scheme, proving the assertion. $\square$

Proof of Theorem 8.1. For $k < d$, equations (8.4) and (8.2) give

$$\mathcal{O}_{Z_R^{[k]}} = \text{Sym } E^* / \mathfrak{a}^{k+1}.$$

A linear identification $S_R \simeq S_{R'}$ identifies B , its tautological bundle, and E . It therefore identifies every one of these lower-order neighbourhoods.

An abstract isomorphism at order d induces an isomorphism of reductions, of conormal bundles, and, by Lemma 8.2, of complete graded normal cones. We spell out why the reconstruction in Theorem 7.1 requires no information outside this cone. Its reduced support is the square-matrix determinant cone; its iterated singular loci give the rank-one cone. The relative scheme of maximal linear spaces in that cone has the two rulings $\mathbb{P}(V) \times B$ and $\mathbb{P}_B(\mathcal{U}^*)$. Exactly the first is projectively trivial, as is seen by restriction to a Schubert line. Thus it is intrinsically oriented. The induced projective transformation of the trivial factor is constant on the connected projective base. Local lifts have the form $T \mapsto gTh^{-1}$ for this one $[g]$. The determinant ideal line is intrinsic, so its ideal quotient in (8.4) recovers the residual ideal. Finally Lemma 7.2 identifies the entire left coefficient line $\mathbb{C}\beta_R$, and exterior duality (7.8) recovers the Pluecker line of R . These are operations on the recovered normal cone; we do not assume that its isomorphism extends to X_R or $\widehat{D}_R$. They imply $R' = gR$.

Conversely such a projective transformation induces the algebra and failure-scheme isomorphisms, carrying the socle Grassmannian to the socle Grassmannian and preserving all its ideal powers. This proves (8.3), also using Theorem 7.3. For sharpness, there are inequivalent pencils in every dimension under consideration: $\langle x_1^2, x_1x_2 \rangle$ has a common linear factor, whereas $\langle x_1^2, x_2^2 \rangle$ does not. Their neighbourhoods are nevertheless isomorphic at every order below d . Thus no smaller order classifies all pencils. $\square$

8.2. A closed immersion of the full parameter scheme. The finite-order result has a scheme-theoretic parameter version, which records infinitesimal variation as well as closed points. Here tensor coordinates are fixed. This additional qualification is important: we are not asserting an equivalence of unmarked moduli stacks.

Fix an auxiliary n -dimensional vector space U , and put

$$F(V) = \bigwedge^p \mathrm{Sym}^2 V, \quad F(U) = \bigwedge^p \mathrm{Sym}^2 U, \quad M = \dim F(U) = \binom{N}{2}, \quad H = \mathrm{Hom}(U, V).$$

Choose a nonzero scalar representative δ of the determinant polynomial on H ; its scalar has no effect on the subspaces below. Let $K_R \subset \mathrm{Sym}^d H^*$ be the degree- d space of $(\det T)I_p(\gamma_R \mathrm{Sym}^2 T)$ in these fixed coordinates.

Theorem 8.3 (Faithful finite-order parameterization). *The assignment*

$$\Phi : \mathrm{Gr}(2, \mathrm{Sym}^2 V) \longrightarrow \mathrm{Gr}(M, \mathrm{Sym}^d H^*), \quad R \longmapsto K_R, \quad (8.7)$$

is a closed immersion. It is defined on the whole Grassmannian, including singular pencils. Its construction and its inverse on the image commute with arbitrary base change over $\mathbb{C}$. In particular it separates families over nonreduced bases and has injective differential at every pencil.

Lemma 8.4 (Tensoring a line with a fixed space). *For finite-dimensional vector spaces A, B_0 , with $b = \dim B_0 > 0$, the map $\mathbb{P}(A) \rightarrow \mathrm{Gr}(b, A \otimes B_0)$, $\ell \mapsto \ell \otimes B_0$, is a closed immersion.*

Proof. Choose bases $a_0, \dots, a_t$ and a basis of B_0 . Above the projective chart $\ell = \mathbb{C}(a_0 + \sum x_i a_i)$, the subspace is the graph of the block matrices $x_i I_b$. The corresponding Grassmannian chart has arbitrary $b \times b$ blocks. The image is cut out by their off-diagonal entries and the differences between their diagonal entries; an inverse reads one diagonal entry in each block. These are scheme-theoretic equations over any $\mathbb{C}$ -algebra, not just pointwise conditions. The preimage of this Grassmannian chart is exactly the stated projective chart, since the projection has determinant a_0^b . Such charts cover the image. The map is proper because its source is projective, so these chartwise closed immersions give a global closed immersion. $\square$

Proof of Theorem 8.3. The coefficient map (7.6) becomes

$$\mu : F(V)^* \otimes F(U) \longrightarrow \mathrm{Sym}^{2p} H^*.$$

It is injective: by Lemma 7.2 its source is irreducible for $\mathrm{GL}(V) \times \mathrm{GL}(U)$, and evaluation at an isomorphism $U \simeq V$ shows that the equivariant map is nonzero. Multiplication by the nonzero polynomial δ is injective. Thus $\nu = \delta\mu$ is a fixed linear inclusion and

$$K_R = \nu(\mathbb{C}\beta_R \otimes F(U)). \quad (8.8)$$

In particular $\dim K_R = M$, also on every singular pencil. By exterior duality the quotient-volume map $R \mapsto \mathbb{C}\beta_R$ is the Pluecker closed immersion $\mathrm{Gr}(2, W) \hookrightarrow \mathbb{P}(F(V)^*)$. Factor Φ through this map, the closed immersion of Lemma 8.4, and the Grassmannian closed immersion induced by ν . This proves the theorem. All are morphisms of schemes defined by the displayed polynomial and linear equations. A closed immersion is a monomorphism and remains one after arbitrary base change. This proves the family assertion, not merely separation of complex-valued points. $\square$

Corollary 8.5 (A flat family of finite normal algebras). *Over $\mathrm{Gr}(2, W)$, the algebras*

$$Q_R = \mathrm{Sym} H^* / ((K_R) + \mathfrak{m}^{d+1}), \quad \mathfrak{m} = \bigoplus_{j>0} \mathrm{Sym}^j H^*, \quad (8.9)$$

form a finite locally free family of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}.$$

In fixed coordinates this family determines every base-changed family of pencils. All its truncations below degree d are constant families.

Proof. Theorem 8.3 makes the spaces K_R a rank- M subbundle of the constant degree- d bundle, with locally free quotient. In degree $j < d$ the algebra is $\text{Sym}^j H^*$, and its degree- d piece is $\text{Sym}^d H^*/K_R$; higher degrees vanish. This proves local freeness and base-change compatibility. Summing the ranks gives the displayed binomial identity. The kernel of the coordinate quotient in degree d recovers K_R , and Φ recovers the pencil family. This is a statement about coordinate quotient algebras; it does not discard their coordinate maps and then infer a family reconstruction theorem. $\square$

8.3. Classical moduli at a sharp order.

Corollary 8.6 (Hyperelliptic classes in finite neighbourhoods). *Let $g \geq 2$, $n = 2g + 2$, and $d = 4g^2 + 12g + 4$. For pencils with simple discriminant, $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic if and only if their associated hyperelliptic curves are isomorphic. Every lower-order neighbourhood is independent of the curve. Thus the same $(2g - 1)$ -dimensional family of curve classes is invisible to all these lower-order neighbourhoods and is separated at order d .*

Proof. Combine Theorem 8.1 with Corollary 7.6, and substitute $n = 2g + 2$ in (8.1). $\square$

The order here concerns a neighbourhood of the entire projective socle Grassmannian. Its global ruling distinguishes the constant V -factor from the tautological factor. A single unmarked Artinian normal slice does not supply this orientation; the coordinate statement of Corollary 8.5 is deliberately different. The result is therefore not a claim that one unmarked local algebra replaces the projective family. The matrix-pencil classification and the hyperelliptic correspondence remain classical. The assertion proved here is that this classical information first appears at the specified order in the intrinsic multiplication-failure neighbourhoods.

9. SPECTRAL TORSION AND RANK-PRESERVING SPECIALIZATION

The finite-neighbourhood theorem recovers more than the discriminant or the corank of each member of a regular pencil. We identify its classical spectral invariant and then give a flat specialization on which the entire discriminant and every reduced rank locus remain constant, while the recovered invariant changes. The two specializations are separated at exactly the order of Theorem 8.1. The parameter space throughout is the usual Grassmannian of pencils; the failure map is the original multiplication map, not an added coefficient-evaluation block.

9.1. The spectral torsion sheaf. Let $R \subset \text{Sym}^2 V$ be a regular pencil, meaning that it contains a nonsingular quadratic form. On its parameter line $L_R = \mathbb{P}(R)$, the tautological quadratic form gives an exact sequence

$$0 \longrightarrow V^* \otimes \mathcal{O}_{L_R}(-1) \xrightarrow{q_R} V \otimes \mathcal{O}_{L_R} \longrightarrow \mathcal{T}_R \longrightarrow 0. \quad (9.1)$$

The first arrow is injective because its determinant is nonzero at the generic point of the smooth integral curve L_R . Thus $\mathcal{T}_R$ is a torsion sheaf of length n , whose zeroth Fitting divisor is the full discriminant. The pair $(L_R, \mathcal{T}_R)$ is considered up to isomorphism of the parameter line. Its definition involves no choice of a basis of R . The relation with elementary divisors is classical; see [40, §2, Corollary 2.1 and equation (5)]. We record the argument to make precise which datum is recovered.

Lemma 9.1 (Sheaves on the affine spectral line). *Let A, B be symmetric complex matrices with A invertible. On the affine line with coordinate z , the cokernel of $zA - B$ is the finite-dimensional $\mathbb{C}[z]$ -module on $\mathbb{C}^n$ for which multiplication by z is $C = A^{-1}B$. An isomorphism of such sheaves, in this fixed coordinate, is equivalent to similarity of the corresponding matrices.*

Proof. Multiplication on the target by A^{-1} identifies the presentation with $zI - C$. Reduction of a polynomial vector modulo these relations sends $\sum_j z^j v_j$ to $\sum_j C^j v_j$; constant vectors give the inverse identification. The equivalence between coherent sheaves on an affine scheme and finitely generated modules identifies a finite-length sheaf supported in this affine line with this module. The support is finite, so its underlying complex vector space is finite dimensional. A module isomorphism is a linear isomorphism G satisfying $GC = C'G$, which is exactly similarity. Extension by zero across a point outside the support loses no sheaf data. $\square$

Lemma 9.2 (Polynomial square roots of invertible operators). *For every invertible complex matrix K there is $h \in \mathbb{C}[x]$ with $h(K)^2 = K$. If K is self-adjoint for a nondegenerate symmetric form and commutes with C , then $h(K)$ has both properties.*

Proof. Write the minimal polynomial as $\prod_\lambda (x - \lambda)^{a_\lambda}$, where $\lambda \neq 0$. In its local Artinian factor choose a square root $\mu_\lambda^2 = \lambda$, and put

$$h_\lambda(x) = \mu_\lambda \sum_{j=0}^{a_\lambda-1} \binom{1/2}{j} \left(\frac{x - \lambda}{\lambda} \right)^j.$$

The formal binomial identity, truncated modulo $(x - \lambda)^{a_\lambda}$, gives $h_\lambda(x)^2 = x$. The Chinese remainder theorem joins these polynomials to a single class modulo the minimal polynomial. Evaluation at K proves the assertion even when K is not semisimple. Polynomials in one self-adjoint operator remain self-adjoint, and every polynomial in K commutes with every operator commuting with K . $\square$

Theorem 9.3 (Spectral reconstruction at the first relation). *Let $n \geq 3$, $d = n^2 + 2n - 4$, and let R, R' be regular pencils. The following are equivalent:*

- (1) *the unmarked schemes $Z_R^{[d]}$ and $Z_{R'}^{[d]}$ are isomorphic;*
- (2) *there is an isomorphism $f : L_R \rightarrow L_{R'}$ with $\mathcal{T}_R \simeq f^* \mathcal{T}_{R'}$;*
- (3) *the pencils R, R' are projectively equivalent.*

In particular the finite scheme determines the full root-labelled spectral torsion sheaf, not only its Fitting divisor or its support. This is an equivalence of isomorphism classes of objects; it is not an assertion identifying their automorphism groups or moduli stacks.

Proof. The equivalence of the first and third conditions is Theorem 8.1. A congruence of pencils induces an isomorphism of (9.1), so the third implies the second.

For the converse choose a common affine coordinate whose point at infinity avoids the two spectral supports, using f to identify the lines. Write the corresponding symmetric pairs as (A, B) and (A', B') , with A, A' invertible, and set $C = A^{-1}B$, $C' = A'^{-1}B'$. Choose the presentation $zA - B$ for the affine coordinate on both lines. Lemma 9.1 identifies the two sheaves with the modules of C, C' , so their isomorphism gives a similarity. After applying that change of coordinates, assume $C' = C$. Both A and A' are nondegenerate symmetric forms for which C is self-adjoint. Then $K = A^{-1}A'$ commutes with C and is self-adjoint for A . Lemma 9.2 provides a polynomial h with $h(K)^2 = K$. Put $H = h(K)$. It commutes with C , is self-adjoint for A , and satisfies

$$H^t A H = A', \quad H^t B H = H^t A C H = A' C = B'.$$

Thus the pairs, and hence the pencils, are congruent. This also shows why the spectral sheaf alone suffices here; no additional pairing on that sheaf is required over $\mathbb{C}$. $\square$

For $x \in \text{Supp } \mathcal{T}_R$, put

$$\ell_a(x) = \text{length}(\mathcal{T}_{R,x} / \mathfrak{m}_x^a \mathcal{T}_{R,x}), \quad a \geq 0. \tag{9.2}$$

If the elementary-divisor exponents at x are $e_1, \dots, e_b$, the structure theorem over the discrete valuation ring $\mathcal{O}_{L_R, x}$ gives

$$\ell_a(x) = \sum_{j=1}^b \min(a, e_j), \quad \ell_a(x) - \ell_{a-1}(x) = \#\{j : e_j \geq a\}. \quad (9.3)$$

Thus these intrinsic local lengths recover every exponent, including repeated roots. The corank of the pencil at x gives only $\ell_1(x)$, and the multiplicity of the discriminant gives only the eventual value $\ell_a(x) = \sum_j e_j$. The intermediate values are genuinely additional information.

9.2. A specialization with constant discriminant and coranks. The following family keeps both ends of this length profile fixed. Fix $i \in \mathbb{C}$ with $i^2 = -1$, and set

$$N_0 = \begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad P_t = \frac{t}{2} N_0, \quad A_0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad C_t = \begin{pmatrix} P_t & I_2 \\ 0 & P_t \end{pmatrix}. \quad (9.4)$$

Put

$$B_t = A_0 C_t = \begin{pmatrix} 0 & P_t \\ P_t & I_2 \end{pmatrix}, \quad R_t = \langle A_0, B_t \rangle \subset \text{Sym}^2 \mathbb{C}^4. \quad (9.5)$$

Both matrices are symmetric and A_0 is invertible. They are independent for every t . For $n > 4$, append I_{n-4} to A_0 and append $\text{diag}(\mu_5, \dots, \mu_n)$ to B_t , where the μ_j are fixed distinct nonzero numbers. Denote the resulting matrices by $A^{(n)}, B_t^{(n)}$ and their pencil again by R_t .

Theorem 9.4 (Separation inside fixed reduced rank data). *For every $n \geq 4$, the family R_t , $t \in \mathbb{A}^1$, has the following properties.*

- (1) *Its full discriminant, with multiplicities, is independent of t :*

$$\det(sA^{(n)} + uB_t^{(n)}) = s^4 \prod_{j=5}^n (s + \mu_j u). \quad (9.6)$$

For every minor size, the reduced determinantal locus on the marked parameter line is also independent of t .

- (2) *At the root $s = 0$, the spectral module has exponents $(3, 1)$ for $t \neq 0$ and $(2, 2)$ for $t = 0$. Consequently $Z_{R_t}^{[d]} \not\cong Z_{R_0}^{[d]}$ for every $t \neq 0$, whereas all the punctured fibres are isomorphic.*
- (3) *The schemes $Z_{R_t}^{[d]}$ form a flat projective family over $\mathbb{A}^1$, with fixed reduction. After one identification of the socle quotient bundles, every truncation of order $k < d$ is a constant family. Thus the first separating order remains exactly $d = n^2 + 2n - 4$ even under the constraints in part (1).*

Proof. The identities $N_0^2 = 0$ and $\text{rank } N_0 = 1$ give

$$C_t^2 = \begin{pmatrix} 0 & tN_0 \\ 0 & 0 \end{pmatrix}, \quad C_t^3 = 0, \quad C_t = \begin{pmatrix} I_2 \\ P_t \end{pmatrix} (P_t \ I_2). \quad (9.7)$$

The right factor in the last expression is surjective and the left factor is injective. Hence $\text{rank } C_t = 2$ for every t , whereas $\text{rank } C_t^2 = 1$ precisely when $t \neq 0$. The nilpotent partitions are therefore $(3, 1)$ and $(2, 2)$. Also $\det A_0 = 1$ and $\det(sI_4 + uC_t) = s^4$, proving (9.6). At $s = 0$ the full pencil has corank two for all t ; at each remaining root it has corank one, and elsewhere corank zero. Equality of these functions on $\mathbb{P}^1(\mathbb{C})$ implies equality of the reduced closed determinantal subschemes for each minor size.

At the unique multiplicity-four root, formula (9.3) yields respectively

$$(\ell_1, \ell_2, \ell_3, \dots) = (2, 3, 4, \dots), \quad (2, 4, 4, \dots).$$

No automorphism of the parameter line can exchange this root with a simple one. Theorem 9.3 separates the special fibre from the punctured fibres and identifies the latter with one another. This is a statement about geometric fibres; no choice of a simultaneous trivializing congruence on $\mathbb{A}^1 \setminus 0$ is needed.

For flatness we use the relative gluing of Theorem 10.1; the following coordinates identify its pullback in the present family. First note that $t \mapsto R_t$ is a morphism to $\mathrm{Gr}(2, \mathrm{Sym}^2 V)$, not merely a collection of pencils. In symmetric matrix coordinates the two fixed linear functionals

$$a(Q) = Q_{13} + iQ_{14}, \quad b(Q) = Q_{33}$$

satisfy $a(A^{(n)}) = b(B_t^{(n)}) = 1$ and $a(B_t^{(n)}) = b(A^{(n)}) = 0$. Their common kernel is a fixed complement to every R_t . It trivializes $S_{R_t} = \mathrm{Sym}^2 V / R_t$, and hence identifies the relative socle Grassmannian with $B \times \mathbb{A}^1$.

On a frame chart of its tautological bundle, the graph description (8.4) identifies the truncated algebra with

$$\bigoplus_{j=0}^{d-1} \mathrm{Sym}^j E^* \oplus (\mathrm{Sym}^d E^* / K_{R_t}). \quad (9.8)$$

By Theorem 8.3, K_{R_t} is a subbundle of constant rank $\binom{N}{2}$, with locally free quotient. This statement commutes with pullback to the present base and with right changes of frame, so the local descriptions glue. Thus (9.8) is locally free on $B \times \mathbb{A}^1$, of rank $\binom{n^2+d}{d} - \binom{N}{2}$. The graph projection is a finite flat morphism to that base. Since B is projective, its composition with projection to $\mathbb{A}^1$ is flat and projective. Below degree d there is no relation, and the fixed complement identifies the entire lower truncated algebra, including multiplication, with a constant family. These arguments prove flatness of the finite neighbourhoods; they make no assertion about flatness of the unrestricted failure schemes. $\square$

The qualifier “reduced” in part (1) is necessary. At $s = 0$, in a local parameter $z = s/u$, the gcd of the three-row minors of the four-dimensional block is a unit times z for $t \neq 0$ and a unit times z^2 for $t = 0$. Both radicals are (z) . Thus the family retains exactly the higher Fitting information discarded by reduction; it does not contradict the classical classification by the full determinantal ideals.

9.3. An independently visible change of reciprocal geometry. For a regular pencil let its reciprocal curve be the projective closure of the inverses of its nonsingular members. These curves and their strata are studied independently of multiplication-failure schemes in [40, §3].

Corollary 9.5 (A conic specializes to a line). *For the four-dimensional family (9.5), the reciprocal curve is a smooth plane conic when $t \neq 0$ and a projective line when $t = 0$. The unmarked twentieth neighbourhood distinguishes these two cases, while all earlier neighbourhoods, the full discriminant, and all reduced rank loci agree.*

Proof. For $s \neq 0$, multiplication on the right by the fixed invertible matrix A_0^{-1} identifies the projective inverse with

$$(sI_4 + uC_t)^{-1} = s^{-1}I_4 - us^{-2}C_t + u^2s^{-3}C_t^2.$$

If $t \neq 0$, the matrices I_4, C_t, C_t^2 are linearly independent: their span has dimension equal to the degree three of the minimal polynomial. The closure is therefore the quadratic Veronese image $[s : u] \mapsto [s^2 : -su : u^2]$, a smooth conic in its spanning plane. At $t = 0$ the last term vanishes, and I_4, C_0 are independent, giving a line. The remaining assertions follow from Theorem 9.4, with $d = 20$. The degree drop does not assert that the reciprocal curves themselves form a flat family; the flat family proved above consists of the finite failure neighbourhoods. $\square$

10. THE UNIVERSAL FINITE NEIGHBOURHOOD AND SPECTRAL STRATA

The finite-order construction is a family of actual neighbourhoods in multiplication-failure schemes. This fact is needed when passing from individual pencils to degenerations and infinitesimal parameter spaces. We establish it before discussing the spectral incidence schemes.

Put $P = \text{Gr}(2, \text{Sym}^2 V)$, and write $\mathcal{R}$ and $\mathcal{S} = (\text{Sym}^2 V \otimes \mathcal{O}_P)/\mathcal{R}$ for its universal relation and quotient bundles. Let

$$\pi : \mathcal{B} = \text{Gr}_P(n, \mathcal{S}) \longrightarrow P, \quad \mathcal{E} = \text{Hom}(\mathcal{U}, V \otimes \mathcal{O}_{\mathcal{B}}),$$

where $\mathcal{U}$ is tautological of rank n . As before, $N = n(n+1)/2$, $p = N - 2$, $d = n + 2p$, and $M = \binom{N}{2}$.

Theorem 10.1 (Universal neighbourhood and arbitrary base change). *The universal multiplication-failure scheme has a canonical order- d neighbourhood $\mathcal{Z}^{[d]}$ of $\mathcal{B}$. There is a rank- M vector subbundle $\mathcal{K} \subset \text{Sym}^d \mathcal{E}^*$ such that*

$$\mathcal{Z}^{[d]} = \text{Spec}_{\mathcal{B}} \left(\text{Sym} \mathcal{E}^* / (\mathcal{K}, \text{Sym}^{d+1} \mathcal{E}^*) \right). \quad (10.1)$$

This is finite locally free over $\mathcal{B}$, of rank

$$\binom{n^2 + d}{d} - \binom{N}{2}, \quad (10.2)$$

and is projective and flat over P . Its construction, its ideal-adic filtration, and its identification with the actual multiplication-failure neighbourhood commute with every morphism of complex schemes $T \rightarrow P$, including nonreduced ones. For $k < d$, its order- k truncation is $\text{Spec}_{\mathcal{B}} \bigoplus_{j=0}^k \text{Sym}^j \mathcal{E}^$, with truncated multiplication. After trivializing $\mathcal{S}$ on the parameter base, these lower-order families are constant.*

Proof. The locus in $\text{Gr}_P(n, V \oplus \mathcal{S})$ on which projection to $\mathcal{S}$ is injective is canonically $\text{Tot } \mathcal{E}$. It is an open neighbourhood of $\mathcal{B}$, not a choice of formal coordinates. On a frame chart of $\mathcal{U}$, let $T : \mathcal{U} \rightarrow V$ be the universal graph map and let $\gamma : \text{Sym}^2 V \rightarrow \mathcal{S}$ be the quotient. The same block elimination that proves (8.4) works over its coordinate ring: the ideal of the multiplication-failure scheme is

$$(\det T) I_p(\gamma \text{Sym}^2 T).$$

Every generator is homogeneous of degree d .

Here is also a coordinate-free description of its first relation bundle. The coefficient map, followed by determinant multiplication, is a morphism

$$\det V^* \otimes \det \mathcal{U} \otimes \det \mathcal{S}^* \otimes \bigwedge^p \text{Sym}^2 \mathcal{U} \longrightarrow \text{Sym}^d \mathcal{E}^*. \quad (10.3)$$

The inclusion used in this expression is $\det \mathcal{S}^* \hookrightarrow (\bigwedge^p \text{Sym}^2 V)^*$. The determinant factors evaluate $\det T$, and the remaining factors evaluate $\bigwedge^p \text{Sym}^2 T$ followed by the quotient. Thus the image is exactly the span of the displayed maximal minors, not a separately inserted coefficient module.

On a frame chart this map is the fixed injective linear coefficient map ν of Theorem 8.3, restricted to a line subbundle tensored with the fixed space $\bigwedge^p \text{Sym}^2 \mathbb{C}^n$. Its image is a subbundle of rank M : locally split the line in the source, and split the fixed complex linear inclusion ν . Both splittings remain splittings over any complex algebra. Denote its image by $\mathcal{K}$. A right change of frame of $\mathcal{U}$ changes the graph coordinates and this coefficient map contragrediently. The determinant character in (10.3) is exactly the one arising from $\det T$. Hence these local ideals and their degree- d subbundles agree on overlaps. In particular, they glue as ideals of the given graph neighbourhood.

Let $\mathfrak{b}$ be the ideal of the zero section inside the multiplication-failure scheme. In graph coordinates its preimage is the positive-degree ideal. Quotienting by $\mathfrak{b}^{d+1}$ therefore gives precisely (10.1). This identifies the gluing with the actual ideal-adic neighbourhood, not merely with algebras having the correct geometric fibres. Its underlying module is

$$\bigoplus_{j=0}^{d-1} \mathrm{Sym}^j \mathcal{E}^* \oplus (\mathrm{Sym}^d \mathcal{E}^* / \mathcal{K}).$$

All terms are locally free. Counting monomials in n^2 variables gives (10.2). The morphism $\mathcal{B} \rightarrow P$ is smooth and projective, so the asserted flatness and projectivity follow. Finally the graph description, the split coefficient inclusion, and the homogeneous quotient all commute with base change. Below degree d the ideal contributes no relation. This proves the final assertion as well. $\square$

Remark 10.2. Over a nonreduced base, the theorem concerns the specified relative socle subscheme $\mathcal{B}_T$ and its ideal. It does not identify that ideal with the absolute nilradical of $\mathcal{Z}_T^{[d]}$, which can also contain nilpotents from the base. On geometric fibres $\mathcal{B}_t$ is exactly the reduction. Nor is this theorem an equivalence of unmarked deformation groupoids: extra automorphisms of truncated algebras are not being identified with pencil congruences.

10.1. Spectral incidence schemes from the first relation space. On the regular-pencil locus, pass locally to ordered generators (A, B) with A invertible, and put $C = A^{-1}B$. For a marked point with affine coordinate z , set $X = C - zI$. The next elementary presentation works over an arbitrary coefficient ring; no diagonalization or division by eigenvalue differences is involved.

Lemma 10.3 (Truncated spectral presentation). *Let D be a commutative ring, $X \in \mathrm{Mat}_n(D)$, and $a \geq 1$. There is a natural isomorphism*

$$\mathrm{coker}(wI - X) \otimes_{D[w]} D[w]/(w^a) \simeq \mathrm{coker}(X^a : D^n \rightarrow D^n). \quad (10.4)$$

Equivalently, the a -square block presentation with diagonal blocks X and subdiagonal blocks $-I$ is equivalent, by invertible row and column operations over D , to $I_{n(a-1)} \oplus X^a$.

Proof. Map the class of $\sum_{j=0}^{a-1} w^j v_j$ to the class of $\sum_{j=0}^{a-1} X^j v_j$ modulo $X^a D^n$. The relation $wv = Xv$ and the relation $w^a v = 0$ show that this map is well-defined. Sending a constant vector to its class is inverse. Eliminating successively the blocks with coefficient $-I$ gives the stated presentation equivalence, using only unit pivots. This also proves naturality under ring homomorphisms. $\square$

Theorem 10.4 (Scheme-theoretic spectral readout). *In the fixed tensor presentation of Theorem 8.3, the first relation subspace determines, on the regular-pencil locus, every closed spectral incidence scheme*

$$\mathfrak{S}_{a,h} = V(I_{n-h+1}((C - zI)^a)), \quad 1 \leq a, h \leq n. \quad (10.5)$$

This determination commutes with arbitrary complex base change. At a geometric point, membership is equivalent to $\mathrm{length}(\mathcal{T}_z / \mathfrak{m}_z^a \mathcal{T}_z) \geq h$. Consequently the schemes of specified spectral length profiles, and not only their reduced supports, can be read from the order- d relation subspace.

Proof. The closed immersion Φ of Theorem 8.3 has an inverse on its scheme-theoretic image. Thus a family of relation subspaces in that image recovers its morphism to P , over any base, and hence its universal pencil. In the displayed chart its spectral sheaf has presentation $(z+w)I - C = wI - X$. Lemma 10.3 gives the presentation X^a for restriction to the order- a fat point centred at z . Its $(h-1)$ -st Fitting ideal is exactly the ideal in (10.5); Fitting ideals of finitely presented modules

commute with base change. On a field the cokernel has dimension at least h exactly when these minors vanish.

These descriptions are independent of the chosen frames. A change of basis in V conjugates X ; a change of affine coordinate replaces the parameter ideal at the marked point by a unit multiple locally. Alternatively, restriction of the spectral sheaf to the fat diagonal on its parameter line defines the same finite module without a coordinate. Thus the Fitting ideals glue. Rank equality is imposed by the corresponding rank inequality together with the open complement of the next inequality. Intersecting these locally closed conditions gives each specified length profile. Formula (9.3) then identifies its geometric fibres with the prescribed elementary-divisor partition. $\square$

10.2. Realizing the complete Segre stratification. Let P_σ denote the reduced closure in P of a regular Segre stratum, and let $J_\sigma = \Phi(P_\sigma)$. Because Φ is a closed immersion, it gives isomorphisms

$$P_\sigma \simeq J_\sigma, \quad P_{\sigma_1} \cap_P \cdots \cap_P P_{\sigma_s} \simeq J_{\sigma_1} \cap_{\Phi(P)} \cdots \cap_{\Phi(P)} J_{\sigma_s}, \quad (10.6)$$

where the intersections carry their scheme structures. In particular their local rings and formal completions are preserved. Pulling back Theorem 10.1 realizes these parameter schemes by flat projective families of the actual finite failure neighbourhoods. The classical inclusions among Segre-stratum closures [40, Theorem 5.1] are therefore all realized, rather than just the one degeneration of Theorem 9.4.

There are two distinct assertions here. The Segre closure order is classical, and transport of closed subschemes through a closed immersion is formal. The geometric input proved here is the base-change-compatible realization by actual multiplication-failure neighbourhoods and the recovery of their spectral Fitting schemes. The ambient parameter space in (10.6) is the image of Φ , not a moduli space of every possible unmarked deformation of a finite scheme.

11. A TOTALLY REAL RECIPROCAL LIKELIHOOD FIBRE

The spectral data also lead to a question independent of reconstruction: can every critical point of reciprocal likelihood on a definite real pencil be made real by a choice of real data? The assertion was formulated as Conjecture 4.5 in [40]. We give an explicit construction valid for every number of distinct eigenvalues and every multiplicity pattern. Its proof counts the critical points directly; it does not assume the complex likelihood-degree formula.

A real symmetric pencil containing a positive definite matrix can be written, after a real congruence and a choice of generators, as $\langle I, \text{diag}(a_1, \dots, a_n) \rangle$. Indeed, make the definite matrix the identity and orthogonally diagonalize the other symmetric matrix. Group its distinct eigenvalues as $\alpha_1 < \cdots < \alpha_r$, with multiplicities $m_1, \dots, m_r > 0$ and $n = \sum_i m_i$. A two-dimensional pencil has $r \geq 2$. If σ_i is the sum of the data entries in the i -th eigenspace, the reciprocal score equations are the critical equations of

$$\tilde{\ell}(x, y) = - \sum_{i=1}^r m_i \log(\alpha_i x + y) - \sum_{i=1}^r \frac{\sigma_i}{\alpha_i x + y} \quad (11.1)$$

on the complement of $\prod_i (\alpha_i x + y) = 0$. The derivatives are rational and independent of the local branches of the logarithms. On the real complement one can equivalently use $\log |\alpha_i x + y|$.

Theorem 11.1 (Explicit totally real likelihood data). *Choose any real numbers*

$$\alpha_r < \beta_1 < \cdots < \beta_{r-1}, \quad D(t) = \prod_{i=1}^r (t - \alpha_i), \quad Q(t) = \prod_{j=1}^{r-1} (t - \beta_j),$$

and prescribe

$$\sigma_i = \frac{Q(\alpha_i)}{D'(\alpha_i)}. \quad (11.2)$$

Then (11.1) has exactly $2r - 3$ critical points on its complex domain. Every one is real and nondegenerate. There is one critical point with $t = -y/x$ in each interval (α_i, α_{i+1}) , and one in each interval (β_j, β_{j+1}) ; there are no others. The same conclusion holds on a nonempty Euclidean open subset of the real data space. Thus the assertion of Conjecture 4.5 in [40] holds for all definite real pencils.

The data here are unrestricted real data, exactly as in that assertion. Neither positivity of the data matrix nor the assertion that all the critical points lie in the positive definite cone is part of the theorem. For example, the residues in (11.2) alternate in sign.

Lemma 11.2 (Two separated strings of roots). *With the notation of the theorem, put*

$$H(t) = \sum_{i=1}^r (n - m_i) \frac{D(t)}{t - \alpha_i}, \quad F(t) = nD(t)Q'(t) - H(t)Q(t). \quad (11.3)$$

The polynomial F has degree exactly $2r - 3$. It has a simple root in each of the intervals in the theorem, has no other root over $\mathbb{C}$, and is relatively prime to DQ .

Proof. Since $\sum_i (n - m_i) = n(r - 1)$, the terms of degree $2r - 2$ in F cancel. At a zero of D ,

$$F(\alpha_i) = -(n - m_i)D'(\alpha_i)Q(\alpha_i).$$

The factors $n - m_i$ are positive, all the values $Q(\alpha_i)$ have the same sign, and consecutive values $D'(\alpha_i)$ have opposite signs. Thus F changes sign in each of the $r - 1$ intervals between the α_i . At a zero of Q ,

$$F(\beta_j) = nD(\beta_j)Q'(\beta_j).$$

Here $D(\beta_j) > 0$, and consecutive values of Q' have opposite signs. There are therefore a further $r - 2$ roots in the intervals between the β_j . These intervals are pairwise disjoint and are disjoint from all the former intervals. The upper bound $2r - 3$ on the degree forces exactly one root in each interval, all simple, and excludes any remaining real or complex root. The displayed endpoint values are nonzero, proving coprimality. This argument includes $r = 2$, when the second list of intervals is empty. $\square$

Proof of Theorem 11.1. Partial fractions give

$$\frac{Q(t)}{D(t)} = \sum_i \frac{\sigma_i}{t - \alpha_i}, \quad S(t) := \sum_i \frac{\sigma_i}{\alpha_i - t} = -\frac{Q(t)}{D(t)}. \quad (11.4)$$

On the chart $x \neq 0$, put $y = -xt$. Locally, up to a constant, the function becomes

$$-n \log x - \sum_i m_i \log(\alpha_i - t) - \frac{S(t)}{x}.$$

The first critical equation gives $x = S(t)/n$. If $S(t) = 0$ it cannot be satisfied on this chart. When $S(t) \neq 0$, the remaining critical equation is

$$\left. \frac{\partial \tilde{\ell}}{\partial t} \right|_{x=S(t)/n} = -\sum_i \frac{m_i}{t - \alpha_i} - n \frac{S'(t)}{S(t)} = -\frac{F(t)}{D(t)Q(t)} = 0. \quad (11.5)$$

The lemma now gives exactly $2r - 3$ solutions in this chart, with the real coordinates

$$x = -\frac{Q(t)}{nD(t)}, \quad y = -xt. \quad (11.6)$$

At each of these points $x \neq 0$ and $\alpha_i x + y \neq 0$, so none is an extraneous solution introduced by clearing denominators.

We must also exclude the omitted chart $x = 0$. Comparing the coefficients of t^{-1} and t^{-2} in (11.4) gives

$$\sum_i \sigma_i = 1, \quad \sum_i \alpha_i \sigma_i = \sum_i \alpha_i - \sum_j \beta_j.$$

At $x = 0$, the y -equation would imply $y = 1/n$. The x -equation would then require

$$\sum_i (n - m_i) \alpha_i = n \sum_j \beta_j. \quad (11.7)$$

But the left side is strictly smaller than $n(r - 1)\alpha_r$, whereas the right side is strictly larger. Thus there is no critical point on $x = 0$. This proves the assertion about the entire complex domain, not just the chosen coordinate chart.

For nondegeneracy, at a critical point in (x, t) -coordinates one has $\tilde{\ell}_{xx} = -n/x^2 \neq 0$. The derivative of the eliminated equation (11.5) is $-F'(t)/(D(t)Q(t)) \neq 0$. It is the Schur complement of $\tilde{\ell}_{xx}$ in the Hessian. Hence the Hessian is nonsingular. The coordinate change $(x, t) \mapsto (x, -xt)$ has nonzero Jacobian at these points, so nondegeneracy also holds in the original coordinates.

Finally allow Q to have any positive leading coefficient and $r - 1$ distinct real roots to the right of α_r . These polynomials form a nonempty Euclidean open set in their r -dimensional coefficient space. The linear map $Q \mapsto (Q(\alpha_i)/D'(\alpha_i))_{i=1}^r$ is an isomorphism, by polynomial interpolation. Scaling Q does not change its critical t -values and scales (x, y) by the same nonzero factor. Thus its image is the required open set of grouped data. The map from the original n data entries to their r group sums is surjective, so its inverse image is open and nonempty as well. $\square$

Example 11.3. For $(\alpha_1, \alpha_2, \alpha_3) = (1, 2, 3)$, $m_1 = m_2 = m_3 = 1$, and $(\beta_1, \beta_2) = (4, 5)$, the data are $(6, -6, 1)$ and

$$F(t) = 15t^3 - 130t^2 + 345t - 278.$$

Its three roots lie respectively in $(1, 2)$, $(2, 3)$, and $(4, 5)$. The theorem proves the same assertion for arbitrary r , not just for examples reached by root finding.

Remark 11.4 (Relation to the intrinsic invariant). The finite failure neighbourhood recovers the complex projective pencil and its spectral sheaf. To apply the real theorem one specifies a real structure and a definite member; these are not asserted to be recoverable from an unmarked complex scheme. After that specification, the construction uses only the ordered spectral points, their multiplicities, and the freely chosen β_j . The result is an independently stated consequence in reciprocal likelihood geometry, rather than a new name for Weierstrass–Segre classification. It proves the cited conjectural assertion without claiming that every real critical point is a statistically admissible maximum.

12. RECOVERING THE POLARIZED RAMIFICATION VARIETY

Let $A = \mathcal{O}_Y(1)$ and $B = \mathcal{O}_{\mathbb{P}(S)}(1)$. In the graph description (4.3) write

$$Z = Y \times \mathbb{P}(S), \quad W = \mathcal{H}^* \otimes Q, \quad Q = (S \otimes \mathcal{O})/\Lambda, \quad \pi : E = \text{Tot}_Z W \rightarrow Z,$$

and let $\rho : E \rightarrow Y$ be the composite projection. Constant one-dimensional vector-space factors in determinant lines below are suppressed; they change none of the isomorphism classes or section rings.

Lemma 12.1 (No extra fibrewise functions). *The natural map $\mathcal{O}_Y \rightarrow \rho_* \mathcal{O}_E$ is an isomorphism.*

Proof. The assertion is local on Y , where $\mathcal{H}$ can be trivialized. If H_0 is its fibre space, the affine bundle projection gives

$$\pi_* \mathcal{O}_E = \bigoplus_{d \geq 0} \text{Sym}^d(H_0 \otimes Q^*).$$

We claim that

$$H^0(\mathbb{P}(S), \text{Sym}^d(H_0 \otimes Q^*)) = 0 \quad (d > 0). \quad (12.1)$$

The injection $Q^* \hookrightarrow S^* \otimes \mathcal{O}$ embeds a global section in the constant space $\text{Sym}^d(H_0 \otimes S^*)$. Such an element is a homogeneous polynomial on $\text{Hom}(H_0, S)$. At every line $\Lambda \subset S$, membership in $\text{Sym}^d(H_0 \otimes (S/\Lambda)^*)$ says this polynomial factors through $\text{Hom}(H_0, S/\Lambda)$; it is invariant under translations by $\text{Hom}(H_0, \Lambda)$. Taking the lines of a basis of S shows invariance under every translation of $\text{Hom}(H_0, S)$. In characteristic zero such a polynomial is constant, so its positive-degree part is zero. This proves (12.1). The degree-zero part is $\mathbb{C}$.

The same calculation tensors with the coordinate ring of each affine trivializing open of Y : cohomology on the projective space commutes with this flat extension from $\mathbb{C}$, and with the displayed direct sum in degree zero. Hence the pushforward is precisely $\mathcal{O}_Y$, with its natural multiplication. The local identifications glue. $\square$

Proposition 12.2 (The canonical nilradical line). *As line bundles on E ,*

$$\mathcal{N} \simeq \pi^*(A^{-1} \boxtimes B^{-1}), \quad \omega_E \simeq \pi^*(A^{-(p-1)} \boxtimes B^{-(p+e-1)}). \quad (12.2)$$

Consequently

$$\omega_E \otimes \mathcal{N}^{\otimes -(p+e-1)} \simeq \rho^* A^e. \quad (12.3)$$

Proof. The determinant defining $\hat{\Delta}$ is a section of $(\det \mathcal{K})^* \otimes \det V$. On E the exact sequence $0 \rightarrow \Lambda \rightarrow \mathcal{K} \rightarrow \mathcal{H} \rightarrow 0$ gives $\mathcal{I}_\Delta|_E = \det \mathcal{H} \otimes \Lambda$, up to a constant factor. Since $\det \mathcal{H} = A^{-1}$ and $\Lambda = B^{-1}$, Theorem 4.2 gives the first formula.

The determinant of W is $(\det \mathcal{H}^*)^{p-1} \otimes (\det Q)^{e-1} = A^{p-1} \boxtimes B^{e-1}$. The canonical bundle of a vector-bundle total space is $\pi^*(\omega_Z \otimes \det W^{-1})$, as follows by taking the determinant of its relative cotangent sequence. Here $\omega_Z = \mathcal{O}_Y \boxtimes B^{-p}$ because $\omega_Y = \mathcal{O}_Y$. This proves the second formula. Subtracting the two exponents gives (12.3): the B -exponent vanishes and the A -exponent is $-(p-1) + (p+e-1) = e$. $\square$

Proof of Theorem 1.2. An isomorphism of failure schemes preserves the nilradical and its annihilator, hence E , its invertible module $\mathcal{N}$, and its canonical bundle. It therefore preserves $\mathcal{L} = \omega_E \otimes \mathcal{N}^{-(p+e-1)}$ and its graded section ring. Proposition 12.2, Lemma 12.1, and the projection formula give

$$H^0(E, \mathcal{L}^n) = H^0(Y, A^{en} \otimes \rho_* \mathcal{O}_E) = H^0(Y, A^{en})$$

compatibly with multiplication. The full section ring of the ample line bundle A^e recovers (Y, A^e) by Proj. In this case it is also the e th Veronese of the homogeneous coordinate ring of a smooth hypersurface, so there is no ambiguity about the grading line bundle.

For a K3 surface the Picard group has no torsion. Here is the short argument needed: a nontrivial torsion line bundle T and its inverse have no nonzero sections, since a nonzero effective torsion divisor has zero intersection with an ample divisor, which is impossible. Riemann–Roch and Serre duality would then give $2 = \chi(T) = -h^1(T) \leq 0$, a contradiction. Thus an isomorphism carrying A^4 to $(A')^4$ carries A to A' . This recovers the quartic polarization, and its complete linear system recovers the quartic embedding up to projective equivalence.

Finally Proposition 4.1 recovers D from $\hat{D}$ by restricting to the smooth locus of the reduction. This operation is intrinsic and is preserved by every abstract scheme isomorphism. The preceding proof applies to that restriction. $\square$

Remark 12.3. For $e = 3$ the theorem recovers (Y, A^3) directly. A chosen cubic root in the Picard group need not be unique. As polarized genus-one curves, however, the possible degree-three roots are isomorphic: after choosing an origin, the map $y \mapsto t_y^* A \otimes A^{-1}$ onto $\text{Pic}^0(Y)$ is multiplication by three, hence surjective. No choice of an origin is needed for the stated section-ring reconstruction. For $e = 4$ the torsion-free argument gives uniqueness without translations.

Remark 12.4 (Relation to the earlier birational argument). The compactification $\mathbb{P}_Z(\mathcal{O} \oplus W)$ is birational to $Y \times \mathbb{P}^{e(p-1)}$. Thus the earlier cancellation proof is a special case of the maximal rationally connected quotient: for a non-uniruled base, that quotient of a product with projective space is the base, birationally. See the rational-quotient construction in [14, Chapter IV, §5]. This explains the earlier abstract recovery, which is retained in the companion. The present proof does not rely on that argument: the section ring gives the actual projective model and its line bundle, not merely a birational class.

13. NINE-DIMENSIONAL VARIATION IN THE K3 CASE

We prove Theorem S5.3. Work on the standard chart of $\mathrm{Gr}(4, \mathrm{Sym}^2 \mathbb{C}^4)$ where the four square coefficients are the identity. Write $x = (x_0, x_1, x_2, x_3)$ and order the six cross terms as

$$m = (x_0x_1, x_0x_2, x_0x_3, x_1x_2, x_1x_3, x_2x_3).$$

For $C \in \mathrm{Mat}_{6 \times 4}$ set

$$q_i = x_i^2 + \sum_{k=1}^6 C_{ki} m_k, \quad J_C = (\partial q_i / \partial x_j), \quad F_C = \det J_C. \quad (13.1)$$

This parametrizes the relation space, not an additional choice of its basis: the square coefficients fix that basis on this chart.

Lemma 13.1 (An integral differential witness). *At the relation system*

$$C_* = \begin{pmatrix} 2 & -1 & 2 & 0 \\ -2 & 0 & -1 & 1 \\ 2 & -1 & -2 & 0 \\ 1 & -1 & 2 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & -2 & 1 & -2 \end{pmatrix}, \quad (13.2)$$

the differential of $C \mapsto [F_C] \in \mathbb{P}(\mathrm{Sym}^4 \mathbb{C}^4)$ has rank 24. This relation system is basepoint-free and $V(F_{C_*})$ is smooth over $\mathbb{C}$.

Proof. Let $\mathrm{cof}_{ij}(J)$ denote the signed cofactor. The 24 parameter derivatives are

$$G_{ik} = \left. \frac{\partial F_C}{\partial C_{ki}} \right|_{C_*} = \sum_{j=0}^3 \mathrm{cof}_{ij}(J_{C_*}) \frac{\partial m_k}{\partial x_j}. \quad (13.3)$$

Order degree-four monomials by descending lexicographic order of their exponent tuples, and form the 35×25 integral coefficient matrix D_* with columns F_{C_*} , followed by G_{ik} with $i = 0, \dots, 3$ outermost and $k = 1, \dots, 6$ innermost. The determinant of its first 25 rows and all columns is

$$4279473148893659522379284480 \neq 0. \quad (13.4)$$

Its residue modulo 101 is 41. Thus F_{C_*} and the 24 derivatives are linearly independent. Quotienting the target tangent space by the line spanned by F_{C_*} proves the differential assertion.

For completeness, Appendix S16 specifies the full coefficient recipe and two additional finite rank certificates. The Macaulay matrix of the q_i in degree five has rank 56, the dimension of all degree-five polynomials. The analogous matrix of the four partials of F_{C_*} in degree nine has rank 220, again the entire homogeneous piece. Explicit square minors are nonzero modulo 101, with residues 44 and 79. The integer matrices therefore have full row rank over $\mathbb{Q}$, and hence over $\mathbb{C}$. A common projective zero is impossible. This proves basepoint-freeness and smoothness without a numerical tolerance or a probabilistic test. $\square$

Proposition 13.2. *The Jacobian map on G_4° has a 24-dimensional image in the space of embedded smooth quartics. Its image in polarized quartic K3 moduli has dimension nine. Its general fibres after projective coordinate equivalence are finite.*

Proof. Changing a basis of R multiplies the determinant by a nonzero scalar, so the projective Jacobian map is well-defined. A change of coordinates in V acts equivariantly. Lemma 13.1 and the 24-dimensional domain show that its image has dimension 24 and that the map is generically finite onto that embedded image. The locus where the differential is injective is nonempty and open; it is invariant because the map is equivariant. The smooth admissible locus is irreducible by Proposition 3.3.

The projective stabilizer of a smooth quartic K3 is finite. Indeed, contraction with its nonzero two-form identifies its tangent bundle with its cotangent bundle, and $H^0(\Omega^1) = 0$ by $H^1(\mathcal{O}) = 0$ and Hodge symmetry over $\mathbb{C}$. The stabilizer, a closed finite-type subgroup of PGL_4 , therefore has zero-dimensional Lie algebra and is finite. Its orbits have dimension 15. Two smooth quartic embeddings define the same polarized K3 precisely when they are related by PGL_4 , because their embeddings are given by the complete polarizing linear systems. The geometric quotient of the smooth quartic locus gives the quartic K3 moduli space; equivalently one may compute on a generic geometric quotient. The invariant image consequently has dimension $24 - 15 = 9$ after quotienting. Standard background on polarized K3 moduli is given in [12, Chapter 5].

To check the finiteness assertion directly, take a general embedded quartic in the 24-dimensional image where the fibre is finite. Every relation system mapping to a projectively equivalent quartic can be moved by the corresponding projective transformation into that finite fibre. Thus there are only finitely many projective equivalence classes above the general polarized point. This argument also makes clear that the assertion is generic, rather than a statement about all exceptional fibres. $\square$

Proof of Theorem S5.3. Combine Proposition 13.2 with the recovery of the polarized quartic in Theorem 1.2 and the algebra-orbit equivalence of Lemma 3.4. $\square$

Corollary 13.3. *There is a flat family of failure schemes on the corank-one open whose canonical polarized invariant has a nine-dimensional image. In particular the associated K3 surfaces are not isotrivial, and the failure schemes are not all abstractly isomorphic. Their reduced divisors become the same scheme after identifying the vector spaces S_R of rank six.*

Proof. Flatness is Corollary S1.3. A fixed abstract K3 cannot give a positive-dimensional algebraic family of polarization classes on that surface: its Picard scheme is discrete, since $H^1(\mathcal{O}) = 0$. Thus the positive-dimensional polarized image also implies non-isotriviality of the surfaces. Distinct values of the canonical polarized invariant give nonisomorphic failure schemes by Theorem 1.2. The graph description of Δ in (4.2) is independent of R . $\square$

14. THE CLASSICAL INCIDENCE SURFACE AND THE NEW INVARIANT

This section distinguishes three geometric objects associated with the same relation system. Let $e = 4$ and let $q_i(x, y)$ denote the symmetric bilinear polarization of $q_i(x)$, normalized by $q_i(x, x) = q_i(x)$. Define

$$Z_R = \{([x], [y]) \in \mathbb{P}(V^*) \times \mathbb{P}(V^*) : q_i(x, y) = 0 \text{ for all } i\}. \quad (14.1)$$

Proposition 14.1. *For $R \in G_4^\circ$, projection on the first factor identifies Z_R with Y_R . Transposition of the two factors induces a fixed-point-free involution on this K3 surface. The polarization recovered in Theorem 1.2 is the first projection's quartic polarization.*

Proof. The bilinear equations at x are $J_R(x)y = 0$, up to the common nonzero factor two. Off Y_R there is no projective solution; on Y_R the matrix has rank exactly three because Y_R is smooth.

Inverting a nonzero three-minor gives a unique kernel line, with its scheme structure, and these kernel lines glue to an inverse for the projection $Z_R \rightarrow Y_R$. The equations are symmetric in x, y , so transposition acts. A fixed point would have $[x] = [y]$ and hence $q_i(x) = 0$ for every i , contrary to basepoint-freeness. The final assertion follows from the first projection and its hyperplane bundle. $\square$

The discriminant of the web is a different embedded hypersurface. If $q_i(x) = x^t A_i x$, its equation is

$$\det\left(\sum_{i=0}^3 \lambda_i A_i\right) = 0 \quad \text{in } \mathbb{P}(R), \quad (14.2)$$

whereas the ramification equation is $\det J_R(x) = 0$ in $\mathbb{P}(V^*)$. The former varies the quadric in the web; the latter varies a point of the source of the quadratic morphism. The two determinants must not be identified simply because both have degree four. The classical construction recalled in [13, §4] relates the bilinear K3 surface to the resolution of the web's symmetroid and to a nodal Enriques quotient; see also the account of Reye congruences in [9]. Proposition 14.1 places our ramification surface in that construction by an explicit kernel-line isomorphism. No claim is made that the existence of that K3 surface, its involution, or webs of quadrics is new.

The invariant used here starts instead with the abstract multiplication-failure scheme. It obtains E as the annihilator of its nilradical, constructs $\omega_E \otimes \mathcal{N}^{-9}$ without the web or either projective space, and recovers the section ring of $\mathcal{O}_{Y_R}(4)$. The corank-two theorem further recovers an additional associated support from the square of the nilradical. These are statements about nonreduced multiplication-failure schemes, not merely about the classical incidence surface or the discriminant cycle.

15. SUBSPACE VARIETIES AND THE SUPPORT CALCULATION

The support argument has two distinct ingredients. The description of tensors by contraction flattenings, the tangent formula for a Grassmannian, and the Schur decomposition are classical. The additional calculation here is the kernel of

$$\kappa_q : \det V \otimes \operatorname{Sym}^4 V \longrightarrow \bigwedge^3 \operatorname{Sym}^2 V$$

for every nonzero symmetric bilinear form q , now extended to all dimensions in Theorem 2.1, followed by its application to the two component lines recovered from the intrinsic normal cone. We give a geometric formulation that distinguishes this calculation from general rank or orbit questions.

For a vector space W define the exterior subspace variety

$$\operatorname{Sub}_r^\wedge(W) = \{[z] \in \mathbb{P}(\bigwedge^4 W) : z \in \bigwedge^4 S \text{ for some } S \subset W, \dim S \leq r\}.$$

It is the reduced closed set defined by rank at most r of $W^* \rightarrow \bigwedge^3 W$, $q \mapsto \iota_q z$. Indeed the support-annihilator identity (5.26) identifies that rank with the smallest support dimension. Thus the minors give its support; we make no assertion here that their ideal is radical. Similarly, $\operatorname{Sub}_d^{\operatorname{sym}}(V) \subset \mathbb{P}(\operatorname{Sym}^4 V)$ is the reduced closed locus of quartics in at most d essential variables. It is the first-catalecticant rank locus by [31, Proposition 1]. The spaces being measured are different: V for essential variables and $W = \operatorname{Sym}^2 V$ for exterior support.

Proposition 15.1 (Exact reduced pullbacks of support varieties). *Let $j : \mathbb{P}(\operatorname{Sym}^4 V) \hookrightarrow \mathbb{P}(\bigwedge^4 \operatorname{Sym}^2 V)$ be the projectivization of j ; the one-dimensional determinant twist does not affect*

this map. Let $\mathcal{Q} = \{[h^2] : 0 \neq h \in \text{Sym}^2 V\}$, the closed quadratic-square locus. As reduced closed subvarieties,

$$(j^{-1} \text{Sub}_4^\wedge(W))_{\text{red}} = \text{Sub}_1^{\text{sym}}(V), \quad (15.1)$$

$$(j^{-1} \text{Sub}_7^\wedge(W))_{\text{red}} = (j^{-1} \text{Sub}_8^\wedge(W))_{\text{red}} = \text{Sub}_2^{\text{sym}}(V), \quad (15.2)$$

$$(j^{-1} \text{Sub}_9^\wedge(W))_{\text{red}} = \text{Sub}_3^{\text{sym}}(V) \cup \mathcal{Q}. \quad (15.3)$$

In particular a quartic with at least three essential variables has image outside the second secant variety $\sigma_2(\text{Gr}(4, W))$, including its limiting secants.

Proof. The support values in Proposition 5.13 give the first two equalities on complex points. They also give the third: its four-essential-variable part consists exactly of squares of nondegenerate quadratics. A degenerate quadratic and its square have the same essential-variable space, since $\partial(h^2) = 2h\partial h$ in a polynomial domain. Their squares therefore already belong to $\text{Sub}_3^{\text{sym}}(V)$. The locus $\mathcal{Q}$ is closed as the image of the projective morphism $[h] \mapsto [h^2]$, which has no base points. All the displayed loci are closed and have the specified reduced structures, so equality on complex points proves the assertions.

A sum of two decomposable four-vectors lies in $\text{Sub}_8^\wedge(W)$. That rank locus is closed, so it contains the closure of all such secant lines. Equation (15.2) then proves the last assertion. This is an exclusion statement; it does not identify the entire intersection of the binary locus with $\sigma_2(\text{Gr}(4, W))$. $\square$

15.1. The classical inputs and the additional assertion. The tangent-space identity in [30, Lemma 2.1] is precisely $\hat{T}_{w_R} \text{Gr}(4, W) = \wedge^3 R \wedge W$. Its eight-dimensional support bound is the immediate linear-algebra consequence used in Lemma 5.14. We do not claim either formula, or the general study of Grassmannian secant dimensions, as new. The issue here is the position of one specified Schur subspace relative to those secants, not the dimension of a general secant variety.

The directly relevant exterior formulation is Landsberg–Ottaviani’s first skew-flattening discussion [34, §6.1], incorporated into the merged paper [35, §10.1]. It defines the exterior subspace varieties by support, gives their set-theoretic minors description, and records $\sigma_r(\text{Gr}(k, W)) \subset \text{Sub}_{rk}(\wedge^k W)$. Sheridan’s enclosing-space formulation [36, §2.4, Definitions 2.10–2.11 in the cited preprint] identifies the same minimal subspace by contraction. These are antecedents of precisely the support notion used here, not merely analogies for symmetric tensors. We do not claim the containment, the enclosing space, or its interpretation as a rank locus as new.

Those cited statements concern arbitrary exterior tensors. They do not state the rank-dependent kernel formula for the restriction $\iota_q j_n$, the even-dimensional quadratic-power exception, or the displayed reduced pullbacks along j_4 . Theorems 2.1 and 2.5 supply that restricted calculation; the secant exclusion is then its immediate consequence using the classical containment. This identifies the additional assertion relative to the statements actually compared, without purporting to certify that no other historical source contains it.

Landsberg–Weyman’s subspace varieties are defined for products of tensor factors; their Theorem 3.1 describes their equations and singularities [32]. That framework explains the use of flattening ranks as support conditions. It is not a theorem about the contraction of our determinant-twisted quartic submodule, nor does it identify the reduced pullbacks (15.1)–(15.3). The exterior rank identity needed here was proved directly, so no scheme-theoretic conclusion for alternating tensors is imported from the tensor-product statement.

The occurrence of $\mathbb{S}_{(5,1,1,1)} V$ is a plethysm calculation, not a novelty claim. Its multiplicity is one in $\wedge^4 \text{Sym}^2 V$, whose two highest weights are $(4, 3, 1, 0)$ and $(5, 1, 1, 1)$. For example its

character identity is

$$[t^4] \prod_{i \leq j} (1 + tx_i x_j) = s_{(4,3,1,0)}(x) + s_{(5,1,1,1)}(x).$$

The highest-weight and exterior-duality verification is given in Lemma 5.7; polarization fixes the normalization of its unique quartic embedding. The skew Cauchy and SL_2 character calculations used for the scalar kernel are likewise standard representation operations, expanded in Lemma 5.11 rather than cited as a new decomposition method.

The precise additional assertions proved here are the dimension-uniform all-ranks contraction kernels of Theorem 2.1 (and their four-dimensional proof in Lemma 5.12), the resulting exact support table, and the reconstruction consequence of combining that table with the intrinsic common-coordinate readout. Exterior support is neither the least number of decomposable summands nor a classification of $GL(W)$ -orbits in $\wedge^4 W$. The reduced support pullbacks above, and the containment of the ambient reconstruction obstruction over binary quartics, must not be mistaken for either such classification. The primary comparisons just made establish which cited tools are classical; they are not an exhaustive historical priority certificate. In particular the six-way theorem comparison with the unread complete Ballico source [2] remains separately unresolved; the publisher first-page definitions do not settle its later theorem statements.

15.2. Exterior symmetric powers, harmonics, and cofactor identities. There are closer representation-theoretic antecedents than general support varieties. Cheng–Wang [37, §2, Theorem 3.5] record the classical exterior-symmetric-square decomposition and its superalgebra extension. Specializing the odd dimension to zero gives the multiplicity-free summands indexed by nested hooks, including $\mathbb{S}_{(n+1, 1^{n-1})} V = \det V \otimes \text{Sym}^n V$. Thus its occurrence, its multiplicity, and the two four-dimensional character identities used here are classical plethysm. The specified pure-power formula fixes a scalar normalization; it does not create a new representation or a new branching rule.

The harmonic decomposition is likewise classical. De Bie–Eelbode–Roels [38, §1, equations (1)–(3)] describe its one-variable form; their Theorem 2.1 develops the two-variable Howe picture under their dimension convention. We do not import that convention into the present low-dimensional cases. Lemma 2.4 proves exactly the full O_n assertion, including O_2 , used in the kernel calculation. The reduction of a kernel to inequivalent harmonic summands is a standard representation argument. What must still be computed for this particular map is which restrictions are zero.

The identity $\sum_i \partial_i A_i = 0$ in Lemma 2.3 is the exterior-valued cofactor cancellation underlying the Piola identity; compare [39, equation (1.1), §3.1]. Commutation of mixed partials and the algebraic Gaussian integration-by-parts recurrence are classical tools. Their use here establishes the scalar exceptional kernel. The nonzero coefficient $m(n+m-2)/(n(n-1))$ establishes injectivity on every other harmonic summand. Neither cofactor cancellation nor harmonic irreducibility is claimed as a new identity.

These comparisons separate three levels of assertion. The module occurrence, harmonic branching, and the ambient skew-flattening bound are classical. The all-rank restriction $\iota_q j_n$ is calculated explicitly here, using those tools; the cited branching statements alone do not specify its singular-radical kernel or its nondegenerate parity alternative. The geometric assertion is that intrinsic nonreduced normal cones recover the required coefficient data through one common projective transformation. Theorem 6.1 separates that assertion from recombination, and Theorem 6.3 verifies it for another class of Fitting schemes. This is a theorem-by-theorem boundary relative to the inspected statements, not certification of exhaustive priority.

15.3. The restricted contraction versus classical operators. The relation with classical invariant theory is more precise than the shared occurrence of a representation. Howe’s [42, Theorem 9 and §4(a), pp. 555–557] gives harmonic irreducibility and the orthogonal decomposition

through the commuting quadratic multiplication and Laplacian operators. Thus the passage to harmonic summands in the proof of Theorem 2.1 is classical, including its full-orthogonal-group interpretation. The additional step is the restriction of the particular exterior contraction to each summand, not a new harmonic decomposition.

The different operators can be compared without terminology. With a volume fixed, the Jacobian map in two variables is

$$C_2(a \wedge b) = a_{x_1} b_{x_2} - a_{x_2} b_{x_1}.$$

It is the first binary transvectant, with the normalization for quadratic inputs in [43, §2.2, Definition 2.6 and Remark 2.7]. Thus neither that covariant nor its polarization is new. Moreover, when $n = 2$, j_2 is an isomorphism onto $\wedge^2 \text{Sym}^2 V$, so for every nonzero q

$$\ker \kappa_{2,q} = j_2^{-1} \left(\bigwedge^2 \ker[q : \text{Sym}^2 V \rightarrow \mathbb{C}] \right). \quad (15.4)$$

This one-dimensional kernel is elementary exterior algebra.

For general n , a quadratic apolar differential operator has the signature $q(\partial) : \text{Sym}^n V \rightarrow \text{Sym}^{n-2} V$. In orthonormal coordinates its kernel is $\mathcal{H}_n$, of dimension $\binom{2n-1}{n} - \binom{2n-3}{n-2}$. In contrast the operator studied here has signature

$$\kappa_{n,q} : \det V \otimes \text{Sym}^n V \longrightarrow \bigwedge^{n-1} \text{Sym}^2 V, \quad \kappa_{n,q} = \iota_q j_n.$$

For nondegenerate q its kernel is zero in odd dimension and the quadratic-power line in even dimension. These are not the apolar harmonic kernels. For singular q , the proof uses its unipotent radical shears as well as its nondegenerate orthogonal factor; a full-rank harmonic branching statement alone cannot supply that argument. The normalization, nonzero harmonic coefficients, and singular-radical calculation are supplied in Section 2.

For comparison with unrestricted exterior contraction, take $0 \neq q \in W^*$, where $W = \text{Sym}^2 V$. Choose $w_0 \in W$ with $q(w_0) = 1$. The decomposition

$$\bigwedge^n W = \bigwedge^n \ker q \oplus w_0 \wedge \bigwedge^{n-1} \ker q$$

shows that $\ker \iota_q = \bigwedge^n \ker q$. Accordingly, the precise restricted problem is

$$\ker \kappa_{n,q} = j_n^{-1} \left(\text{im } j_n \cap \bigwedge^n \ker q \right). \quad (15.5)$$

Knowing the ambient kernel does not compute this intersection. In the singular case the answer is $\det V \otimes \text{Sym}^n(\text{rad } q)$; in the nondegenerate case it is zero for odd n and the quadratic-power line for even n . The calculation in Section 2 keeps the normalization $C_n j_n = 2 \text{id}$, the full orthogonal stabilizer, and its singular radical shears. These specify an equivalent-map comparison that is sharper than comparing representation names.

Finally, the universal coefficient map (7.6) is classical matrix-coefficient representation theory. What is proved geometrically is that its specified left subspaces can be recovered from an abstract failure scheme, with one common projective transformation. In the natural pencil family there is only one such subspace, and exterior duality recovers the full pencil. The classical Weierstrass–Segre classification and the pencil–hyperelliptic correspondence classify objects after a pencil has been supplied; they do not enter the proof that the abstract scheme supplies that pencil. These comparisons identify the exact additional arguments relative to the inspected statements, without claiming exhaustive historical nonanticipation.

ACKNOWLEDGMENT OF ASSISTANCE

AI assistance was used in developing and drafting the arguments and finite verification scripts. All structural proofs are supplied for independent mathematical scrutiny; computational receipts are not formal proof certificates. No AI system is listed as an author.

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